

The main goal of nonparametric regression is the flexible modeling of effects of continuous covariates on a dependent variable. We have already seen in several practical applications that a purely linear model is not always sufficient. This insufficiency could either result from theoretical considerations about the given application or simply from uncertainty about the specific form of an effect that a covariate has on the response. In Sect. 3.1.3, we considered two possible approaches for modeling the nonlinear effect of a continuous covariate: simple transformations and polynomials. We found that scatter plots of the residuals, in particular partial residual plots, are useful diagnostic devices for identifying nonlinear relations between covariates and response. Even though these methods may be sufficient for simple relationships, they quickly become intractable in situations with a larger number of potential covariates. Moreover, due to the limited number of transformations that are actually used in practice, the functional forms associated with covariate effects are both limited and not very flexible. In this chapter, we will therefore introduce flexible regression techniques that enable the automatic, data-driven estimation of nonlinear effects. We will first concentrate on models for the effect of one single continuous covariate on a response variable, having an approximately normal distribution. The methods developed in this context will be the basis for the bivariate smoothing methods presented in Sect. 8.2, and the more complex additive models of Chap. 9.

To motivate the following discussion, we consider two examples. In the first, we analyze the risk of malnutrition of children in developing countries (Tanzania, in this case), similar to Example 1.2 (p. 5). We consider chronic malnutrition measured in terms of a Z-score as the response variable and examine its dependence on the age of the child. Figure 8.1 shows the corresponding scatter plot for only one of the districts in Tanzania (Ruvuma). We will limit our considerations in this chapter to this single district, since this yields clearer graphical presentations due to the relatively small sample size. However, in general, the results carry over to the entire data set. Figure 8.1 suggests a nonlinear relationship, where young children have a higher Z-score than older children, implying that young children suffer from a lower risk of malnutrition than older ones. At first glance, a simple polynomial or



Fig. 8.1 Malnutrition in Tanzania: scatter plot of the Z-score for chronic malnutrition versus the age of the child in months for one of the districts in Tanzania (Ruvuma)

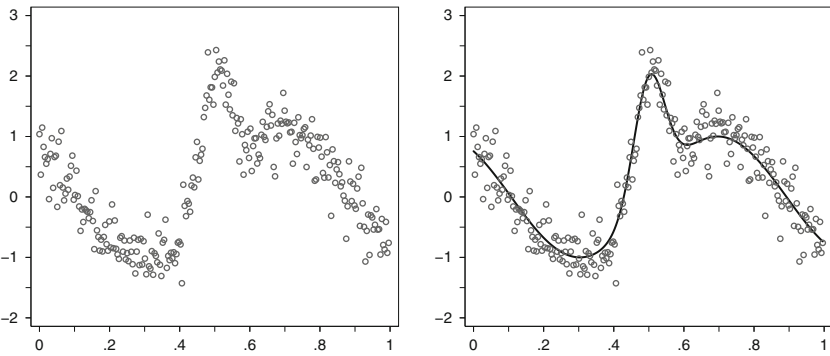


Fig. 8.2 Scatter plot of a simulated data set with nonlinear effect of the covariate: The *right panel* additionally shows the true covariate effect. The data have been simulated according to the model $y = f(x) + \varepsilon$ where $f(x) = \sin(2(4x - 2)) + 2 \exp(-(16^2)(x - 0.5)^2)$ and $\varepsilon \sim N(0, 0.3^2)$

the inverse transformation of age appears adequate for modeling the age effect. In the approaches presented in Sect. 8.1, we will, however, learn that a more flexible model provides additional insights.

As a second example, we use the simulated data set presented in Fig. 8.2. The left panel shows a scatter plot of the simulated data, whereas the right panel also provides the underlying true effect of the covariate. Despite the fact that the scatter plot indicates the structure of the true function fairly well, it is rather difficult to approximate this function through a transformation or in terms of simple polynomials.

8.1 Univariate Smoothing

In this first basic section, we examine several nonparametric regression approaches, which allow flexible modeling of the effect of one continuous covariate on a continuous dependent variable. Such approaches are called *scatter plot smoothers* since the data can be visualized best in a scatter plot, and the goal is to determine a smooth function representing the effect of the covariate. More precisely, we assume that data are given in the form (y_i, z_i) , $i = 1, \dots, n$, where the y_i are observations of the response variable and the z_i represent the corresponding values of the continuous covariate. In the standard univariate nonparametric regression model, we assume that we are able to explain the response variable through a deterministic function of the covariate plus an additive error term:

$$y_i = f(z_i) + \varepsilon_i.$$

Different assumptions regarding the function f then lead to different modeling possibilities. In order to obtain a simpler estimation problem, we often impose qualitative constraints concerning the smoothness of the function f , e.g., regarding its continuity or differentiability. We also should point out that some nonparametric smoothing approaches are, despite their name, in fact purely parametric. In contrast to the regression models we have examined thus far, these models are usually determined by a large number of parameters, so that individual parameters no longer have a meaningful interpretation. Nevertheless, we will continue to use the term nonparametric regression in our presentation, but will also exchangeably use the notion of univariate smoothing.

In nonparametric regression models, we make the same assumptions about the error term as in the classical linear model of Chap. 3: The errors are independent and identically distributed with

$$E(\varepsilon_i) = 0 \quad \text{and} \quad \text{Var}(\varepsilon_i) = \sigma^2, \quad i = 1, \dots, n.$$

Similar as in the linear model, it follows

$$E(y_i) = f(z_i) \quad \text{and} \quad \text{Var}(y_i) = \sigma^2, \quad i = 1, \dots, n,$$

i.e., the expected value of the response variable is to be modeled through the function f . In some situations, we will additionally assume that the errors follow a normal distribution, especially when constructing confidence intervals for f .

8.1.1 Polynomial Splines

As a first approach for nonparametric regression, we consider *polynomial splines* or *regression splines* which are closely related to the idea of polynomial regression modeling. Therefore, we revisit the polynomial model

$$f(z_i) = \gamma_0 + \gamma_1 z_i + \dots + \gamma_l z_i^l,$$

8.1 Univariate Smoothing

Data

Measurements (y_i, z_i) , $i = 1, \dots, n$, for a continuous response variable y and a continuous covariate z .

Model

$$y_i = f(z_i) + \varepsilon_i$$

with independent and identically distributed errors and

$$E(\varepsilon_i) = 0 \quad \text{and} \quad \text{Var}(\varepsilon_i) = \sigma^2.$$

In some cases, we additionally assume that the errors are i.i.d. normally distributed, so that

$$\varepsilon_i \sim N(0, \sigma^2).$$

which models the effect of covariate z on responses y as a polynomial of degree l . This model is within the scope of linear models discussed in Chap. 3 and, thus, we are able to determine the regression coefficients of the polynomials and the complete function f using ordinary least squares. In contrast to Chap. 3, we denote the regression coefficients as γ_j , which will allow us to distinguish the more flexible models discussed in this chapter from linear models.

Based on the simulated data example, Fig. 8.3 shows why a purely polynomial model is often insufficient to estimate nonlinear functions $f(z)$. We considered polynomials up to a degree of $l = 15$ for estimating f . As seen, polynomials are not able to properly reproduce the local maximum at $z = 0.5$, even with (the very high) degree of $l = 11$. For $l = 15$ this maximum is nearly obtained, but the estimated relationship is very wiggly and not quite satisfactory in other regions.

To make the polynomial model more flexible, we next partition the domain of the covariate into intervals and estimate separate polynomials in each interval. In other words, we use several locally defined polynomials instead of one global model. Yet we can still estimate each of the polynomial parts separately using least squares. The left panel in Fig. 8.4 illustrates this approach. Here we split up the domain of z into ten intervals of width 0.1 and fit separate polynomials in each. This actually leads to a more flexible estimate of the function, which reflects the form of the true function. However, the estimated function also exemplifies the major disadvantage of piecewise polynomial models: Since the polynomials are fit separately on the intervals, the polynomial pieces do not lead to an overall smooth function and, in particular, they show different function values at the interval boundaries. Therefore, it would be desirable to impose further smoothness restrictions on the function

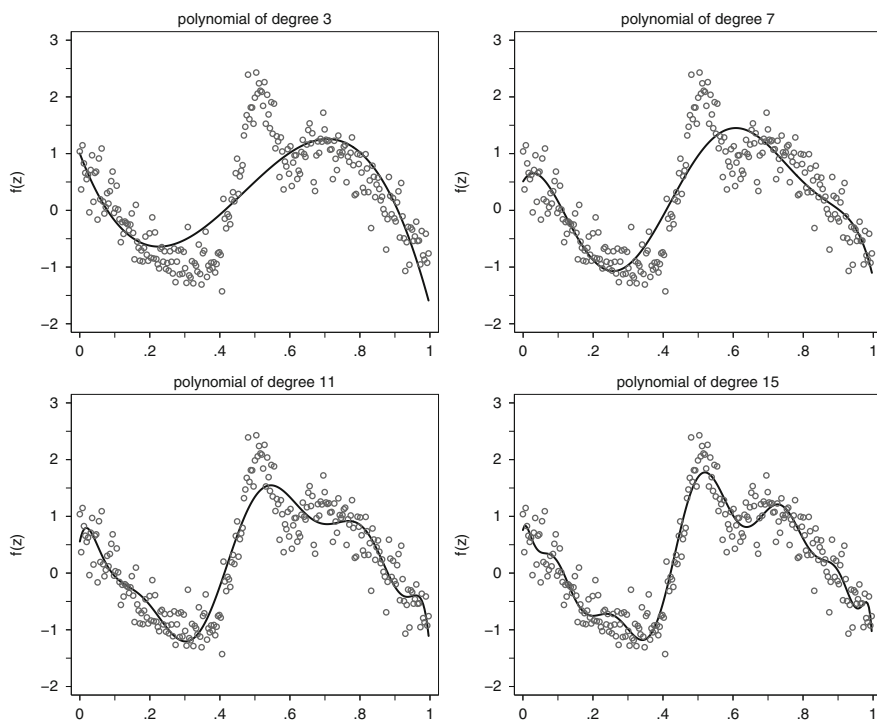


Fig. 8.3 Polynomial regression models for the simulated data set

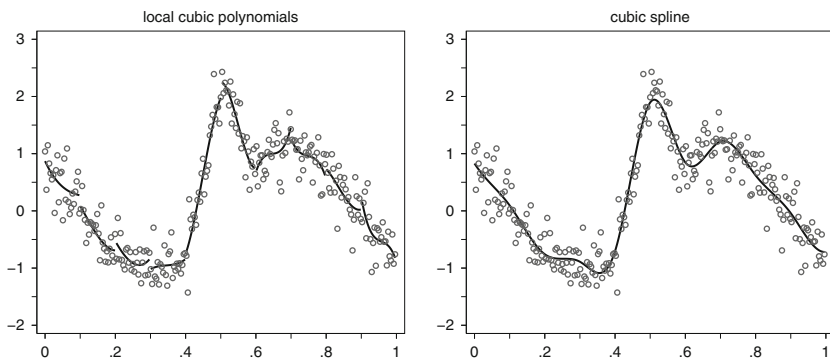


Fig. 8.4 Piecewise polynomial regression (*left*) and polynomial splines (*right*)

at the interval boundaries to obtain a function estimate similar to the one shown in the right panel of Fig. 8.4. Introducing these additional smoothness constraints leads to the class of *polynomial splines*. The main idea still is to define local polynomials on intervals in the domain of the covariate; however, to guarantee

sufficient smoothness, we additionally require that the resulting function should be $(l - 1)$ -times continuously differentiable at the interval boundaries. This yields the following definition:

8.2 Polynomial Splines

A function $f : [a, b] \rightarrow \mathbb{R}$ is called a polynomial spline of degree $l \geq 0$ with knots $a = \kappa_1 < \dots < \kappa_m = b$, if it fulfills the following conditions:

1. $f(z)$ is $(l - 1)$ -times continuously differentiable. The special case of $l = 1$ corresponds to $f(z)$ being continuous (but not differentiable). We do not state any smoothness requirements for $f(z)$ when $l = 0$.
2. $f(z)$ is a polynomial of degree l on the intervals $[\kappa_j, \kappa_{j+1})$ defined by the knots.

This summarizes our initial considerations, yielding a piecewise polynomial function where the partition of the covariate domain results from the specification of the *knots* $\kappa_1 < \dots < \kappa_m$. Moreover, the function is assumed to be $(l - 1)$ -times continuously differentiable to ensure the desired smoothness restrictions at the knots. Figure 8.5 shows some simple polynomial splines with a small number of knots to demonstrate the impact of the chosen spline degree. While the degree l of the spline determines the global smoothness, the diversity of available functions is mainly driven by the number of knots. The more knots we use the higher the number of piecewise polynomials that constitute the polynomial spline. In the following sections, we will discuss the influence of the spline degree and the number (as well as the location) of knots in more detail.

Prior to using polynomial splines in nonparametric regression, we need a representation of the set of polynomial splines for a given degree and knots configuration. This can be achieved with different but equivalent approaches. In the following two sections, we will discuss the two most popular variants: the truncated power series and B-splines.

Polynomial Splines and the Truncated Power Series

Consider the regression model

$$y_i = \gamma_1 + \gamma_2 z_i + \dots + \gamma_{l+1} z_i^l + \gamma_{l+2} (z_i - \kappa_2)_+^l + \dots + \gamma_{l+m-1} (z_i - \kappa_{m-1})_+^l + \varepsilon_i$$

with

$$(z - \kappa_j)_+^l = \begin{cases} (z - \kappa_j)^l & z \geq \kappa_j, \\ 0 & \text{otherwise.} \end{cases}$$

The first part of this model is a global polynomial of degree l , as presented at the beginning of this chapter (but with a different way of indexing where the intercept is denoted as γ_1 instead of γ_0 for reasons that will become clearer later on). In contrast, the coefficient of the highest polynomial changes at every single knot $\kappa_2, \dots, \kappa_{m-1}$.

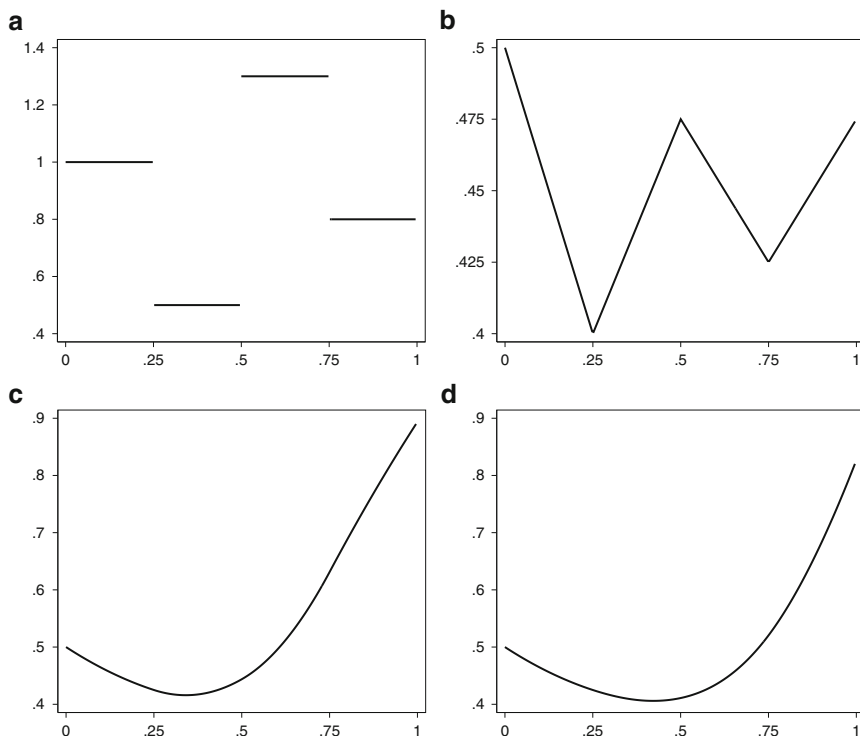


Fig. 8.5 Examples of polynomial splines of degree 0, 1, 2, and 3 with knots $\kappa_1 = 0$, $\kappa_2 = 0.25$, $\kappa_3 = 0.5$, $\kappa_4 = 0.75$, and $\kappa_5 = 1$. (a) Spline of degree 0, (b) Spline of degree 1, (c) Spline of degree 2, (d) Spline of degree 3

On the one hand, this specification allows the use of local polynomials in every interval defined by the knots, and on the other hand it fulfills the demand for global smoothness. Figure 8.6 illustrates the concept for a polynomial spline with $l = 1$. Panel (a) shows the functions that define the model, i.e., the global polynomial function of degree $l = 1$ (dashed line) and the additional truncated portions (solid lines). We scale these functions with estimated regression coefficients according to the given data, yielding panel (b). We used equally spaced knots with a distance of 0.1 to define the basis functions.

The horizontal line at $y \approx 0.8$ then corresponds to the global constant γ_1 . In the first interval $[0, 0.1)$, we obtain a decreasing function starting from this global level, represented by the slope parameter γ_2 . From the knot $\kappa_2 = 0.1$ onward, γ_3 superimposes this slope. In our example, γ_3 is also negative resulting in a somewhat steeper decreasing trend of the function. The positive coefficient γ_4 decreases the negative slope from $\kappa_3 = 0.2$ onward. Nevertheless, we still have a negative trend. When considering the further trend of the function, the coefficients γ_j indicate the change of the slope that occurs at the corresponding knot κ_{j-1} . Due to the special

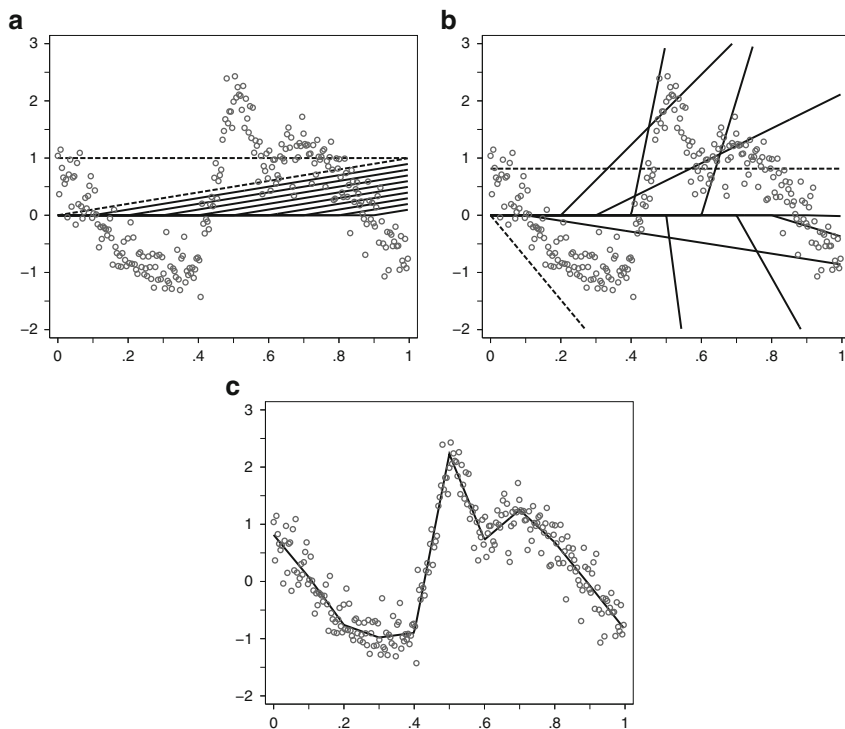


Fig. 8.6 Polynomial spline fit with linear truncated polynomials. (a) Basis functions, (b) Scaled basis functions (c) Sum of scaled basis functions

construction of the truncated function $(z - \kappa_j)_+^l$, we ensure that the change in slope is smooth enough so that the properties of a polynomial spline are preserved. Finally, when adding up all of the scaled functions, we obtain the fit of $f(z)$ illustrated in Fig. 8.6c.

More formally, it can be shown that each polynomial spline of degree l with knots $\kappa_1 < \dots < \kappa_m$ can be uniquely determined as a linear combination of the $d = m + l - 1$ functions:

$$\begin{aligned} B_1(z) &= 1, \quad B_2(z) = z, \quad \dots, \quad B_{l+1}(z) = z^l, \\ B_{l+2}(z) &= (z - \kappa_2)_+^l, \quad \dots, \quad B_d(z) = (z - \kappa_{m-1})_+^l \end{aligned}$$

(see Figs. 8.7 and 8.8, in which these functions are shown for $l = 0$ and $l = 2$ based on a small number of knots). This yields the following representation of the nonparametric regression problem:

$$y_i = f(z_i) + \varepsilon_i = \sum_{j=1}^d \gamma_j B_j(z_i) + \varepsilon_i.$$

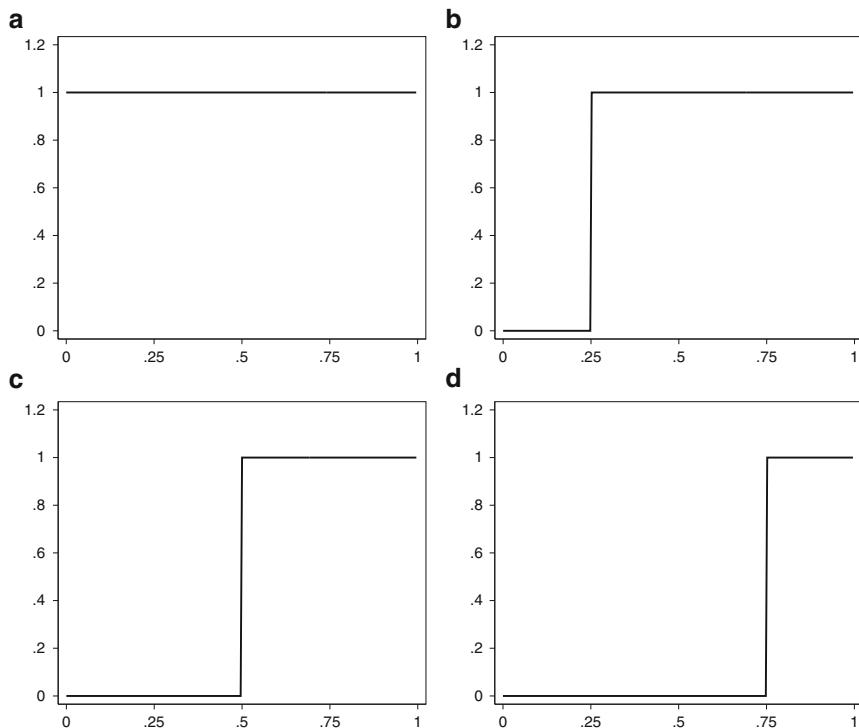


Fig. 8.7 TP basis for splines of degree 0 based on the knots $\{0, 0.25, 0.5, 0.75, 1\}$. (a) Basis function B_1, (b) Basis function B_2, (c) Basis function B_3, (d) Basis function B_4

We call the functions $B_1, \dots, B_d$ *basis functions*, since we can uniquely represent all polynomial splines by using these functions. To distinguish between various bases for the function space of all splines, we call this basis the *truncated power series basis* (TP basis). One can show that polynomial splines indeed form a d -dimensional vector space so that it is justified to call the TP basis a basis.

Modeling $f(z)$ as a polynomial spline has the advantage that we can still understand the nonparametric regression model as a linear model, but with a possibly large number of parameters. If we define the vectors of the observed response variables $\mathbf{y}$ and the errors $\boldsymbol{\varepsilon}$, as well as the design matrix

$$\mathbf{Z} = \begin{pmatrix} B_1(z_1) & \dots & B_d(z_1) \\ \vdots & & \vdots \\ B_1(z_n) & \dots & B_d(z_n) \end{pmatrix} = \begin{pmatrix} 1 & z_1 & \dots & z_1^l & (z_1 - \kappa_2)_+^l & \dots & (z_1 - \kappa_{m-1})_+^l \\ \vdots & & & & & & \vdots \\ 1 & z_n & \dots & z_n^l & (z_n - \kappa_2)_+^l & \dots & (z_n - \kappa_{m-1})_+^l \end{pmatrix},$$

we obtain the equation

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon},$$

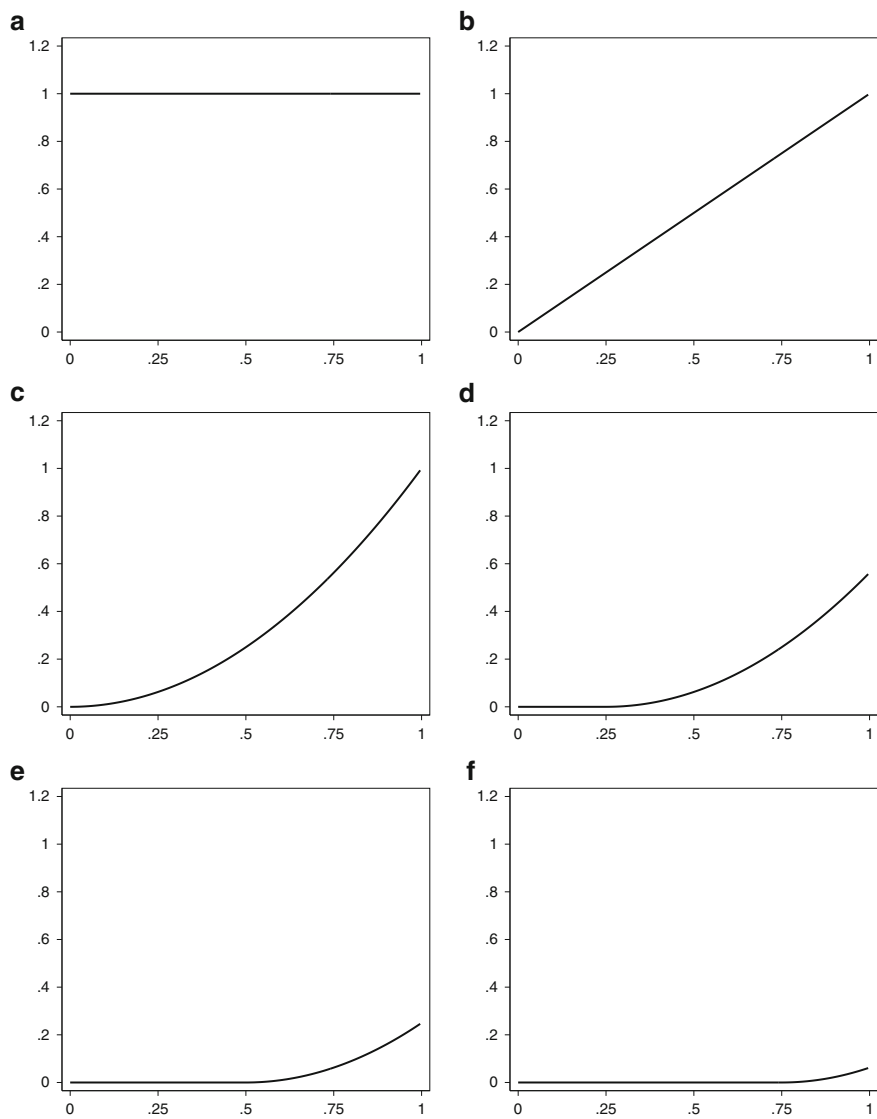


Fig. 8.8 TP basis for splines of degree 2 based on the knots $\{0, 0.25, 0.5, 0.75, 1\}$. (a) Basis function B_1 , (b) Basis function B_2 , (c) Basis function B_3 , (d) Basis function B_4 , (e) Basis function B_5 , (f) Basis function B_6

with the coefficient vector $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_d)'$. This is indeed a linear model with regression coefficients $\boldsymbol{\gamma}$. The usual least squares estimate is thus

$$\hat{\boldsymbol{\gamma}} = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{y}.$$

However, in contrast to the linear models described in Chap. 3, an interpretation of the individual estimated parameters $\hat{\gamma}_j$ is not very informative. Rather, we are interested in the form of the estimated function, which results from the estimated coefficients. In other words, we consider

$$\hat{f}(z) = \mathbf{z}'\hat{\boldsymbol{\gamma}}$$

with $\mathbf{z} = (B_1(z), \dots, B_d(z))'$ depending on the chosen covariate value z . We can then assess the quality of the model fit in a scatter plot of the data, using the estimated curve. We will return to appropriate measures for assessing the model fit in a subsequent section.

Example 8.1 Malnutrition in Tanzania—Modeling with Polynomial Splines

In our first illustration of nonparametric regression, we compare different specifications for the spline degree and the number of knots using the Tanzania data. Figure 8.9 displays various corresponding fits. Whereas the spline degree mainly determines the overall smoothness of the estimated function, the number of knots influences the flexibility of the estimated curve. For example, with the spline degree $l = 0$, we obtain a piecewise constant function. However, such a fit is inconsistent with the suggestions of a smooth trend for the Z-score with varying age. Increasing the spline degree accordingly, we move from a continuous ($l = 1$) to a continuous and differentiable function ($l = 2$).

For comparison, the numbers of knots are varied (11 and 21). The larger number of knots leads to a “wigglier” estimate for the function, irrespective of the chosen spline degree. It appears that the estimation of the function with 11 knots corresponds better to the theoretical understanding regarding the relationship between age and malnutrition than the one with 21 knots where the additional roughness is difficult to interpret in a meaningful way. It is interesting to note that, in addition to the low risk of malnutrition of young children, we see an unexpected increase in the estimated function after 24 months. This increase actually has an assignable cause due to a change in the reference population to which the Tanzanian children are compared. For young children up to 24 months age, the reference population consists of US American children from white parents of a high socioeconomic status, whereas for older children the reference population is a representative sample from all US American children. This change in reference population is what produces an artificially higher Z-score or a perceived lower risk of malnutrition for the age of 24 months. It is unlikely that we would have been able to identify this effect of the change in the reference population with the use of simple transformations or low-order polynomials.

△

Influence of the Knots

Example 8.1 showed that the choice of the number and the position of the knots, as well as the spline degree, have a definite effect on function estimation. Cubic splines are often used as a default, since they lead to a smooth, twice continuously differentiable function. In contrast, it is much more difficult to provide a “rule of thumb” for the number of knots. As already shown in Example 8.1, a higher

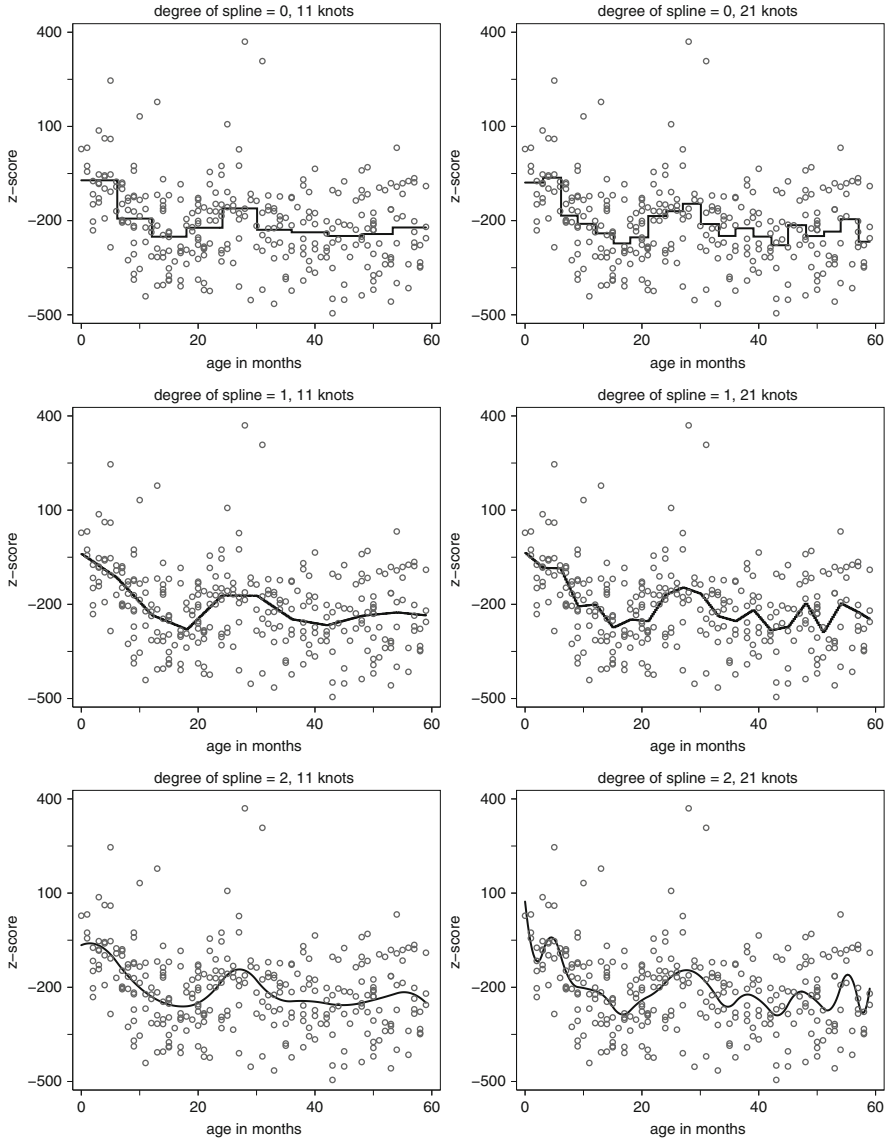


Fig. 8.9 Nonparametric estimates for the age effect based on polynomial splines

(lower) number of knots usually leads to a more (less) flexible estimated function. Figure 8.10 illustrates this effect for simulated data, using a cubic spline while varying the numbers of knots. Since we know the true function for the simulated data, it is easier to assess the quality of the estimates in relation to the number of knots in this case.

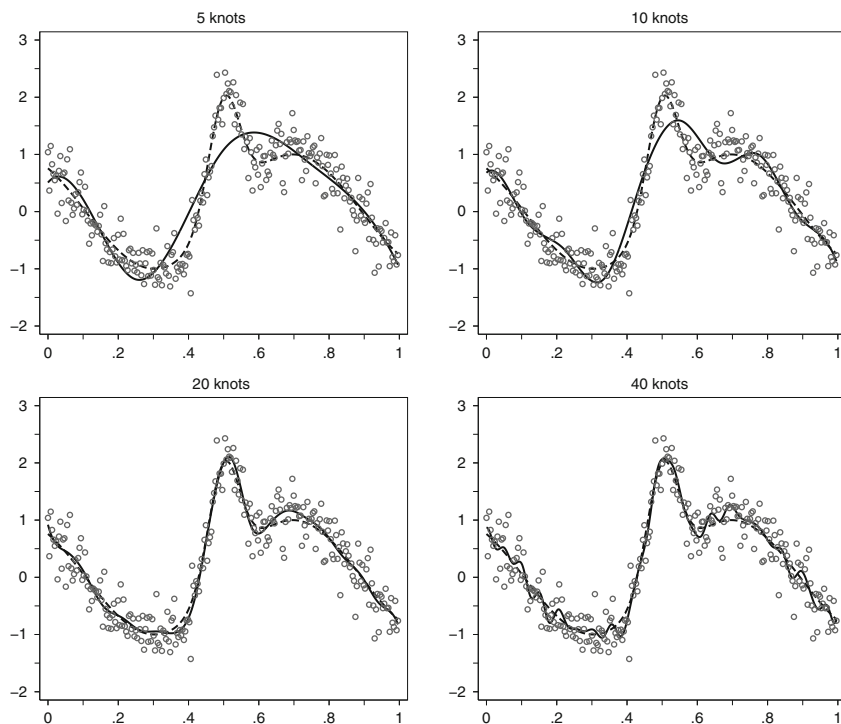


Fig. 8.10 Impact of the number of knots on cubic spline fits: The estimated function is represented as a *solid line* while the true function is superimposed as a *dashed line*

We find results similar to those found in Example 8.1: For a small number of knots, the resulting function estimate is very smooth, and, in our example, does not fit the true function well. In particular, the maximum at $z = 0.5$ is not achieved. We do get more satisfactory results with 20 knots, while further increasing the number of knots produces additional artifacts and results in very rough estimates that are difficult to interpret.

In addition to the number of knots, we also have to choose the distribution or the position of the knots along the covariate axis. The following three approaches are commonly used in practice:

- *Equidistant knots*: The domain $[a, b]$ of z is split into $m - 1$ intervals of width

$$h = \frac{b - a}{m - 1}$$

in order to obtain the knots

$$\kappa_j = a + (j - 1) \cdot h, \quad j = 1, \dots, m.$$

In all examples considered so far, we have always tacitly assumed equidistant knots.

- *Quantile-based knots:* Use the $(j - 1)/(m - 1)$ -quantiles ($j = 1, \dots, m$) of the observed covariate values $z_1, \dots, z_n$ as knots. By doing so, we place many knots in areas where we have a large number of observations and therefore the distribution of the knots better adapts to the distribution of the explanatory variable.
- *Visual knot choice based on a scatter plot:* Studying the scatter plot of the data allows us to (subjectively) target the placement of the knots either to adapt the knot density to the variability of data or to account for the specific context of the estimation problem.

None of these strategies, however, answers the main question regarding the number of knots. To overcome this problem, we basically have two possibilities: the regularization of the estimation problem through the introduction of a penalty (similar to ridge regression, see Sect. 3.4.4), or the adaptive (i.e., the automatic, data-driven) selection of knots with the help of model choice strategies. Penalization approaches will be the main focus of this book and will be discussed in Sects. 8.1.2–8.1.6. Adaptive methods will be sketched in Sect. 8.1.10. First, we consider an alternative representation of polynomial splines.

B-Splines

Although the TP basis is easy to understand, alternative bases for polynomial splines with numerically favorable properties exist. These alternatives will also be useful in the construction of penalization approaches in the subsequent sections. Due to their construction from truncated polynomials, the calculation of TP basis functions can lead to numerical instabilities for covariates with large values. Moreover, the basis functions of the TP basis are nearly collinear (nearly linear dependent), especially in cases when two knots are very close to each other. Consequently, we will choose the *basic spline* or *B-spline basis* as an alternative basis for polynomial splines.

Prior to giving an exact mathematical definition of B-splines, we first informally motivate the construction of the basis functions. We start with the observations that we made at the beginning of the chapter about the approximation of $f(z)$ with piecewise polynomials. There we realized that additional smoothness conditions had to be imposed on the function $f(z)$. B-spline basis functions are now constructed from piecewise polynomials that are fused smoothly at the knots to achieve the desired smoothness constraints. More specifically, a B-spline basis function consists of $(l + 1)$ polynomial pieces of degree l , which are joined in an $(l - 1)$ -times continuously differentiable way. Figure 8.11 displays the resulting basis function for the spline degrees $l = 0, 1, 2, 3$. All B-spline basis functions are set up based on a given knot configuration. Figure 8.12 displays such complete B-spline bases for equidistant and unevenly distributed knots. Using the complete basis, the function $f(z)$ can again be represented through a linear combination of $d = m + l - 1$ basis functions, i.e.,

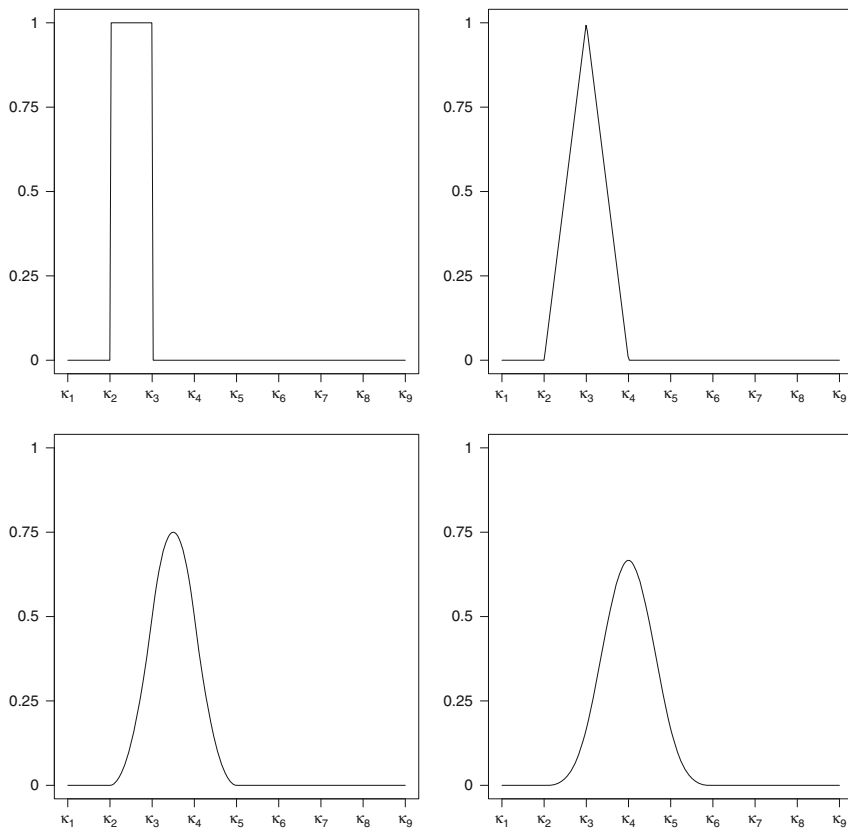


Fig. 8.11 Single B-spline basis functions for degrees $l = 0, 1, 2, 3$ and equidistant knots

$$f(z) = \sum_{j=1}^d \gamma_j B_j(z).$$

The main advantage of the B-spline basis is its local definition. In contrast to the truncated polynomials of the TP basis with positive values starting from a certain knot, B-spline basis functions are only positive on an interval based on $l + 2$ knots. Moreover, B-spline basis functions are bounded from above so that the numerical problems of the TP basis do not occur.

From Fig. 8.11, we can immediately deduce the definition for B-splines of order $l = 0$ as

$$B_j^0(z) = I(\kappa_j \leq z < \kappa_{j+1}) = \begin{cases} 1 & \kappa_j \leq z < \kappa_{j+1} \\ 0 & \text{otherwise} \end{cases} \quad j = 1, \dots, d-1,$$

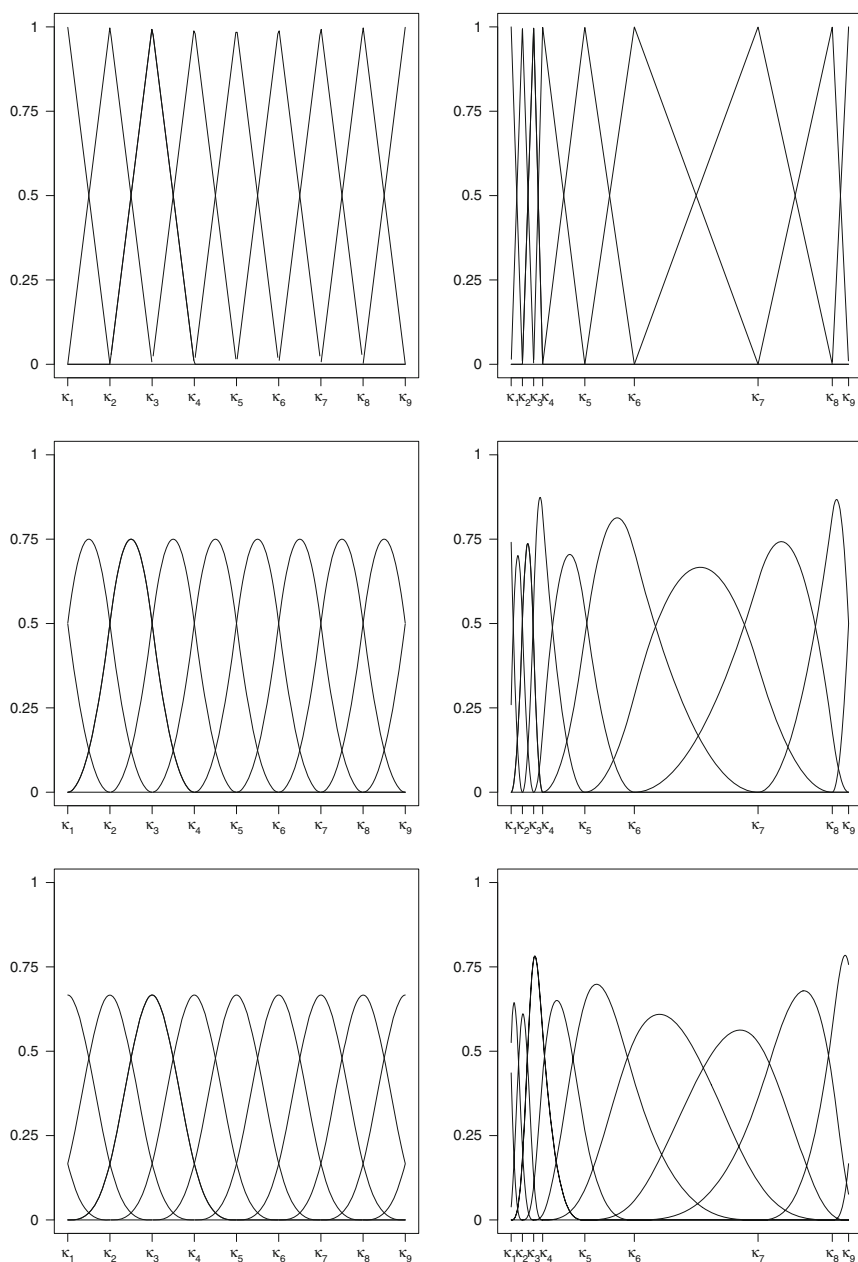


Fig. 8.12 B-spline bases of degree $l = 1, 2, 3$ with equidistant knots (*left panel*) and unevenly distributed knots (*right panel*)

where $I(\cdot)$ denotes the indicator function. In this case, it is also easy to show the equivalence to the TP basis. If we build successive differences of the TP basis functions of degree $l = 0$, we obtain functions that are constant over intervals formed by two adjacent knots, and therefore form a B-spline basis. For B-splines of a higher order, analogous representations based on combinations of piecewise polynomials of degree l can be derived. For example, we obtain the basis functions of degree $l = 1$ as

$$B_j^1(z) = \frac{z - \kappa_{j-1}}{\kappa_j - \kappa_{j-1}} I(\kappa_{j-1} \leq z < \kappa_j) + \frac{\kappa_{j+1} - z}{\kappa_{j+1} - \kappa_j} I(\kappa_j \leq z < \kappa_{j+1}),$$

i.e., each basis function is defined by two linear segments on the intervals $[\kappa_{j-1}, \kappa_j)$ and $[\kappa_j, \kappa_{j+1})$, which are continuously combined at the knot κ_j . In general, higher-order B-splines are defined recursively:

$$B_j^l(z) = \frac{z - \kappa_{j-l}}{\kappa_j - \kappa_{j-l}} B_{j-1}^{l-1}(z) + \frac{\kappa_{j+1} - z}{\kappa_{j+1} - \kappa_{j+1-l}} B_j^{l-1}(z).$$

Applying this formula, for example, for B-splines of order $l = 1$, we obtain exactly the same expression as above. To use the recursive definition of B-splines for the calculation of the basis functions, we need $2l$ *outer knots* outside of the domain $[a, b]$ in addition to the *interior knots* $\kappa_1, \dots, \kappa_m$. This leads to the expanded knots sequence $\kappa_{1-l}, \kappa_{1-l+1}, \dots, \kappa_{m+l-1}, \kappa_{m+l}$. The definition of this enlarged number of knots is straightforward for equidistant knots, since we can use the same distance h when defining knots outside of the domain $[a, b]$. This, however, is not possible for unequally distributed knots. Here, it is common to use the distance between the two smallest or largest knots within the covariate domain.

The recursive definition of B-splines makes them appear more complicated than TP-splines; however, this apparent disadvantage is easily balanced with the better numerical properties associated with B-splines. Moreover, most statistical programs (e.g., R or STATA) offer built-in implementations for B-splines so that the user often does not have to deal with such details. For equidistant knots B-splines can also be easily computed from truncated polynomials; see [Eilers and Marx \(2010\)](#) for details.

We next summarize some of the basic characteristics of B-spline basis functions, which can informally be derived from Figs. 8.11 and 8.12:

1. *B-splines form a local basis.* Each basis function is positive only in an interval formed by $l + 2$ adjacent knots. When using equidistant knots, all basis functions have the same form and are only shifted along the z -axis. At any point $z \in [a, b]$, $l + 1$ basis functions are positive.
2. *Unity decomposition.* For every point $z \in [a, b]$, we have:

$$\sum_{j=1}^d B_j(z) = 1.$$

3. *Overlapping with $2l$ adjacent basis functions.* Every basis function (within the domain $[a, b]$) overlaps with exactly $2l$ adjacent basis functions.
4. *Bounded basis functions.* The domain of the individual basis functions is bounded upwards.
5. *Derivatives.* Since the individual basis functions are composed of polynomial parts, it is straightforward to determine derivative formulae. For every single basis function we have

$$\frac{\partial}{\partial z} B_j^l(z) = l \cdot \left(\frac{1}{\kappa_j - \kappa_{j-l}} B_{j-l}^{l-1}(z) - \frac{1}{\kappa_{j+1} - \kappa_{j+1-l}} B_j^{l-1}(z) \right).$$

Therefore, we obtain the derivative for the entire polynomial spline as

$$\frac{\partial}{\partial z} \sum_j \gamma_j B_j^l(z) = l \cdot \sum_j \frac{\gamma_j - \gamma_{j-1}}{\kappa_j - \kappa_{j-l}} B_{j-l}^{l-1}(z). \quad (8.1)$$

As a consequence, we are able to express the derivative of a polynomial spline in terms of the differences of adjacent basis coefficients and B-spline basis functions of one lower degree. Thus, by estimating the coefficients γ_j , we do not only obtain an estimate for the function itself but also for its derivative. Analogously, we can express higher-order derivatives using higher-order differences of the coefficients and basis functions of lower order; see [De Boor \(2001\)](#), p. 115.

Similar to the TP basis, the estimation of a polynomial spline in B-spline representation can be traced back to the estimation of a linear model with a large number of parameters and design matrix

$$\mathbf{Z} = \begin{pmatrix} B_1^l(z_1) & \dots & B_d^l(z_1) \\ \vdots & & \vdots \\ B_1^l(z_n) & \dots & B_d^l(z_n) \end{pmatrix}.$$

Due to the properties of the B-spline basis, this design matrix has some special characteristics. It is apparent that the matrix does not contain an explicit intercept term. Since the rows of the design matrix sum to one, due to the unit decomposition of the B-spline basis, the intercept is however implicitly contained in the span of the basis. Specifying an additional intercept would lead to an unidentifiable model. Since the B-spline basis is locally defined, $\mathbf{Z}$ has a special structure and mainly consists of zeros. This results in a band matrix structure for $\mathbf{Z}'\mathbf{Z}$ with bandwidth l . We can use this property to solve the normal equation $\mathbf{Z}'\mathbf{Z}\boldsymbol{\gamma} = \mathbf{Z}'\mathbf{y}$ in a numerically efficient way.

Figure 8.13 illustrates the estimation of a B-spline fit for the simulated data example. First, we calculate a complete B-spline basis (in this case of degree 3) for a given number of knots [panel (a)]. The least squares estimate $\hat{\boldsymbol{\gamma}}$ then yields an

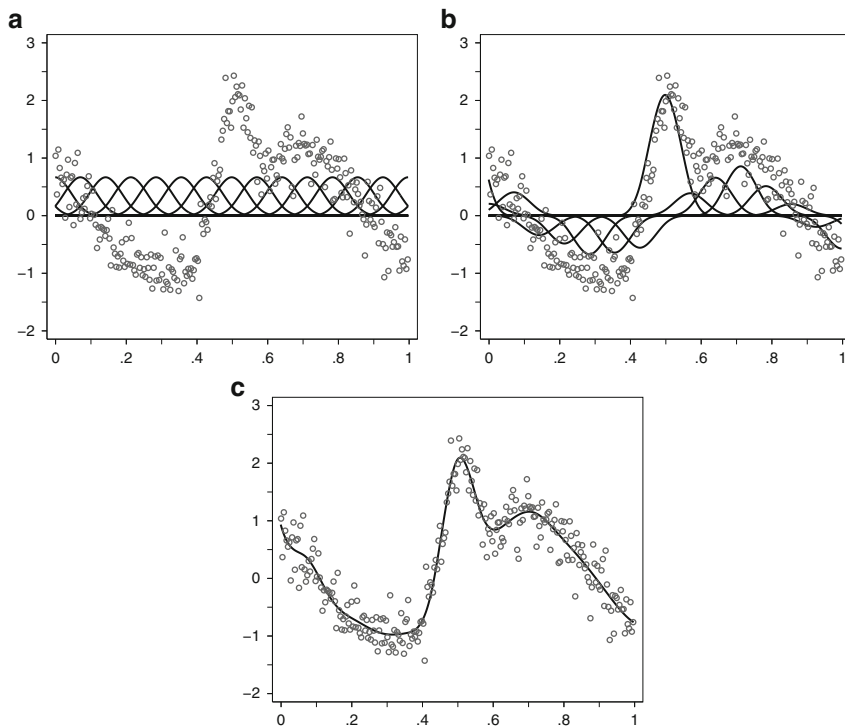


Fig. 8.13 Schematic representation of a nonparametric fit with cubic B-splines. (a) B-spline basis (b) Scaled B-spline basis (c) Sum of scaled B-spline basis functions

amplitude $\hat{\gamma}_j$ for the scaling of every basis function [panel (b)]. When summing the scaled basis functions, we obtain the final estimate [panel (c)].

8.1.2 Penalized Splines (P-Splines)

As we have seen in the previous section, the quality of a nonparametric function estimated by polynomial splines crucially depends on the number of knots. There are basically two strategies to overcome this problem:

- The adaptive choice of knots based on model choice strategies
- The regularization of the estimation problem through the introduction of roughness penalties

We will mostly concentrate on approaches based on penalties and, as mentioned, will only give a brief overview of adaptive methods in Sect. 8.1.10.

The main idea of *penalized splines (P-splines)* can be summarized as follows:

- Approximate the function $f(z)$ with a polynomial spline that uses a generous number of knots (usually about 20–40). This ensures that $f(z)$ can be approximated with enough flexibility to represent even highly complex functions.
- Introduce an additional penalty term that prevents overfitting and minimize a *penalized least squares (PLS) criterion* instead of the usual least squares criterion.

P-Splines Based on a TP Basis

We first consider P-splines based on the TP basis, i.e.,

$$f(z) = \gamma_1 + \gamma_2 z + \dots + \gamma_{l+1} z^l + \gamma_{l+2} (z - \kappa_2)_+^l + \dots + \gamma_d (z - \kappa_{m-1})_+^l.$$

This basis consists of two different parts: the first $l + 1$ basis functions, which describe a global polynomial in z , and the truncated powers describing deviations from this polynomial. Estimated functions will be rough when there is a lot of variability in the second part of the model. Therefore, regularized function estimates may be defined by introducing a penalty for the coefficients of the corresponding basis functions. We can create such a penalty, e.g., using the sum of squared coefficients

$$\sum_{j=l+2}^d \gamma_j^2,$$

so that large coefficients associated with the truncated powers are penalized. Instead of the usual residual sum of squares that underlies the basis function approaches of the previous section,

$$\text{LS} = \sum_{i=1}^n (y_i - f(z_i))^2 = \sum_{i=1}^n \left(y_i - \sum_{j=1}^d \gamma_j B_j(z_i) \right)^2,$$

we now minimize the penalized residual sum of squares

$$\text{PLS}(\lambda) = \sum_{i=1}^n \left(y_i - \sum_{j=1}^d \gamma_j B_j(z_i) \right)^2 + \lambda \sum_{j=l+2}^d \gamma_j^2.$$

The penalty is constructed to discourage estimated functions that are too rough, thereby preventing overfitting to the data. The *smoothing parameter* $\lambda \geq 0$ controls the influence of the penalty. As $\lambda \rightarrow 0$, the effect of the penalty disappears so that the penalized residual sum of squares approximately corresponds to the standard residual sum of squares and we have an estimate for $\boldsymbol{\gamma}$ that is close to the least squares estimate. As $\lambda \rightarrow \infty$, the estimation criterion is dominated by the penalty such that $\hat{\gamma}_j = 0$ for $j = l + 2, \dots, d$, and the estimate for $f(z)$ approaches a

polynomial of degree l . Varying the value of the smoothing parameter allows us to choose continuously between these two extremes.

The main advantage of the penalization is that the smoothness of the estimated fit is no longer controlled by the number and the position of knots but rather by one single real-valued parameter, namely the smoothing parameter. Since we use a large number of knots, the exact positioning of these knots is of minor importance. To simplify matters, equidistant knots are often used in practice, but knots based on quantiles are used as well. Figure 8.14 shows that penalized function estimates depend on the number of knots only very moderately (when an optimal value is chosen for the smoothing parameter using one of the methods discussed in Sect. 8.1.9). For a very small number of knots, the estimate may not be flexible enough. However, once a sufficiently large number of knots are used, the estimated function only slightly changes when the number of knots is increased further, provided that the smoothing parameter is adapted appropriately.

All smoothing techniques presented in the following will depend on some kind of smoothing parameter that works similarly as the smoothing parameter of penalized splines. i.e., controls the trade-off between smoothness of the resulting estimate and fidelity to the data. Of course, it will then be important to determine a suitable value for the smoothing parameter from the given data. In Sect. 8.1.9, we will discuss the choice of the smoothing parameter in more detail and will offer different possibilities to estimate it along with the nonparametric function $f(z)$.

P-Splines Based on B-Splines

If we represent $f(z)$ using B-spline basis functions instead of a TP basis, an appropriate penalty is less obvious since the decomposition in a parametric polynomial part and a deviation from the polynomial part does not occur in the basis function definition. In order to create a penalty for B-splines, we therefore take a different approach. To characterize the smoothness of any type of function, the use of (squared) derivatives is appropriate, since these represent measures for the variability of a function. Penalties based on the second derivative, such as

$$\lambda \int (f''(z))^2 dz, \quad (8.2)$$

are particularly attractive since they measure the curvature of a function. This type of penalty forms the basis of the implementation of penalized splines in the R package `mgcv`, see Wood (2006), and can also be used in more general approaches, see Sect. 8.1.3. A simple approximation to the derivative is available for B-splines, which we can use for the construction of specialized penalties. From Eq. (8.1), we know that the first derivative of a B-spline can be written as a function of the first differences of the corresponding coefficient vector. In order to obtain a smooth function and to avoid too large values of the first derivative, we introduce penalties that are based on exactly these differences. In the following, we will limit ourselves to equidistant knots, since they result in very simple expressions for the difference penalty.

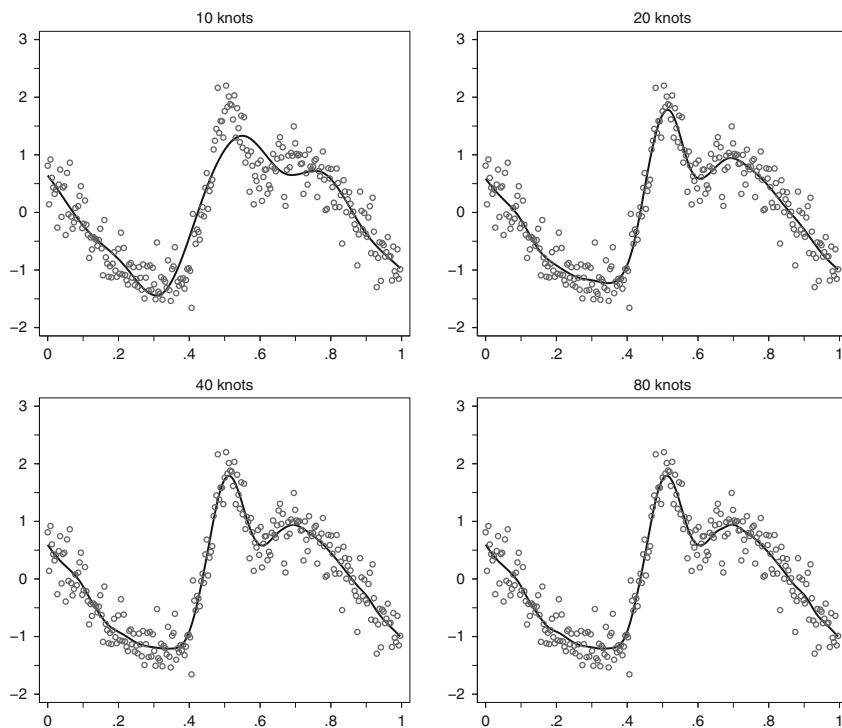


Fig. 8.14 Influence of the number of knots on estimated P-splines

Analogous to first-order differences, we can use differences of a higher order r if we aim at a smooth function in terms of r th-order derivatives. This leads to the penalized residual sum of squares

$$\text{PLS}(\lambda) = \sum_{i=1}^n \left(y_i - \sum_{j=1}^d \gamma_j B_j(z_i) \right)^2 + \lambda \sum_{j=r+1}^d (\Delta^r \gamma_j)^2,$$

where Δ^r denotes r th-order differences, which are recursively defined by

$$\begin{aligned} \Delta^1 \gamma_j &= \gamma_j - \gamma_{j-1}, \\ \Delta^2 \gamma_j &= \Delta^1 \Delta^1 \gamma_j = \Delta^1 \gamma_j - \Delta^1 \gamma_{j-1} = \gamma_j - 2\gamma_{j-1} + \gamma_{j-2}, \\ &\vdots \\ \Delta^r \gamma_j &= \Delta^{r-1} \gamma_j - \Delta^{r-1} \gamma_{j-1}. \end{aligned}$$

This type of smoother has been proposed in [Eilers and Marx \(1996\)](#) and is currently one of the most popular smoothers. In order to obtain a better understanding of the impact of the difference penalty on the resulting estimates, we visualize penalized spline estimates for different values of the smoothing parameter, similar as in Fig. 8.13. We will discuss in detail below exactly how these estimates can be obtained. Figure 8.15 shows the results with a second-order difference penalty and cubic B-splines. In most of our applications, second-order differences are chosen, while other orders, e.g., first or third, are possible.

We find that, for a large value of the smoothing parameter ($\lambda \rightarrow \infty$), the function estimate for $f(z)$ is close to linear in case of second-order differences. More generally, as $\lambda \rightarrow \infty$, the fit approaches a polynomial of degree $r - 1$ with r -th order differences, provided that the degree of the spline is at least as large as the order of the differences, i.e., $l \geq r$. For first-order differences, this can easily be shown based on the formula for the first derivative of a B-spline in Eq. (8.1): The first derivative is equal to zero and the corresponding function is a constant if and only if all first-order differences of coefficients are zero. This can be achieved with a large smoothing parameter that places a large weight on the penalty. For penalties with a larger difference order, analogous results can be obtained from formulae for higher-order derivatives.

In comparison to penalized splines that are based on the TP basis, we obtain the additional freedom to separately choose the degree of the limiting polynomial obtained with heavy penalization as $\lambda \rightarrow \infty$ and the degree of the splines to be used for the modeling of $f(z)$. In contrast, the unpenalized polynomial is always of degree l for the TP basis since only the coefficients of the truncated powers are penalized. When using B-splines, we are able to choose the degree of the polynomial that results with heavy penalization, through the choice of the difference order k , which is independent of the degree l of the spline. This provides us with additional flexibility in the modeling process and also with a clear separation of the two different model features.

Penalized Least Squares Estimation

For the derivation of the PLS estimate, it is advantageous to write the penalty in matrix notation. For the TP basis, this is simply achieved via

$$\lambda \sum_{j=l+2}^d \gamma_j^2 = \lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}$$

with the vector $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_d)'$ and the *penalty matrix*

$$\mathbf{K} = \text{diag}(\underbrace{0, \dots, 0}_{(l+1)}, \underbrace{1, \dots, 1}_{(m-2)}).$$

For the B-spline penalty based on differences, we start by writing the vector of first differences using the difference matrix

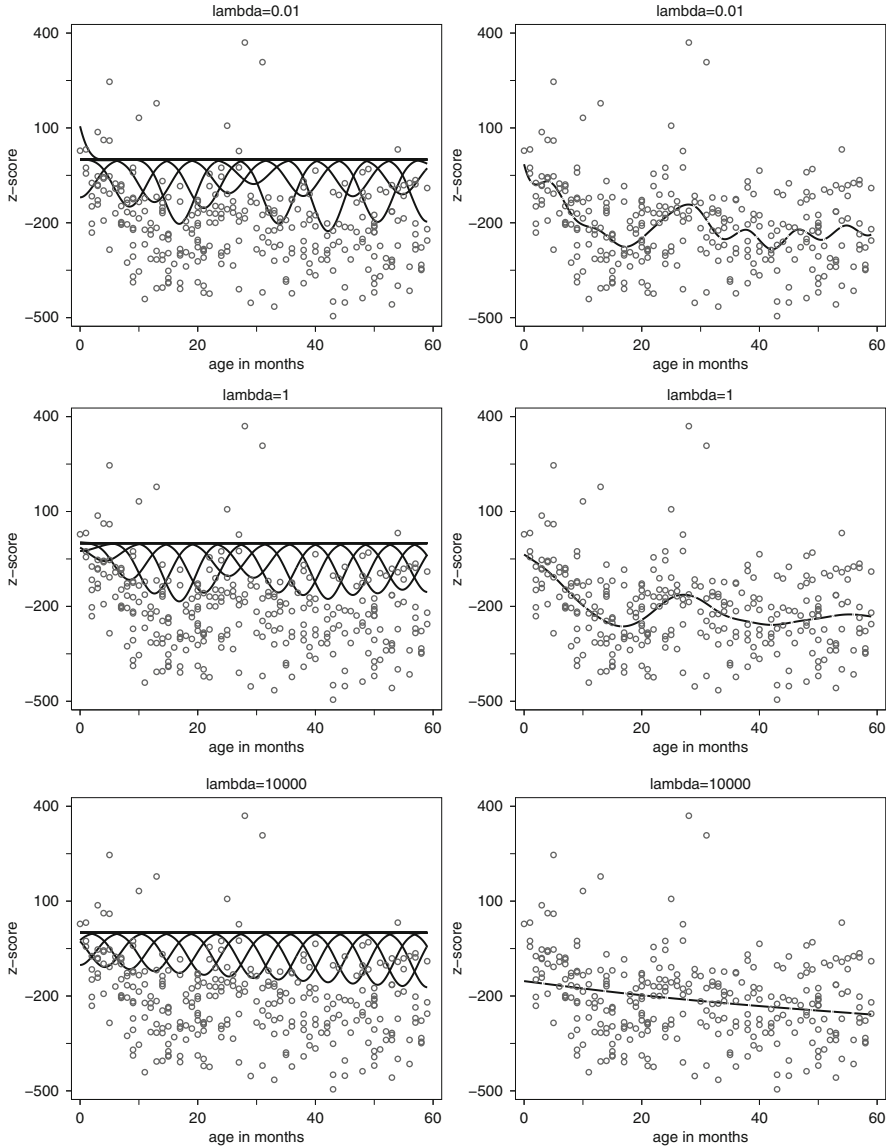


Fig. 8.15 Malnutrition in Tanzania: impact of the smoothing parameter on estimated P-splines with second-order difference penalty

$$D_1 = \begin{pmatrix} -1 & 1 & & & \\ & -1 & 1 & & \\ & & \ddots & \ddots & \\ & & & -1 & 1 \end{pmatrix}$$

of dimension $(d - 1) \times d$, yielding

$$\mathbf{D}_1 \boldsymbol{\gamma} = \begin{pmatrix} \gamma_2 - \gamma_1 \\ \vdots \\ \gamma_d - \gamma_{d-1} \end{pmatrix}.$$

Higher differences can be expressed recursively with the help of difference matrices

$$\mathbf{D}_r = \mathbf{D}_1 \mathbf{D}_{r-1}.$$

For example, with $r = 2$, we obtain the $(d - 2) \times d$ -difference matrix

$$\mathbf{D}_2 = \begin{pmatrix} 1 & -2 & 1 & & & \\ & 1 & -2 & 1 & & \\ & & \ddots & \ddots & \ddots & \\ & & & 1 & -2 & 1 \end{pmatrix}.$$

This yields the penalty

$$\lambda \sum_{j=r+1}^d (\Delta^r \gamma_j)^2 = \lambda \boldsymbol{\gamma}' \mathbf{D}_r' \mathbf{D}_r \boldsymbol{\gamma} = \lambda \boldsymbol{\gamma}' \mathbf{K}_r \boldsymbol{\gamma},$$

with first- and second-order difference penalty matrices

$$\mathbf{K}_1 = \begin{pmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 1 \end{pmatrix}, \quad \mathbf{K}_2 = \begin{pmatrix} 1 & -2 & 1 & & & \\ -2 & 5 & -4 & 1 & & \\ 1 & -4 & 6 & -4 & 1 & \\ & \ddots & \ddots & \ddots & \ddots & \ddots \\ & & 1 & -4 & 6 & -4 & 1 \\ & & & 1 & -4 & 5 & -2 \\ & & & & 1 & -2 & 1 \end{pmatrix},$$

respectively.

If we define the penalty based on the integral of the squared second derivatives as in Eq. (8.2), we obtain another quadratic penalty

$$\begin{aligned} \int (f''(z))^2 dz &= \int \left(\sum_{j=1}^d \gamma_j B_j''(z) \right)^2 dz \\ &= \int \sum_{r=1}^d \sum_{j=1}^d \gamma_r \gamma_j B_r''(z) B_j''(z) dz \end{aligned}$$

8.3 Penalized Splines

Model

We approximate function f using polynomial splines so that we are able to write the nonparametric regression model as a linear model

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}.$$

Penalized Least Squares Criterion

Rather than using the standard residual sum of squares, we estimate $\boldsymbol{\gamma}$ by minimizing the PLS criterion

$$\text{PLS}(\lambda) = (\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma})'(\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma}) + \lambda\boldsymbol{\gamma}'\mathbf{K}\boldsymbol{\gamma}.$$

The smoothing parameter $\lambda \geq 0$ controls the compromise between fidelity to the data and smoothness of the resulting function estimate. For splines in a TP basis representation, we penalize the sum of squared coefficients of the truncated powers. For B-splines, we construct the penalty based on the sum of squared differences of neighboring coefficients or based on the integral of the function's squared second derivative.

Penalized Least Squares Estimation

In either case, the PLS estimate has the form

$$\hat{\boldsymbol{\gamma}} = (\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})^{-1}\mathbf{Z}'\mathbf{y}.$$

$$\begin{aligned} &= \sum_{r=1}^d \sum_{j=1}^d \gamma_r \gamma_j \int B_r''(z) B_j''(z) dz \\ &= \boldsymbol{\gamma}'\mathbf{K}\boldsymbol{\gamma}, \end{aligned}$$

with $\mathbf{K}[r, j] = \int B_r''(z) B_j''(z) dz$. The entries of the penalty matrix $\mathbf{K}$ result from the integrated products of second derivatives of the B-spline basis functions.

In general, we can write the penalized residual sum of squares as

$$\begin{aligned} \text{PLS}(\lambda) &= (\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma})'(\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma}) + \lambda\boldsymbol{\gamma}'\mathbf{K}\boldsymbol{\gamma} \\ &= \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{Z}\boldsymbol{\gamma} - \boldsymbol{\gamma}'\mathbf{Z}'\mathbf{y} + \boldsymbol{\gamma}'\mathbf{Z}'\mathbf{Z}\boldsymbol{\gamma} + \lambda\boldsymbol{\gamma}'\mathbf{K}\boldsymbol{\gamma} \\ &= \mathbf{y}'\mathbf{y} - 2\boldsymbol{\gamma}'\mathbf{Z}'\mathbf{y} + \boldsymbol{\gamma}'(\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})\boldsymbol{\gamma} \end{aligned}$$

using an appropriate penalty matrix $\mathbf{K}$. Taking the derivative with respect to $\boldsymbol{\gamma}$ (compare Theorem A.33 in Appendix A.8) and setting the derivative equal to zero yields the system of equations

$$-2\mathbf{Z}'\mathbf{y} + 2(\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})\boldsymbol{\gamma} = \mathbf{0},$$

and, therefore, the PLS estimate

$$\hat{\boldsymbol{\gamma}} = (\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})^{-1}\mathbf{Z}'\mathbf{y}. \quad (8.3)$$

Consequently, the vector of estimated function evaluations $\hat{\mathbf{f}} = (\hat{f}(z_1), \dots, \hat{f}(z_n))'$ is calculated as

$$\hat{\mathbf{f}} = \mathbf{Z}\hat{\boldsymbol{\gamma}} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})^{-1}\mathbf{Z}'\mathbf{y}.$$

Note that the form of the PLS estimate is very similar to the unpenalized least squares estimate, as it only differs by the additional term $\lambda\mathbf{K}$. This additional term represents the influence of the penalty within the PLS criterion. For $\lambda = 0$, we again obtain the solution of the unpenalized optimization problem. Based on similar considerations as presented for the least squares estimate in section “Statistical Properties without Specific Distributional Assumptions” of Sect. 3.2.3, we obtain the covariance matrix of the PLS estimate:

$$\text{Cov}(\hat{\boldsymbol{\gamma}}) = \sigma^2(\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})^{-1}\mathbf{Z}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})^{-1}.$$

In contrast to the covariance matrix of the least squares estimate, this covariance matrix consists of three parts and has the form of a “sandwich” matrix (known from Sects. 4.1.3, p. 190 and 7.3.3, p. 378). The two additional expressions containing the penalty matrix are responsible for the stabilization of the PLS estimator and, in comparison to the unpenalized least squares estimator, typically lead to smaller variances. However, this achieved efficiency implies that the PLS estimator is no longer unbiased. Provided that the smoothing parameter is chosen appropriately, the PLS estimator generally yields a smaller mean squared error. Thus, from a statistical point of view, the estimator has better properties than the unpenalized least squares estimator.

We find very similar forms when comparing the PLS estimate with that of the posterior expectation

$$\mathbf{E}(\boldsymbol{\gamma} | \mathbf{y}) = (\mathbf{X}'\mathbf{X} + \mathbf{M}^{-1})^{-1}\mathbf{X}'\mathbf{y}$$

of a linear model with prior distribution $\boldsymbol{\gamma} \sim \mathbf{N}(\mathbf{0}, \sigma^2\mathbf{M})$ (derived in Sect. 4.4, pp. 225ff.). We can thus supplement penalized splines with a Bayesian interpretation as detailed below. Additionally, penalized splines are closely related to models containing random effects, as illustrated by a comparison with the estimating equations in Box 7.3 on p. 375. We will use both the Bayesian interpretation and the relation to mixed models in Sect. 8.1.9 when we elaborate on choosing the smoothing parameter λ .

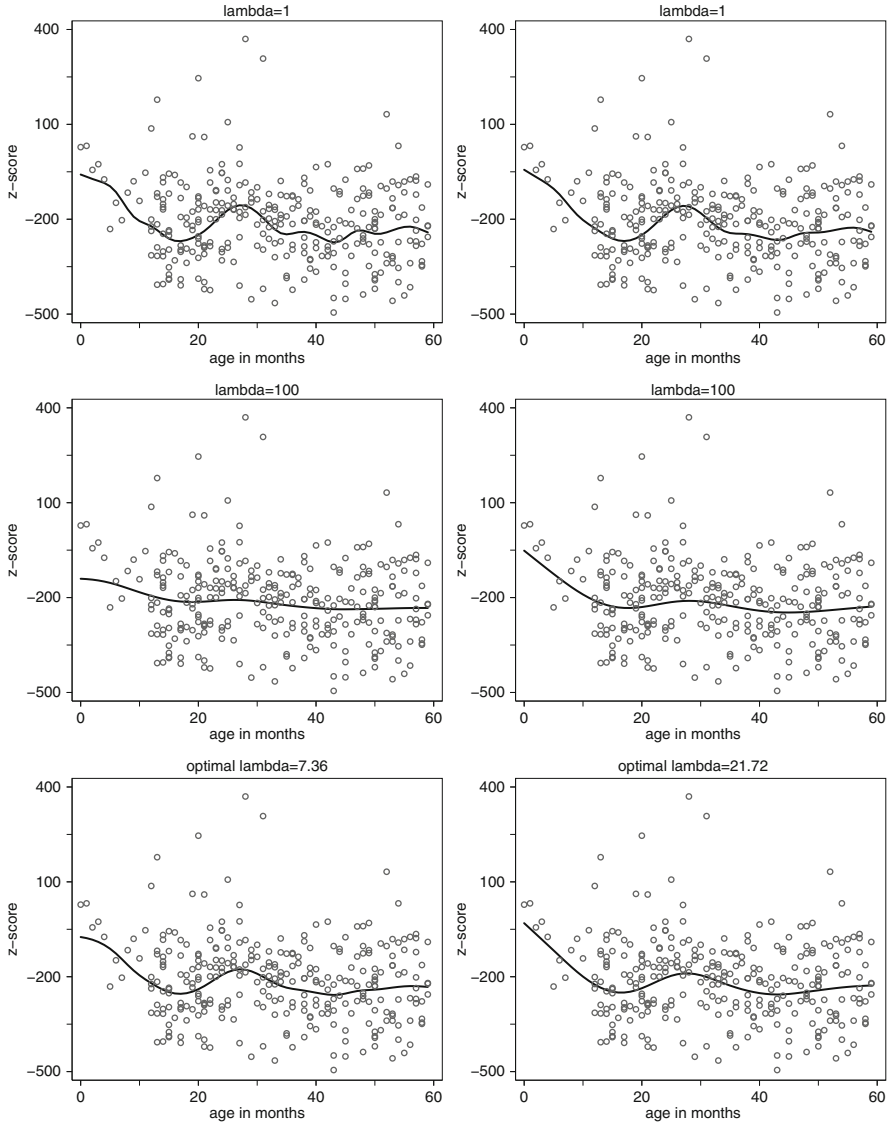


Fig. 8.16 Malnutrition in Tanzania: P-spline estimates based on first (*left panel*) and second (*right panel*) order differences and different smoothing parameters

Example 8.2 Malnutrition in Tanzania: P-Splines

Similar to Example 8.1, we compare several specifications for penalized splines using the data on malnutrition in Tanzania. We focus on P-splines in B-spline representation with 20 inner knots, and examine several smoothing parameter choices as well as penalties that are based on first and second differences. Figure 8.16 shows some of the corresponding results.

In the first row, we can see the estimated function $\hat{f}(z)$, which results from choosing a rather small smoothing parameter. As a consequence, the estimated functions are relatively rough with somewhat smoother results in case of second-order differences. If we choose a larger smoothing parameter (middle row in Fig. 8.16), we obtain very smooth fits. In particular, first-order differences yield an almost constant fit, i.e., the estimated function is close to the limiting case obtained with $\lambda \rightarrow \infty$. For second-order differences, we still find some deviation from a linear fit. The bottom row shows estimates resulting from “optimal” smoothing parameters, which were chosen with the help of one of the approaches presented in Sect. 8.1.9. This results in a (at least visually) plausible compromise between fidelity to the data and smoothness. The two aforementioned main features of the relationship between age and malnutrition are easily seen with either penalty: a higher Z-score, which is a lower risk for malnutrition of young children, and a local maximum of the Z-score at the age of approximately 24 months, which is due to the change of the reference population. It would be especially difficult to discover the effect of the change of the reference population without an automatic and data-driven choice of the smoothing parameter.

△

Bayesian P-Splines

Besides their motivation based on a PLS criterion, penalized splines can also be derived in a Bayesian framework. In particular, this allows us to employ Bayesian approaches for the estimation of P-splines including the smoothing parameter; see Sect. 8.1.9. We first focus on penalized splines that are based on B-splines, followed by a brief presentation of Bayesian TP-splines.

For the Bayesian formulation, we start with the same observation model as in the previous section, i.e.,

$$y_i = f(z_i) + \varepsilon_i = \sum_{j=1}^d \gamma_j B_j(z_i) + \varepsilon_i,$$

with B-spline basis functions B_j . Instead of imposing a penalty, we will now develop an appropriate prior assumption for $\boldsymbol{\gamma}$ that enforces a smooth function estimation. The stochastic analogue for the difference penalty is *random walks* of order k (RW k). A random walk of first order (RW1) is defined by

$$\gamma_j = \gamma_{j-1} + u_j, \quad u_j \sim N(0, \tau^2), \quad j = 2, \dots, d,$$

or equivalently

$$\gamma_j - \gamma_{j-1} = u_j, \quad u_j \sim N(0, \tau^2), \quad j = 2, \dots, d,$$

so that a connection to the first-order difference penalty is recognizable. We have to make further assumptions for the prior of the starting value γ_1 and a noninformative prior distribution; $p(\gamma_1) \propto \text{const}$ will be our standard option.

Our prior specification corresponds to the conditional distributions

$$\gamma_j \mid \gamma_{j-1}, \dots, \gamma_1 \sim N(\gamma_{j-1}, \tau^2). \quad (8.4)$$

$$\begin{aligned}
& \propto \prod_{j=2}^d \frac{1}{\sqrt{2\pi\tau^2}} \exp\left(-\frac{1}{2\tau^2}(\gamma_j - \gamma_{j-1})^2\right) \\
& = \frac{1}{(2\pi\tau^2)^{(d-1)/2}} \exp\left(-\frac{1}{2\tau^2} \sum_{j=2}^d (\gamma_j - \gamma_{j-1})^2\right) \\
& = \frac{1}{(2\pi\tau^2)^{(d-1)/2}} \exp\left(-\frac{1}{2\tau^2} \boldsymbol{\gamma}' \mathbf{K}_1 \boldsymbol{\gamma}\right).
\end{aligned}$$

The first step is based on using the rule of total probability to factor the joint density. Due to the Markov property, we then obtain simplified expressions for the conditional distributions, so that we can insert the density of the normal distribution that appears in the RW1 prior. The proportionality sign results from the flat prior for the first regression coefficient γ_1 , which we did not specify exactly but only up to a proportionality constant. The joint distribution has the form of a multivariate normal distribution with expectation $\mathbf{0}$ and precision matrix $\mathbf{K}_1/\tau^2$, where $\mathbf{K}_1 = \mathbf{D}_1' \mathbf{D}_1$ is defined by the first-order difference matrix $\mathbf{D}_1$ as in the previous section. However, this precision matrix does not have full rank, since rows and columns sum to zero. Consequently, we cannot obtain the inverse, and the covariance matrix “ $\tau^2 \mathbf{K}_1^{-1}$ ” of the prior distribution does not exist. Thus, we cannot normalize the density of $\boldsymbol{\gamma}$, and the integral of the density diverges. Therefore, the joint prior of $\boldsymbol{\gamma}$ is an improper, singular normal distribution (see Appendix B.3.2). More precisely, we call the distribution partially improper since the precision matrix is not of full rank, but $\text{rk}(\mathbf{K}_1) = d - 1 > 0$ holds. For a first-order random walk, the deterministic part of the prior distribution (as defined in Appendix B.3.2) corresponds to a constant vector representing the level of function f . With the partially improper prior, we define a flat prior for this level. The stochastic part of the prior represents the deviations from the constant. We will use similar ideas in Sect. 8.1.9 in order to create a representation of penalization approaches as mixed models.

The multivariate prior distribution for the vector $\boldsymbol{\gamma}$ also determines the undirected forms of the conditional distribution (8.4). In this case, we condition on all remaining parameters and not only on the preceding values $\gamma_{j-1}, \dots, \gamma_1$. For the RW1, we obtain

$$\gamma_j \mid \cdot \sim \mathcal{N}\left(\frac{1}{2}(\gamma_{j-1} + \gamma_{j+1}), \frac{\tau^2}{2}\right), \quad j = 2, \dots, d-1$$

(with the exception of the boundary parameters γ_1 and γ_d). We again recognize the Markov property (this time, however, in undirected form), according to which the conditional distribution of γ_j , given all remaining parameters, only depends on the two immediate neighbors. More precisely, the conditional expectation is the local average of the two adjacent regression coefficients. A more detailed discussion of the random walk prior that also contains information about the boundary parameters can be found in [Lang and Brezger \(2004\)](#).

Although the joint prior distribution is partially improper, we obtain a proper multivariate normal posterior distribution for $\boldsymbol{\gamma}$. The derivation is similar to the steps to obtain the posterior in Bayesian linear model; see Sect. 4.4. In particular, we obtain the following expressions for the posterior expectation and covariance matrix of $\boldsymbol{\gamma}$:

$$E(\boldsymbol{\gamma} \mid \mathbf{y}, \sigma^2, \tau^2) = \left(\mathbf{Z}'\mathbf{Z} + \frac{\sigma^2}{\tau^2} \mathbf{K}_1 \right)^{-1} \mathbf{Z}'\mathbf{y},$$

i.e., the PLS estimate with smoothing parameter $\lambda = \sigma^2/\tau^2$, and

$$\text{Cov}(\boldsymbol{\gamma} \mid \mathbf{y}, \sigma^2, \tau^2) = \sigma^2 (\mathbf{Z}'\mathbf{Z} + \lambda \mathbf{K}_1)^{-1}.$$

In contrast to the non-Bayesian results of the last section, we obtain a simpler form for the covariance matrix. The reason is due to the stochastic formulation of the Bayesian model: Whereas the frequentist approach considers the regression coefficients $\boldsymbol{\gamma}$ as fixed and unknown parameters, we now consider the regression coefficients as random.

In the Bayesian formulation, the P-spline smoothing parameter λ is given by the ratio of the error variance and the variance of the RW1. This solidifies our previous conjectures regarding the influence of the variance τ^2 of the random walk and further leads to an interesting interpretation of the smoothing parameter λ : The larger the variance of the prior distribution is relative to the variance of the residuals, the less the estimation will be penalized. Consequently, we always have to interpret the value of τ^2 relative to the variance σ^2 that is associated with the measurement error $\boldsymbol{\varepsilon}$. We can then refer to λ as the noise-to-signal ratio.

For random walks of a higher order, we can derive analogous results. We now briefly consider the special case of the RW2. A RW2 prior for $\boldsymbol{\gamma}$ is defined by

$$\gamma_j = 2\gamma_{j-1} - \gamma_{j-2} + u_j, \quad u_j \sim N(0, \tau^2), \quad j = 3, \dots, d,$$

usually in combination with noninformative prior distributions $p(\gamma_1) \propto \text{const}$ and $p(\gamma_2) \propto \text{const}$ for the initial values. When reformulating the RW2 definition to

$$\gamma_j - 2\gamma_{j-1} + \gamma_{j-2} = u_j, \quad u_j \sim N(0, \tau^2), \quad j = 3, \dots, d,$$

we again can identify the similarity to the second-order difference penalty that was presented in the previous section. If we specify the assumption of a RW2 in terms of conditional distributions, we obtain

$$\gamma_j \mid \gamma_{j-1}, \gamma_{j-2}, \dots, \gamma_1 \sim N(2\gamma_{j-1} - \gamma_{j-2}, \tau^2)$$

corresponding to a local linear extrapolation of the conditional expectation; see Fig. 8.18. For the joint distribution of the entire vector $\boldsymbol{\gamma}$, we have

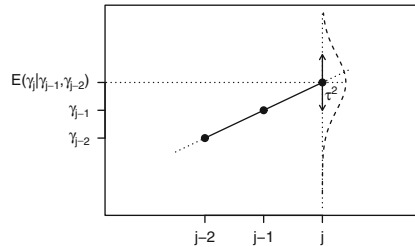


Fig. 8.18 Conditional distribution imposed by a second-order random walk

$$p(\boldsymbol{\gamma} \mid \tau^2) \propto \frac{1}{(\tau^2)^{(d-2)/2}} \exp\left(-\frac{1}{2\tau^2} \boldsymbol{\gamma}' \mathbf{K}_2 \boldsymbol{\gamma}\right),$$

with $\mathbf{K}_2 = \mathbf{D}_2' \mathbf{D}_2$. Again, the precision matrix is not of full rank and the prior distribution for the RW2 is partially improper with $\text{rk}(\mathbf{K}_2) = d - 2$.

Although we can obtain point estimates and covariance matrices for Bayesian P-splines analytically when the smoothing parameter is given, we mainly use the Bayesian formulation to develop a Bayesian inferential framework for penalized splines and the more general penalization approaches in Sect. 8.1.9. This also opens up the possibility to determine the smoothing parameter in a rather simple way. In addition, it is easier to implement the additive extensions, within the scope of Bayesian approaches, discussed in Chap. 9.

P-splines in TP basis representation can also be motivated in a Bayesian framework. In fact, it is even easier to construct the prior distribution, since an i.i.d. prior for the regression coefficients lends itself to the equivalence for the ridge penalty described in the previous section. More precisely, we assume that the coefficients $\gamma_{l+2}, \dots, \gamma_d$ of the truncated polynomials are independently and identically $N(0, \tau^2)$ distributed. For the coefficients of the global polynomials $\gamma_1, \dots, \gamma_{l+1}$, we consider noninformative prior distributions, i.e., $p(\gamma_j) \propto \text{const}$, $j = 1, \dots, l + 1$. Similar to the frequentist penalization approach, the two different types of prior distributions reflect a distinction between parameters to be estimated in a restricted and an unrestricted way. The Bayesian formulation of TP-splines also introduces a connection between nonparametric regression models and mixed models since the distributional assumptions cannot only be interpreted as prior information, but also distinguish between fixed and random effects in a mixed model. Such a connection will also help us to derive an estimation approach for the smoothing parameter.

Similar to P-splines based on B-splines, we can now derive the posterior expectation and covariance matrix for TP-splines. In particular, the PLS estimator is obtained as the posterior expectation.

8.1.3 General Penalization Approaches

Prior to discussing further models of nonparametric regression, we collect some properties of penalized splines that also apply to a number of general penalization approaches. In fact, many of the univariate and bivariate smoothing methods that we discuss in Sects. 8.1 and 8.2 can be included in such a general framework.

Representing polynomial splines with basis functions as in

$$f(z) = \sum_{j=1}^d \gamma_j B_j(z) \quad (8.5)$$

resulted in a “large” linear model of the form

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}, \quad (8.6)$$

in which the design matrix was defined by the evaluations of the basis functions. However, in Eq. (8.5), we are of course not limited to TP-splines or B-splines, but we can rather use any type of functions for the approximation of $f(z)$. The choice of the basis functions B_j then determines the class of functions we employ to describe $f(z)$. For example, either the TP or the B-spline basis results in the space of all polynomial splines for the given knot configuration and degree l . In the following sections we will learn about other models of the form (8.6) that are useful in nonparametric regression. The concept of penalized residual sums of squares used to derive P-splines can also be implemented more generally. Assuming that Eq. (8.5) holds, we can always define a suitable penalty via the integral of the squared second derivative (as shown on p. 437):

$$\int (f''(z))^2 dz = \sum_{i=1}^d \sum_{j=1}^d \gamma_i \gamma_j \int B_i''(z) B_j''(z) dz = \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}.$$

We again find a quadratic penalty for $\boldsymbol{\gamma}$ where the entries of the penalty matrix $\mathbf{K}$ are determined by the second derivatives of the basis functions. As with P-splines, alternative construction mechanisms for penalties may exist for more general types of basis functions, but these often lead to quadratic penalties $\lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}$ as well.

As an optimization criterion, we always obtain a PLS criterion of the form

$$\text{PLS}(\lambda) = (\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma})'(\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma}) + \lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma},$$

differing only in the specification of the design matrix $\mathbf{Z}$ and the penalty matrix $\mathbf{K}$. Minimization of the PLS criterion yields the PLS estimate

$$\hat{\boldsymbol{\gamma}} = (\mathbf{Z}'\mathbf{Z} + \lambda \mathbf{K})^{-1} \mathbf{Z}'\mathbf{y}.$$

8.4 General Penalization Approaches

Model

The function f is represented through a large linear model, such that

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}.$$

Often this is achieved by approximating the function $f(z)$ using basis functions, i.e.,

$$f(z) = \sum_{j=1}^d \gamma_j B_j(z).$$

Penalization

To regularize estimation, we assume a quadratic penalty of the form

$$\lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}$$

or the normal prior distribution

$$p(\boldsymbol{\gamma} \mid \tau^2) \propto \left(\frac{1}{2\pi\tau^2} \right)^{\text{rk}(\mathbf{K})/2} \exp \left(-\frac{1}{2\tau^2} \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma} \right)$$

for $\boldsymbol{\gamma}$. This leads to the PLS criterion

$$(\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma})'(\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma}) + \lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}$$

and the PLS estimate

$$\hat{\boldsymbol{\gamma}} = (\mathbf{Z}'\mathbf{Z} + \lambda \mathbf{K})^{-1} \mathbf{Z}'\mathbf{y}.$$

We can also consider penalty approaches with quadratic penalties in a Bayesian context. Penalties of the form $\lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}$ correspond to the assumption of a multivariate normal prior with density

$$p(\boldsymbol{\gamma} \mid \tau^2) \propto \left(\frac{1}{2\pi\tau^2} \right)^{\text{rk}(\mathbf{K})/2} \exp \left(-\frac{1}{2\tau^2} \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma} \right)$$

for the coefficient vector $\boldsymbol{\gamma}$. However, we should keep in mind that the penalty matrix $\mathbf{K}$ may not be of full rank. In such cases, the density cannot be normalized and is only defined upon proportionality.

Two of the approaches that we will discuss in Sect. 8.1.9 to determine the optimal smoothing parameter will be explicitly tailored to models using general penalty approaches.

At this point, we also want to briefly discuss the possibility to model non-normally distributed response variables within the nonparametric regression framework (which will be discussed in more detail in Chap. 9). Transferring the ideas of nonparametric regression to response variables from exponential families, as considered in Chap. 5, we obtain models of the form

$$E(y_i | \eta_i) = h(\eta_i), \quad \eta_i = f(z_i),$$

for the expectation of the response variables. As in Chap. 5, $h(\cdot)$ defines a known response function, e.g., the logistic cumulative distribution function in the logit model or the exponential function associated with log-linear Poisson models. If we use basis functions for approximating the function f , we obtain the vector of linear predictors as $\boldsymbol{\eta} = \mathbf{Z}\boldsymbol{\gamma}$. Instead of imposing a penalty on the least squares criterion as we did in nonparametric regression for normally distributed responses, we now consider the penalized log-likelihood criterion

$$l_{\text{pen}}(\boldsymbol{\gamma}) = l(\boldsymbol{\gamma}) - \frac{\lambda}{2} \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma} \quad (8.7)$$

for the estimation of the regression coefficients. We denote by $l(\boldsymbol{\gamma})$ the usual log-likelihood function of a generalized linear model; see Sect. 5.4.1. The smoothing parameter has to be divided by 2 to make the likelihood-based approach equivalent to the PLS approach discussed so far. In fact, Eq. (8.7) is equivalent to the PLS criterion in case of normally distributed responses.

In order to determine the penalized ML estimate, we amend the Fisher scoring approach, discussed in Sect. 5.4.2, with the penalty. In particular, the estimated $\hat{\boldsymbol{\gamma}}$ can be determined using iteratively weighted PLS estimation; see Chap. 9 for details. To keep our discussion of further modeling alternatives in nonparametric regression models as simple as possible, we will restrict ourselves to normally distributed responses in this chapter but will return to the more general setting in Chap. 9.

8.1.4 Smoothing Splines

When working with penalized splines, we limit ourselves, right from the start, to a function space spanned by the spline basis functions. However, we can also work with more general function spaces, as we will demonstrate with the example of *smoothing splines*. Here we only assume that the function $f(z)$ is twice continuously differentiable, so that we can use the PLS criterion

$$\sum_{i=1}^n (y_i - f(z_i))^2 + \lambda \int (f''(z))^2 dz \quad (8.8)$$

to determine an estimate of f . In this setting, we define the penalty to be the integrated squared second derivative, which yields a measure for the overall curvature of the function. Surprisingly, a very special class of functions emerges as the solution for the optimization problem, despite the generality of the approach taken. This class of functions are the so-called *natural cubic splines*, a special subset of cubic polynomial splines.

8.5 Natural Cubic Splines

The function $f(z)$ is a natural cubic spline based on the knots, $a \leq \kappa_1 < \dots < \kappa_m \leq b$, if:

- (i) $f(z)$ is a cubic polynomial spline for the given knots.
- (ii) $f(z)$ satisfies the boundary conditions $f''(a) = f''(b) = 0$, i.e., $f(z)$ is linear in the intervals $[a, \kappa_1]$ and $[\kappa_m, b]$.

Clearly, every natural cubic spline is also a cubic polynomial spline. However, due to the boundary conditions, we have additional restrictions such that m basis functions are sufficient to represent a natural cubic spline (in comparison to the $m+2$ basis functions for standard cubic polynomial splines). The boundary conditions are also the motivation for the name “natural cubic spline.” Originally, the word “spline” designated a curve template, which means a flexible ruler that allows one to draw a smooth function that interpolates a given set of points. In our example, these points would be the function values evaluated at the knots. Outside of the knot set, the ruler then again takes its “natural,” linear form.

The optimal solution for Eq. (8.8) is now given by a natural cubic spline with knots at the d ordered and unique covariate values $z_{(1)} < \dots < z_{(d)}$. Such a spline is then also called a *smoothing spline*.

Since natural cubic splines are special polynomial splines, we can represent them using the basis functions already discussed in previous sections. Note, however, that d basis functions are sufficient for describing a natural cubic spline, and therefore the representation through common cubic polynomial spline functions contains two redundant basis functions. We obtain a “real” basis for natural cubic splines by introducing suitable modifications of the polynomial spline basis functions at the boundary of the domain. In the representation

$$f(z) = \sum_{j=1}^d \gamma_j B_j(z),$$

the basis functions $B_3, \dots, B_{d-2}$ remain unchanged, while the basis functions B_1, B_2, B_{d-1} , and B_d are adjusted such that the natural boundary conditions are

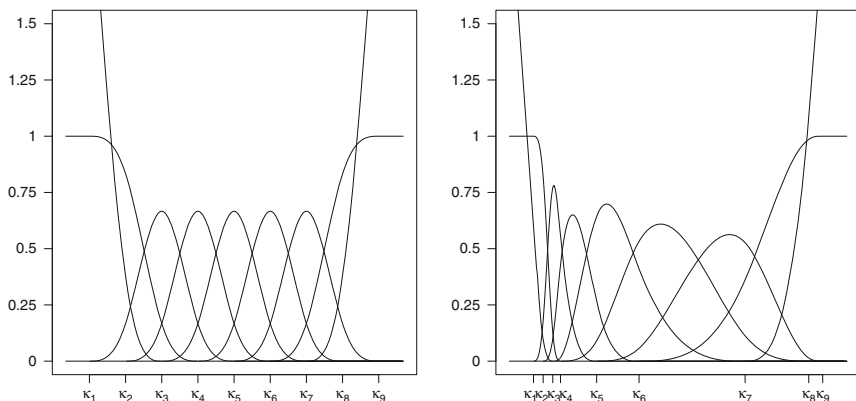


Fig. 8.19 Basis of a natural cubic spline with equidistant knots (*left panel*) and with unevenly distributed knots (*right panel*)

fulfilled. We restrain from an explicit derivation of the resulting basis functions, and rather provide a visualization in Fig. 8.19, using the same knot configuration that was used for constructing the B-spline basis in Fig. 8.12 (p. 428). We find that the basis functions are indeed the same in the interior, whereas the outer basis functions are modified to fulfill the boundary conditions of the natural cubic spline.

For minimizing the PLS criterion (8.8), we need an expression for the penalty based on the integrated squared second derivative. With natural cubic splines using the basis functions presented in Fig. 8.19, we again have

$$\int (f''(z))^2 dz = \sum_{i=1}^d \sum_{j=1}^d \gamma_i \gamma_j \int B_i''(z) B_j''(z) dz = \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}.$$

Thus the PLS criterion to be minimized is

$$(\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma})'(\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma}) + \lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma},$$

where $\mathbf{Z}$ contains the basis functions evaluated at the observed z values. We find that the form of the expression to be minimized corresponds exactly to the PLS criterion discussed in the previous sections of this chapter. Therefore, the PLS estimate is again given by

$$\hat{\boldsymbol{\gamma}} = (\mathbf{Z}'\mathbf{Z} + \lambda \mathbf{K})^{-1} \mathbf{Z}'\mathbf{y}.$$



Alternatively, we can directly write the smoothing spline minimization problem in terms of the function evaluations $f(z_{(j)})$. If we collect the function evaluations in the vector $\mathbf{f} = (f(z_{(1)}), \dots, f(z_{(d)}))'$, and define the respective design matrix $\mathbf{Z}_f$ as an incidence matrix with entries

8.6 Smoothing Splines

We seek the function f , which minimizes the least squares criterion

$$\sum_{i=1}^n (y_i - f(z_i))^2 + \lambda \int (f''(z))^2 dz,$$

among all twice continuously differentiable functions. The solution for this optimality criterion is a natural cubic spline with knots located at the observed covariate values. We represent natural cubic splines using a modified B-spline basis, so that the PLS estimate can be calculated in the same way as for P-splines, i.e.,

$$\hat{\mathbf{y}} = (\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})^{-1}\mathbf{Z}'\mathbf{y}.$$

$$\mathbf{Z}_f[i, j] = \begin{cases} 1 & \text{if } z_i = z_{(j)}, \\ 0 & \text{otherwise,} \end{cases}$$

we obtain the PLS criterion

$$(\mathbf{y} - \mathbf{Z}_f \mathbf{f})'(\mathbf{y} - \mathbf{Z}_f \mathbf{f}) + \lambda \mathbf{f}' \mathbf{K}_f \mathbf{f},$$

where the penalty matrix can be constructed based on the distances $\delta_j = z_{(j+1)} - z_{(j)}$ (see, e.g., [Fahrmeir and Tutz \(2001\)](#), p. 179–180). Minimizing the PLS criterion for $\mathbf{f}$ results in

$$\hat{\mathbf{f}} = (\mathbf{Z}_f' \mathbf{Z}_f + \lambda \mathbf{K}_f)^{-1} \mathbf{Z}_f' \mathbf{y},$$

which corresponds to the PLS estimate $\hat{\mathbf{y}}$.

Smoothing splines are typically based on a high-dimensional basis so that a large set of parameters needs to be estimated. In particular, the basis usually grows with the sample size so that for large data sets the practical calculations become demanding. Consequently, a reduced basis is used in many implementations of smoothing splines. A naive approach, implemented, for example, in PROC GAM in SAS, is to simply use a subset of the distinct covariate values to determine the set of knots. However, for this reduced basis, the optimality property that led us to the consideration of smoothing splines is no longer fulfilled. An alternative, corresponding to the default smoother in the mgcv package in R, is to determine a reduced basis in terms of an optimal approximation of the original basis utilizing a spectral decomposition of the design matrix. We will discuss this possibility in somewhat more detail in Sect. 8.2.2 on thin plate splines in the context of bivariate



smoothing and utilizing a different basis since smoothing splines are a special case of thin plate splines.

8.1.5 Random Walks



Random walks are not only used for the construction of prior distributions in Bayesian regression models but can also be employed to directly model nonlinear effects. In particular, random walks provide a univariate counterpart to Markov random fields that we will discuss in the context of spatial smoothing in Sect. 8.2.4.

Random walk models are especially useful when analyzing time series data, i.e., we are assuming a temporal trend function $f(t)$ that is to be modeled nonparametrically, similarly as with function $f(z)$:

$$y_t = f(t) + \varepsilon_t, \quad t = 1, \dots, T.$$

In a random walk approach, we identify every function evaluation $f(t)$ with the parameter γ_t and then assume a random walk prior for these parameters, i.e.,

$$f(t) = \gamma_t = \gamma_{t-1} + u_t, \quad u_t \sim N(0, \tau^2),$$

or

$$f(t) = \gamma_t = 2\gamma_{t-1} - \gamma_{t-2} + u_t, \quad u_t \sim N(0, \tau^2),$$

with random walks of first and second order, respectively. Random walks can also be considered as special P-splines if we use a B-spline basis of degree zero ($l = 0$) and if we further identify the knots with the observed time points. As with Bayesian P-splines, a PLS criterion results from this setting, which we can explicitly write for the RW2 as

$$\text{PLS}(\lambda) = \sum_{t=1}^T (y_t - \gamma_t)^2 + \lambda \sum_{t=3}^T (\gamma_t - 2\gamma_{t-1} + \gamma_{t-2})^2.$$

In econometrics, the solution of the respective minimization problem is also known as the Hodrick–Prescott filter (see Hodrick & Prescott, 1997), but actually dates back to Whittaker (1923). In Hodrick and Prescott (1997), rules of thumb for the choice of the smoothing parameter are motivated by theoretical considerations regarding the signal-to-noise ratio. In contrast to this, embedding random walks into the general context of nonparametric regression offers the opportunity to use the approaches for an optimal choice of smoothing parameters as discussed in Sect. 8.1.9.

While in time series the assumption of equally spaced time intervals is often satisfied, this is usually not the case when transferring the random walk approach to nonparametric smoothing of general covariate effects $f(z)$. In this case, a weighted random walk definition is often considered with the weights derived from the

distances between covariate values. We will not discuss this case further here. A detailed discussion can be found in [Rue and Held \(2005\)](#) within the scope of more general Gauß–Markov random fields.



8.1.6 Kriging

In this section, we discuss a smoothing approach, which was not directly developed for univariate smoothing, but which has its origin in spatial statistics. The method is referred to as *kriging*, named after the South African mining engineer D. G. Krige, who used the approach when modeling the course of stone cores based on depth drillings. Even though kriging is rarely used for univariate smoothing, we introduce it at this point to discuss the general idea and its close relationship to basis function smoothing.

Classical Kriging

The main idea of kriging is to express (originally spatial) correlations among the data using parametric correlation functions and to employ this information when fitting a regression model. This can be achieved through a stationary and multivariate normally distributed stochastic process, which induces the desired correlations analogous to random effects models. Therefore kriging is often thought of as synonymous to the inclusion of a stationary Gaussian process (or stationary Gaussian field in spatial statistics) in the regression model. Since we focus for the moment on univariate smoothing methods, we consider time series data instead of spatially aligned data, which will be discussed in more detail in Sect. 8.2.3.

We assume that a time series in continuous time $t \in \mathbb{R}$,

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + \epsilon_t$$

is given. In contrast to the classical linear model, the assumption of uncorrelated errors is usually violated in this situation, and it seems plausible to rather consider temporally correlated errors. Therefore we assume that the error ϵ is multivariate normal with expectation $\mathbf{0}$ and covariance matrix $\sigma^2 \mathbf{I} + \tau^2 \mathbf{R}$. The matrix $\mathbf{R}$ is a correlation matrix, described in terms of a parametric correlation function ρ , leading to the elements

$$\mathbf{R}[t, s] = \text{Corr}(\epsilon_t, \epsilon_s) = \rho(t, s).$$

Hence, the covariance matrix is composed of an uncorrelated part $\sigma^2 \mathbf{I}$ and a correlated part $\tau^2 \mathbf{R}$. The former part corresponds to the usual covariance matrix in the classical linear model, while the later part models temporal dependencies. If the parameters σ^2 and τ^2 , as well as the correlation function ρ , are fully known, we obtain a general linear model with covariance matrix $\text{Cov}(\mathbf{y}) = \mathbf{V} = \sigma^2 \mathbf{I} + \tau^2 \mathbf{R}$, which can be estimated as outlined in Sect. 4.1. In particular, the weighted least squares estimate for $\boldsymbol{\beta}$ results as

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}' \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}' \mathbf{V}^{-1} \mathbf{y}.$$

Similar as in Chap. 7, we can estimate the variance parameters σ^2 and τ^2 (and also possibly additional parameters for the correlation function), by maximizing the profile likelihood or the corresponding restricted likelihood function.

Most commonly, *stationary* error distributions are assumed, i.e., $\rho(t, s)$ only depends on the time difference $t - s$, and not on the exact locations of the time points t and s . This yields correlation functions $\rho(h)$ with scalar argument $h = t - s$.

Clearly, we have $\rho(0) = 1$ and $\rho(h) = \rho(-h)$ for a valid correlation function. Moreover, some further properties are often assumed:

- As $h \rightarrow \infty$, we have $\rho(h) \rightarrow 0$, i.e., for observations having large temporal distance, the correlation becomes negligible.
- The correlation function $\rho(h)$ decreases monotonically. This corresponds to the intuitive idea that observations that are temporally close to each other should always show a higher correlation than observations that are more distant apart in time.

Some frequently used parametric correlation functions belong to one of the following classes:

The family of *spherical correlation functions* is defined by

$$\rho(h; \phi) = \begin{cases} 1 - \frac{3}{2}|h/\phi| + \frac{1}{2}|h/\phi|^3 & 0 \leq h \leq \phi, \\ 0 & h > \phi. \end{cases}$$

The only parameter of the spherical correlation function is given by $\phi > 0$, which controls the distance at which the correlation decreases to 0. See Fig. 8.20a which shows spherical correlation functions for different values of ϕ . Due to this interpretation, ϕ is also called the *range* of the correlation function. The non-differentiability of the spherical correlation function in $h = \phi$ can be problematic when considering maximum likelihood estimation of the parameter ϕ . Obviously, the change from positive correlations to non-existent correlations occurs in a continuous but not differentiable form.

The *power exponential family* is defined by

$$\rho(h; \phi, \kappa) = \exp(-|h/\phi|^\kappa)$$

with parameters $\phi > 0$ and $0 < \kappa \leq 2$. The basic exponential correlation function results for $\kappa = 1$. The Gaussian correlation function, which resembles the normal density, follows for $\kappa = 2$. Figure 8.20b shows the power exponential family for various values of κ (and fixed $\phi = 1$). In contrast to the spherical correlation function, the power exponential correlation between two points remains positive regardless of their distance, i.e., the range is unlimited. Therefore, one often considers the *effective range* instead, i.e., the distance where the correlation function falls below a predetermined small value (e.g., 0.01). In applications, correlations between points that are farther apart than the effective range are then usually considered negligible and are therefore set to zero to achieve a sparser correlation matrix $\mathbf{R}$. The effective range of the power exponential family is mostly determined

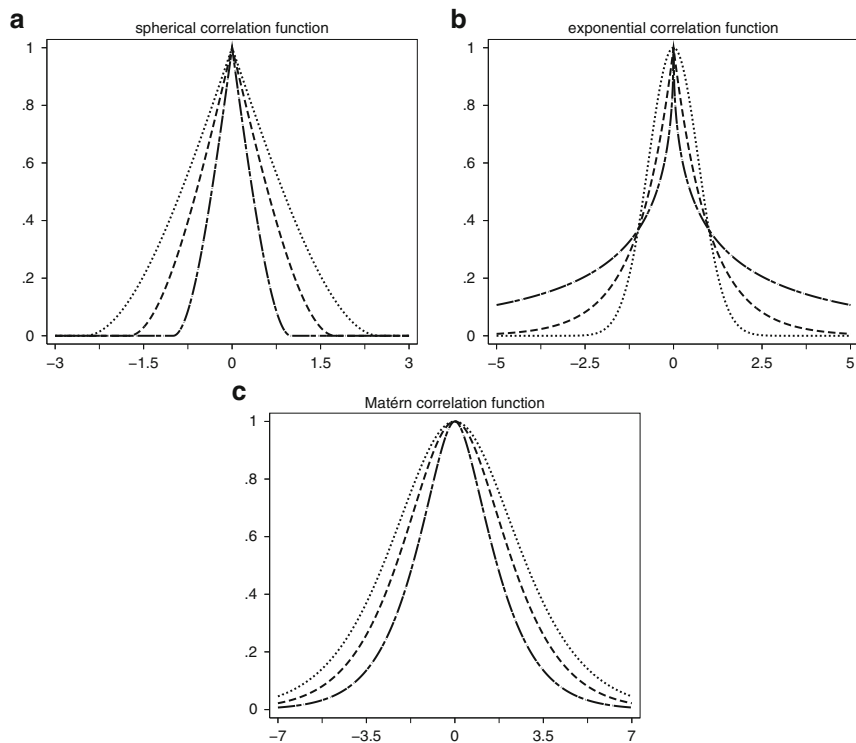


Fig. 8.20 Examples of parametric correlation functions: Panel (a) shows spherical correlation functions for $\phi = 1$ (—), $\phi = 1.75$ (---), and $\phi = 2.5$ (···); panel (b) visualizes exponential correlation functions (with fixed $\phi = 1$) for $\kappa = 0.5$ (—), $\kappa = 1$ (---) and $\kappa = 2$ (···); and panel (c) shows Matérn correlation functions (with fixed $\phi = 1$) for $\kappa = 1.5$ (—), $\kappa = 2.5$ (---), and $\kappa = 3.5$ (···). Note that the range of the correlation functions varies over the different panels, i.e., the range of the *horizontal axis* is adapted to facilitate the comparison of correlation functions within one panel

by the parameter ϕ , which controls the scaling of the time axis, but also varies depending on the specific value of κ .

The basic exponential correlation function also has an interesting connection with linear models having autocorrelated errors; see also Sect. 4.1.4. In this case, the time index t and also the observed time difference h are assumed to be discrete. In this setting, the simplest and the most frequently used model to account for temporal correlations is an AR(1)-process, which assumes

$$\epsilon_t = \rho \epsilon_{t-1} + u_t, \quad u_t \sim N(0, \tau^2), \quad (8.9)$$

for the errors. The parameter ρ controls the influence of the lagged values ϵ_{t-1} on ϵ_t , so that positively correlated errors result for $\rho > 0$, with correlation function

$$\text{Corr}(\epsilon_t, \epsilon_{t+h}) = \rho^h, \quad h = 1, 2, \dots$$

Hence, the correlation decays exponentially in the time distance h similar to the time-continuous exponential correlation function. If we define the parameter $\rho = \exp(-1/\phi)$, we can rewrite the exponential correlation function as

$$\rho(h) = \exp(-h/\phi) = \exp(-1/\phi)^h = \rho^h.$$

Therefore, the autoregressive error process Eq. (8.9) can also be interpreted as a discretized version of a continuous-time process with exponential correlation function.

One of the most flexible families of correlation functions, which in practice became the standard of spatial statistics, is the *Matérn* family $\rho(h; \phi, \kappa)$ with $\phi > 0$, $\kappa > 0$. The general representation of correlation functions from this family is, however, only possible with the help of modified Bessel functions of the order κ , which can only be evaluated numerically, but cannot be derived explicitly. Consequently, one often restricts the attention to a subset of the Matérn correlation functions with $\kappa = 0.5, 1.5, 2.5, \dots$. For these cases, the correlation function can be derived analytically and, for example, takes the form

$$\begin{aligned}\rho(h; \phi, \kappa = 0.5) &= \exp(-|h/\phi|), \\ \rho(h; \phi, \kappa = 1.5) &= \exp(-|h/\phi|)(1 + |h/\phi|), \\ \rho(h; \phi, \kappa = 2.5) &= \exp(-|h/\phi|)(1 + |h/\phi| + \tfrac{1}{3}|h/\phi|^2), \\ \rho(h; \phi, \kappa = 3.5) &= \exp(-|h/\phi|)(1 + |h/\phi| + \tfrac{2}{5}|h/\phi|^2 + \tfrac{1}{15}|h/\phi|^3).\end{aligned}$$

For $\kappa = 0.5$, the Matérn correlation function again simplifies to the basic exponential correlation function. Figure 8.20c shows the remaining three examples. Larger values of κ evidently lead to correlation functions with a larger effective range, and thus stronger correlations for points in time with large temporal distance. If we let κ go to infinity, we obtain the Gaussian correlation function as the limiting case, i.e., yet another special case of the power exponential correlation function.

Kriging as Time Series Smoother

In order to motivate the use of correlation functions and stationary Gaussian processes in nonparametric function estimation, we decompose the error ϵ_t into a temporally correlated component γ_t and the remaining independent and identically distributed component ε_t :

$$y_t = \mathbf{x}_t' \boldsymbol{\beta} + \gamma_t + \varepsilon_t, \quad t \in \{t_{(1)}, \dots, t_{(d)}\}.$$

With adequate distributional specifications for γ_t and ε_t , we obtain exactly the covariance structure for the response variable $\mathbf{y}$ considered above in the previous paragraph.

In matrix notation, the model can be represented as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon},$$

where $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_d)'$ defines the vector of the d unique and temporally ordered errors γ_t and $\mathbf{Z} = \mathbf{I}_d$ is the d -dimensional identity matrix. In a second step, we further reparameterize the model to

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{R} \cdot \mathbf{R}^{-1}\boldsymbol{\gamma} + \boldsymbol{\varepsilon} = \mathbf{X}\boldsymbol{\beta} + \tilde{\mathbf{Z}}\tilde{\boldsymbol{\gamma}} + \boldsymbol{\varepsilon}$$

with $\tilde{\mathbf{Z}} = \mathbf{Z}\mathbf{R} = \mathbf{R}$ and $\tilde{\boldsymbol{\gamma}} = \mathbf{R}^{-1}\boldsymbol{\gamma}$.

This reparameterization evidently leads to the same distribution of $\mathbf{y}$ and, thus, to an equivalent model formulation. However, the interpretation of the design matrix $\tilde{\mathbf{Z}}$ changes. Due to the special structure of $\mathbf{Z}$, we obtain

$$\tilde{\mathbf{Z}}[i, j] = \rho(t_{(i)}, t_{(j)})$$

for the individual entries. If we compare this definition with the construction of the design matrix for B-splines or TP-splines, we find that the correlation function ρ has the role of a basis function, and that the observed time points $t_{(j)}$ adopt the role of knots. Thus, for the temporal trend, we have

$$f(t) = \gamma_t = \sum_{j=1}^d \tilde{\gamma}_j \rho(t, t_{(j)}).$$

The joint distribution of the temporally correlated effects $\tilde{\boldsymbol{\gamma}}$ is

$$\tilde{\boldsymbol{\gamma}} \sim \mathbf{N}(0, \tau^2 \mathbf{R}^{-1})$$

and therefore has a density of the form

$$p(\tilde{\boldsymbol{\gamma}} \mid \tau^2) \propto \exp\left(-\frac{1}{2\tau^2} \tilde{\boldsymbol{\gamma}}' \mathbf{R} \tilde{\boldsymbol{\gamma}}\right).$$

This form exactly corresponds to the smoothing prior that we discussed in connection with the Bayesian formulation of penalty approaches. It also yields an equivalent PLS criterion defined as

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \tilde{\mathbf{Z}}\tilde{\boldsymbol{\gamma}})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \tilde{\mathbf{Z}}\tilde{\boldsymbol{\gamma}}) + \frac{\sigma^2}{\tau^2} \tilde{\boldsymbol{\gamma}}' \mathbf{R} \tilde{\boldsymbol{\gamma}}.$$

In this interpretation, we assume that all parameters and in particular the range parameter ϕ in the correlation function are either given or determined before estimating the regression coefficients $\tilde{\boldsymbol{\gamma}}$ so that the basis functions (and also the penalty matrix) do no longer contain any unknown parameters. In Example 8.3 below, we will provide a simple data-driven rule to determine the range parameter when applying kriging as a smoothing approach.

In summary, even though the kriging method was derived from a purely stochastic model to describe correlated data, the method is formally equivalent to a basis function approach with basis functions ρ and penalty matrix $\mathbf{R}$. This explains why the simultaneous modeling of a temporal trend through a nonparametric approach (or a simple polynomial), together with temporally correlated errors, can lead to identification problems. Due to the temporally correlated errors, a nonparametric trend function is also implicitly estimated, so that the additional assumption of a further trend leads to a nearly non-identifiable model. The representation of kriging as a basis function approach also illustrates that the estimate of the temporal effects inherits the smoothness properties of the correlation function ρ . When the correlation function is differentiable or continuous, the same holds for the estimated function.

Kriging as a Nonparametric Smoothing Procedure

In order to use kriging to estimate nonlinear covariate effects, we return to the model

$$y_i = f(z_i) + \varepsilon_i.$$

If $z_{(1)} < \dots < z_{(d)}$ are the ordered covariate values, we define the parameters $\gamma_j = f(z_{(j)})$ and assume a stationary Gaussian process prior, with expectation 0, variance τ^2 , and correlation function

$$\rho(\gamma_j, \gamma_r) = \rho(|z_{(j)} - z_{(r)}|).$$

Accordingly, we can also view the correlation function as a basis function, and, as with smoothing splines, we can identify the knot positions as the covariate locations $z_{(j)}$. We thus obtain the representation

$$\mathbf{y} = \tilde{\mathbf{Z}} \tilde{\boldsymbol{\gamma}} + \boldsymbol{\varepsilon}$$

with $\mathbf{Z}[i, j] = \rho(|z_i - z_{(j)}|)$ and $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_d)'$. We then have to optimize the penalized residual sum of squares

$$\text{PLS}(\lambda) = (\mathbf{y} - \tilde{\mathbf{Z}} \tilde{\boldsymbol{\gamma}})'(\mathbf{y} - \tilde{\mathbf{Z}} \tilde{\boldsymbol{\gamma}}) + \lambda \tilde{\boldsymbol{\gamma}}' \mathbf{K} \tilde{\boldsymbol{\gamma}}$$

with $\mathbf{K}[j, r] = \rho(|z_{(j)} - z_{(r)}|)$ and smoothing parameter $\lambda = \sigma^2/\tau^2$. Hence, the usual results for PLS estimation carry over to kriging and, in particular, we obtain the known form of the PLS estimate.

Example 8.3 Malnutrition in Tanzania: Kriging

Figure 8.21 shows several kriging estimates for the effect of age on the malnutrition score. These estimates are obtained with the four previously presented special Matérn correlation functions, with scale parameter determined via

$$\hat{\phi} = \max_{j,r} |z_{(j)} - z_{(r)}|/c. \quad (8.10)$$

8.7 Kriging as a Nonparametric Smoothing Procedure

The function f is expanded in terms of basis functions resulting from a parametric correlation function $\rho(h)$. This yields the PLS criterion

$$\text{PLS}(\lambda) = (y - \tilde{Z}\tilde{\gamma})'(y - \tilde{Z}\tilde{\gamma}) + \lambda\tilde{\gamma}'K\tilde{\gamma},$$

with $\tilde{Z}[i, j] = \rho(|z_i - z_{(j)}|)$ and $K[j, r] = \rho(|z_{(j)} - z_{(r)}|)$. Smoothing properties of the chosen correlation function carry over to the estimate $\hat{f}(z)$.

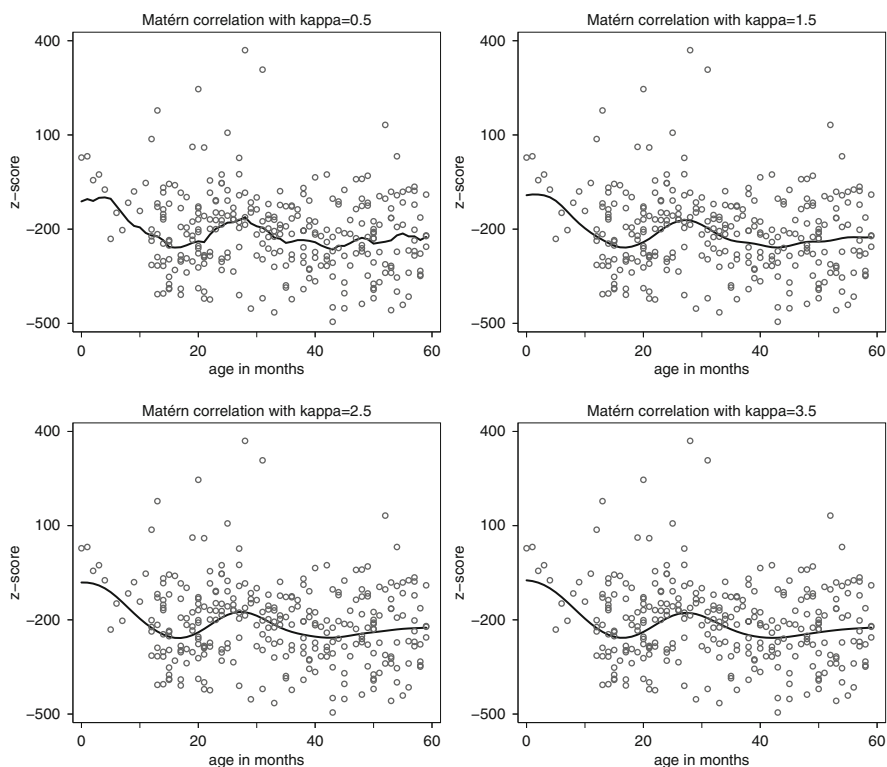


Fig. 8.21 Malnutrition in Tanzania: kriging estimates

The value of the constant c is chosen such that $\rho(c; \phi = 1)$ is small, i.e., in essence, c determines the effective range of the correlation function. Choosing c in this manner ensures that the correlation functions are well distributed across the covariate domain. For polynomial splines, the choice of knots automatically guarantees such a distribution while in the case of kriging an effective range that is too small can result in non-overlapping correlation functions. This would in turn induce unstable estimation of the nonparametric

effects in the corresponding regions. Moreover, Eq. (8.10) guarantees scale invariance of the resulting estimates, i.e., when scaling the covariates z_i with a constant factor, this scaling is compensated by rescaling the range parameter ϕ . The variance τ^2 (and, thus, also the smoothing parameter λ) is determined using REML estimation as already introduced in the context of linear mixed models in Sect. 7.3.2. Section 8.1.9 will show that indeed REML estimation can also be used to determine smoothing parameters in penalized basis function approaches or more generally in nonparametric regression models.

The estimated age effects $\hat{f}(z)$ also show the impact of specific properties of the chosen correlation function: For $\kappa = 0.5$ we obtain a rather rough, non-differentiable function estimate, while for larger values of κ , the function is estimated much smoother. Moreover, the differences between the estimates decrease as the correlation function varies beyond $\kappa = 0.5$.

△

8.1.7 Local Smoothing Procedures

We next examine another class of smoothers, which, in general, cannot be derived from a global regression formulation but are defined *locally*. Many of these estimates are extremely intuitive and easy to understand, and thus are often used in explorative analyses.

Nearest Neighbor Estimates

A widely used method for the (descriptive) smoothing of time series is *running means*. For a time series y_t , $t = 1, \dots, T$, running means of order 3 are, for example, defined by

$$\hat{y}_t = \frac{1}{3}(y_{t-1} + y_t + y_{t+1}),$$

with appropriate modifications at the boundaries. Obviously, taking averages of observed values in time generally smooths the random fluctuations of a time series and gives a first impression of the underlying trend.

Nearest neighbor estimates extend the concept of running means into a more general framework, and also enable the application in nonparametric regression models. In general, a nearest neighbor estimate is defined by

$$\hat{f}(z) = \text{Ave}_{j \in N(z)} y_j,$$

where Ave defines some averaging operator and $N(z)$ is an appropriate neighborhood of z . In the example of running means of order 3, the averaging operator is the arithmetic mean and the two (temporally) adjacent values $t - 1$ and $t + 1$ (as well as t itself) form the neighborhood of time point t . In general, observations are considered to be their own neighbors, so that $z_i \in N(z_i)$ holds.

The following averaging operators are often used for the determination of nearest neighbor estimates:

8.8 Nearest Neighbor Estimates

The general form of a nearest neighbor estimate is

$$\hat{f}(z) = \text{Ave}_{j \in N(z)} y_j,$$

with a user-defined averaging operator Ave and a local neighborhood $N(z)$. As averaging operator, we often use the arithmetic mean, the median, or the linear regression. Neighborhoods can either be defined symmetrically around z or can be based on the k -nearest neighbors.

1. Arithmetic mean (running mean): Determine the arithmetic mean of the response variable in the neighborhood of z , i.e.,

$$\hat{f}(z) = \frac{1}{|N(z)|} \sum_{j \in N(z)} y_j,$$

where $|N(z)|$ is the number of neighbors of z .

2. Median (running median): Determine the median of the response variables in the neighborhood of z , i.e.,

$$\hat{f}(z) = \text{Median}\{y_j, j \in N(z)\}.$$

3. Linear regression (running line): Estimate a linear regression based on the observations in the neighborhood of z and use the prediction from this model as the estimate, i.e.,

$$\hat{f}(z) = \hat{\gamma}_{0,z} + \hat{\gamma}_{1,z}z,$$

where $\hat{\gamma}_{0,z}$ and $\hat{\gamma}_{1,z}$ are the least squares estimates using the data $\{(y_j, z_j), j \in N(z)\}$.

To completely determine a nearest neighbor estimate, we also need an adequate neighborhood definition. Commonly used variants are the following:

1. Symmetric neighborhoods of order k
2. Neighborhoods that consist of k -nearest neighbors

Running means of order 3 provide a simple example for a symmetric neighborhood. When defining symmetric neighborhoods (with an uneven order k), we generally proceed as follows: When estimating $f(z_i)$, we use the nearest $(k - 1)/2$ observations to the right and to the left from z_i (in addition to z_i itself) as the neighborhood. Hence, the order is divided symmetrically to the area left and right of z_i , yielding a justification for the term *symmetric neighborhood*. If, for the sake

of simplicity, we assume ordered observations with $z_1 \leq \dots \leq z_n$, the symmetric neighborhood is given by

$$N(z_i) = \{\max(1, i - (k - 1)/2), \dots, i - 1, i, i + 1, \dots, \min(n, i + (k - 1)/2)\}.$$

This definition already includes the necessary corrections at the boundaries of the defined domain, as there are not enough observations available to apply the standard definition. Note that the neighborhood at the boundaries is no longer symmetric and also does not contain k observations.

Instead of using the order k , symmetric neighborhoods can also be defined based on the fraction $\omega \in (0, 1)$ of observations contained in the neighborhood (also referred to as the *bandwidth*). The corresponding order k is then given by ωn , where some rounding scheme has to be applied for non-integer values of ωn . The order k , as well as the fraction ω , can be interpreted as smoothing parameters of the nearest neighbor estimate. Figure 8.22 demonstrates the effect of varying ω for the Tanzania data set using local arithmetic means. The closer the bandwidth ω is to 1, the more neighbors are included in the estimate, leading to smoother functions. As ω gets smaller and finally approaches 0, the estimate $\hat{f}(z_i)$ is based on only very few observations, resulting in a very rough estimate.

The definition of a neighborhood based on k -nearest neighbors generally leads to asymmetric neighborhoods. In this case, the neighborhood $N(z)$ is given by

$$N(z) = \{i : d_i \in \{d_{(1)}, \dots, d_{(k)}\}\},$$

where $d_{(1)}, \dots, d_{(n)}$ represents the ordered distances, $d_i = |z_i - z|$. In this setting, the same number of neighbors is used, even at the boundaries, i.e., modifications at the boundaries are not necessary. Similar to the symmetric neighborhoods, we can interpret the number of neighbors k as a smoothing parameters. If k is small, a rough estimate results while, on the other hand, a very smooth estimate is obtained with k chosen close to the sample size n . In the latter case, we end up with the value of the averaging operator applied to the complete data set.

Local Polynomial Regression and the Nadaraya–Watson Estimator

Although, at the beginning of this chapter, we already discussed the difficulties arising from the global approximation of nonlinear functions $f(z)$, it is often possible to *locally* approximate $f(z)$ with polynomials (as already illustrated in Fig. 2.12 on p. 46 for a local linear approximation). To motivate such an approximation, we consider the (local) approximation of an l -times continuously differentiable function $f(z_i)$ using a *Taylor series expansion* around z , yielding

$$f(z_i) \approx f(z) + (z_i - z)f'(z) + (z_i - z)^2 \frac{f''(z)}{2!} + \dots + (z_i - z)^l \frac{f^{(l)}(z)}{l!}.$$

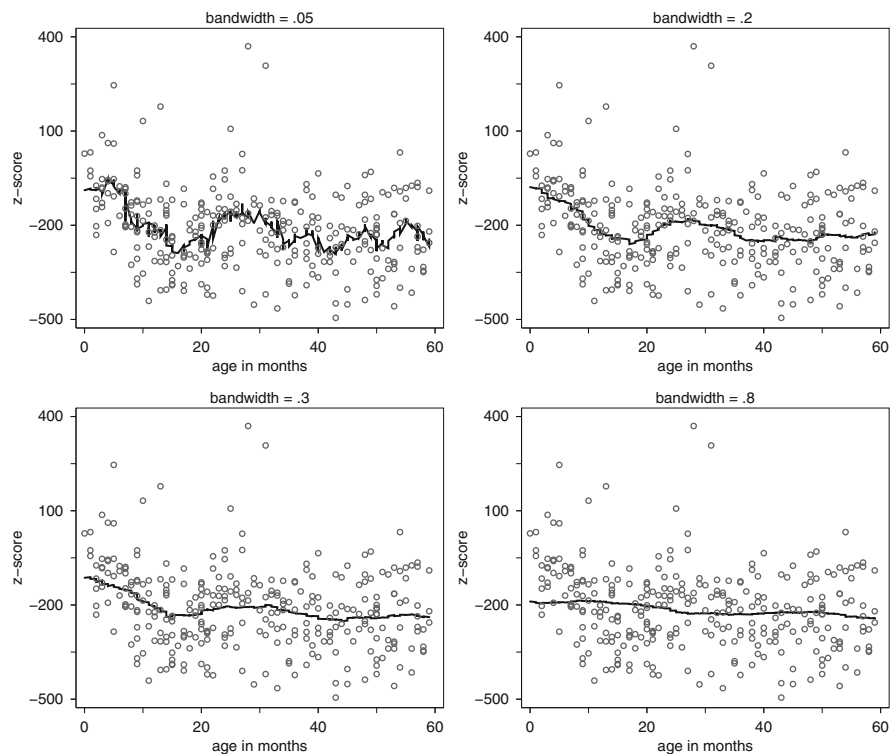


Fig. 8.22 Malnutrition in Tanzania: running mean with different bandwidths

Hence, we approximate the function $f(z_i)$ with polynomials of the form $(z_i - z)^j$ in a neighborhood of z_i . The polynomials are weighted by the derivatives $f^{(j)}(z)/j!$ evaluated at the expansion point z . Note that $f(z)$ is considered as the zero derivative of f . In practice, $l = 1$, i.e., local linear regression is used most frequently.

If we apply the Taylor series approximation to the problem of nonparametric function estimation for $f(z)$ at a given point z , we obtain

$$\begin{aligned}
 y_i &= f(z_i) + \varepsilon_i \\
 &\approx f(z) + (z_i - z)f'(z) + (z_i - z)^2 \frac{f''(z)}{2!} + \dots + (z_i - z)^l \frac{f^{(l)}(z)}{l!} + \varepsilon_i \\
 &= \gamma_0 + (z_i - z)\gamma_1 + (z_i - z)^2\gamma_2 + \dots + (z_i - z)^l\gamma_l + \varepsilon_i,
 \end{aligned}$$

for each observation (y_i, z_i) with expansion point z . Therefore, a polynomial regression model for y_i results, which is based on polynomials of the form $(z_i - z)^j$ and regression coefficients $\gamma_j = f^{(j)}(z)/j!$. If we now estimate the regression parameters from this model, we obtain an implicit estimate for the function value

$f(z)$ through $\hat{\gamma}_0 = \hat{f}(z)$, and more generally, we even obtain estimates for the derivatives through $j!\hat{\gamma}_j = \hat{f}^{(j)}(z)$.

Hence, an estimate for $f(z)$ can in principle be determined by fitting a specific linear model. However, since the Taylor series approximation is only valid locally, i.e., close to the expansion point z , estimation is usually based on a weighted version of the residual sum of squares. This results in a weighted least squares criterion of the form

$$\sum_{i=1}^n \left(y_i - \sum_{j=0}^l \gamma_j (z_i - z)^j \right)^2 w_\lambda(z, z_i)$$

with weights $w_\lambda(z, z_i)$. These are typically constructed based on the distances $|z_i - z|$ such that larger weights result for observations with a small distance. A general class of such weights results with the use of *kernel functions* K in

$$w_\lambda(z, z_i) = K\left(\frac{z_i - z}{\lambda}\right). \quad (8.11)$$

Typical examples of kernel functions include

$$\begin{aligned} K(u) &= \begin{cases} \frac{1}{2} & -1 \leq u \leq 1 \\ 0 & \text{otherwise} \end{cases} && \text{Uniform kernel,} \\ K(u) &= \begin{cases} \frac{3}{4}(1 - u^2) & -1 \leq u \leq 1 \\ 0 & \text{otherwise} \end{cases} && \text{Epanechnikov kernel,} \\ K(u) &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}u^2\right) && \text{Gaussian kernel;} \end{aligned}$$

see Fig. 8.23. The Epanechnikov kernel, as well as the uniform kernel, set all weights outside the interval $[-1, 1]$ to zero, whereas the Gaussian kernel maintains positive weights for observations of arbitrary distance. The additional parameter λ (the bandwidth of the kernel) controls how quickly the weights approach zero and can be viewed as the smoothing parameter of local polynomial regression. Kernel functions are also commonly applied in nonparametric density estimation (see Fahrmeir, Künstler, Pigeot, & Tutz, 2007 or Härdle, 1990) and are therefore also densities themselves. Most commonly, kernel functions are assumed to be symmetric about zero.

In order to determine the weighted least squares estimate in local polynomial regression modeling, we first write the weighted residual sum of squares in matrix notation as

$$(\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma})' \mathbf{W} (\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma})$$

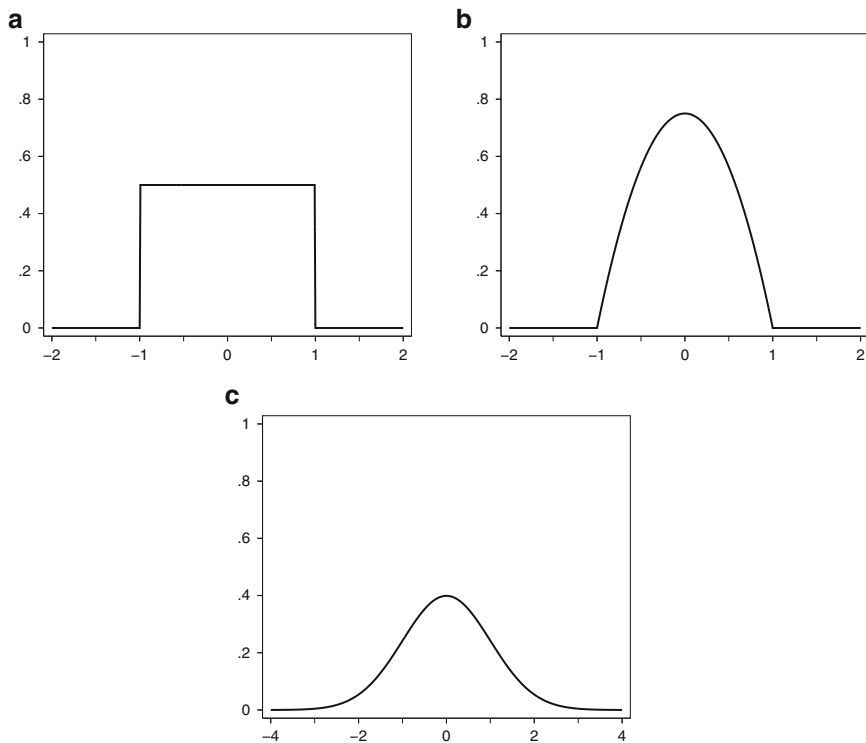


Fig. 8.23 Examples of kernel functions. (a) Rectangular kernel, (b) Epanechnikov kernel, (c) Gaussian kernel

with design matrix

$$\mathbf{Z} = \begin{pmatrix} 1 & (z_1 - z) & \dots & (z_1 - z)^l \\ \vdots & \vdots & & \vdots \\ 1 & (z_n - z) & \dots & (z_n - z)^l \end{pmatrix},$$

vector of the regression coefficients $\boldsymbol{\gamma} = (\gamma_0, \dots, \gamma_l)'$, and weight matrix

$$\mathbf{W} = \text{diag}(w_\lambda(z, z_1), \dots, w_\lambda(z, z_n)).$$

Minimizing this residual sum of squares corresponds to the estimation of a general linear model as discussed in Sect. 4.1. Therefore, we obtain the weighted least squares estimate

$$\hat{\boldsymbol{\gamma}} = (\mathbf{Z}'\mathbf{W}\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{W}\mathbf{y} \quad \text{and} \quad \hat{f}(z) = \hat{\gamma}_0. \quad (8.12)$$

8.9 Local Polynomial Regression

Based on the Taylor series expansion around the point z , we obtain a local representation

$$y_i \approx \gamma_0 + (z_i - z)\gamma_1 + (z_i - z)^2\gamma_2 + \dots + (z_i - z)^l\gamma_l + \varepsilon_i$$

with regression coefficients $\gamma_j = f^{(j)}(z)/j!$. To determine the corresponding estimate, we minimize the weighted least squares criterion,

$$\sum_{i=1}^n \left(y_i - \sum_{j=0}^l \gamma_j (z_i - z)^j \right)^2 w_\lambda(z, z_i),$$

where the weights $w_\lambda(z, z_i)$ are defined based on a kernel function. From the resulting weighted least squares estimate, we obtain $\hat{f}(z) = \hat{\gamma}_0$. The Nadaraya–Watson estimate results as a special case with $l = 0$.

Since both the design matrix $\mathbf{Z}$ and the weight matrix $\mathbf{W}$ depend on the location of z where we aim to estimate f , each point requires a separate model fit. In practice, one usually considers estimates at an equidistant grid of covariate values.

An interesting special case of local polynomial regression results for the local constant polynomial model. In this case, we obtain the weighted residual sum of squares

$$\sum_{i=1}^n (y_i - \gamma_0)^2 w_\lambda(z, z_i),$$

and the estimate $\hat{f}(z)$ has the explicit representation

$$\hat{f}(z) = \frac{\sum_i w_\lambda(z, z_i) y_i}{\sum_i w_\lambda(z, z_i)}. \quad (8.13)$$

Alternatively, we can also derive the same estimate from nonparametric density estimation ideas. The estimate is then called the Nadaraya–Watson estimator; see, for example, [Fahrmeir and Tutz \(2001, Chap. 5\)](#) or [Härdle \(1990\)](#).

Loess

Nearest neighbor estimates often result in relatively rough function estimates, even if the chosen smoothing parameter is based on a suitable optimality criterion. This is due to the definition of local estimates based on neighborhoods: While the observations within the neighborhood have relatively large impact on the estimates,

8.10 Loess Estimation

1. Determine $N(z)$, the set of the k -nearest neighbors of z .
2. Determine the largest distance between any two data points within the neighborhood

$$\Delta(z) = \max_{i,j \in N(z)} |z_i - z_j|.$$

3. Define weights

$$w_{\Delta(z)}(z, z_i) = K\left(\frac{|z - z_i|}{\Delta(z)}\right)$$

with the tri-cubic kernel function

$$K(u) = \begin{cases} (1 - |u|^3)^3 & -1 \leq u \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

4. Determine $\hat{f}(z)$ using weighted linear regression based on the data points in the neighborhood $N(z)$.

all of the other observations are completely neglected. If we combine the use of weights as introduced for local polynomial regression with nearest neighbor estimates, we may expect smoother estimates due to the smoothly decaying weights instead of weights dropping to zero outside the neighborhood. This combination leads to *locally weighted regression (loess)*, originally proposed by [Cleveland \(1979\)](#) and summarized in [Box 8.10](#). The term loess is derived both from an abbreviation of LOcal regrESSion as well as from the German word *Löß*. In geology, the term *Löß* defines sediment that resulted from the erosion of rocks and that was accumulated by the wind. Levels of *Löß* form a smooth surface, a fact that has led to the corresponding term for the smooth function estimates resulting from the loess procedure. The term *lowess* is also often used instead of loess, being an abbreviation of LOcally WEighted Scatter plot Smoothing. This highlights that loess is a weighted estimate for nonparametric regression. In the original loess proposal, asymptotic considerations motivated the use of the tri-cubic kernel function but of course other kernel functions can be used as well. In statistical software packages, different loess variants are available either under the name loess or lowess. Even though they differ from each other in detail, their main approach corresponds to the algorithm provided in [Box 8.10](#).

Sometimes, a robustified version of loess is also applied where the definition of weights is iterated to reduce the influence of observations with large residuals. More precisely, in each iteration, new weights are computed as

$$w_{\Delta(z)}(z, z_i) = \delta_i K\left(\frac{|z - z_i|}{\Delta(z)}\right)$$

where δ_i is derived from the residuals of the previous iterations such that δ_i is small for large residuals. For example, $\delta_i = \exp(-|\hat{\varepsilon}_i|)$ is a possible choice.

8.1.8 General Scatter Plot Smoothing

To enable a thorough comparison of different smoothing procedures, this section introduces a general formulation of scatter plot smoothers, followed by the presentation of properties derived from this formulation.

Linear Smoothers

When reviewing the various smoothing procedures discussed so far, we find, with very few exceptions, that most of the approaches can be represented as

$$\hat{f}(z) = \sum_{i=1}^n s(z, z_i) y_i, \quad (8.14)$$

with weights $s(z, z_i)$. Accordingly, we obtain the estimate $\hat{f}(z)$ as a weighted sum of the observations y_i , where the weights depend on the location z , as well as on the observed values of the covariates $z_1, \dots, z_n$. Since $\hat{f}(z)$ can be expressed as a linear combination of the observed response values, we also refer to smoothing procedures with property (8.14) as *linear smoothing procedures*.

It is particularly straightforward to verify property (8.14) for nearest neighbor estimates (see p. 460ff). The estimated function $\hat{f}(z)$ was, in this case, defined as a local average of all observed dependent variables within the neighborhood of z . If we use a linear averaging operator, we also obtain a linear smoother. This implies that both the local arithmetic mean and local linear regression are linear smoothers. For the local arithmetic mean, we have

$$s(z, z_i) = \begin{cases} \frac{1}{|N(z)|} & \text{if } i \in N(z), \\ 0 & \text{otherwise.} \end{cases}$$

Since the median is a nonlinear averaging operator, it is not possible to determine the weights required in sum (8.14) and, thus the local median is an example of a nonlinear smoother.

Linear smoothers can also be written in matrix notation as

$$\hat{f}(z) = s(z)' \mathbf{y},$$

with vector $s(z) = (s(z, z_1), \dots, s(z, z_n))'$. In particular, we have $\hat{f}(z_i) = s(z_i)' \mathbf{y}$ for the observed covariate values. If we apply this equation to the n covariate values, we obtain the linear smoother

$$\hat{\mathbf{f}} = \mathbf{S} \mathbf{y},$$

with $\hat{\mathbf{f}} = (\hat{f}(z_1), \dots, \hat{f}(z_n))'$ and the $n \times n$ - *smoother matrix* $\mathbf{S}$ consisting of the row vectors $s(z_i)'$. The smoother matrix will form the basis for many of the tools to characterize nonparametric smoothers; see below.

The linear model discussed in Chap. 3, as well as the general linear model introduced in Sect. 4.1, can actually be viewed as linear smoothers. In these cases, the smoother matrix coincides with the prediction or hat matrix $\mathbf{H}$ (see p. 107) and is therefore given by

$$\mathbf{S} = \mathbf{H} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$$

for the classical linear model and by

$$\mathbf{S} = \mathbf{Z}(\mathbf{Z}'\mathbf{W}\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{W}$$

for the general linear model. As a consequence, polynomial splines and any other basis function approach (without penalization) are linear smoothers. In order to estimate the function f at a specific (possibly unobserved) covariate value z , we obtain

$$s(z)' = \mathbf{z}'(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}',$$

where $\mathbf{z}$ is the vector of basis functions evaluated at z (see p. 423).

For any roughness penalty approach derived from a model of the form

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$$

with penalty matrix $\mathbf{K}$ and smoothing parameter λ , we only have to replace the least squares estimate with the PLS estimate $\hat{\boldsymbol{\gamma}}$; see formula (8.3) on p. 439. Thus, we obtain

$$\mathbf{S} = \mathbf{Z}(\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})^{-1}\mathbf{Z}'$$

and

$$s(z)' = \mathbf{z}'(\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})^{-1}\mathbf{Z}'.$$

It follows that all penalization approaches derived from polynomial splines, but also smoothing splines, random walks, and kriging methods are linear smoothers.

Moreover, the two remaining methods (local polynomial regression and loess) can also be cast in the framework of linear smoothing methods. For local polynomial regression, the function estimate $\hat{f}(z)$ was derived as the weighted least squares estimate (8.12). Thus, we obtain the corresponding weight vector $s(z)$ as

$$\hat{f}(z) = \hat{\gamma}_0 = \mathbf{e}_1'(\mathbf{Z}'\mathbf{W}\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{W}\mathbf{y}, = s(z)'\mathbf{y}$$

with $\mathbf{e}_1 = (1, 0, 0, \dots)'$. The vector $\mathbf{e}_1$ extracts the first component of the weighted least squares estimate that corresponds to the function estimate and drops all values corresponding to derivatives. The linearity of the loess estimate directly follows from its construction as a weighted nearest neighbor estimate.

8.11 Confidence Intervals and Confidence Bands for Linear Smoothers

1. Pointwise confidence intervals with level α :

$$\hat{f}(z) \pm z_{1-\alpha/2} \sigma \sqrt{s(z)'s(z)},$$

where $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ -quantile of the standard normal distribution.

2. Simultaneous confidence bands at $\{z_1, \dots, z_j, \dots, z_r\}$ with Bonferroni correction:

$$\hat{f}(z_j) \pm z_{1-\alpha/(2r)} \sigma \sqrt{s(z_j)'s(z_j)}.$$

3. Simultaneous confidence bands at $\{z_1, \dots, z_j, \dots, z_r\}$ based on the joint distribution of $(f(z_1), \dots, f(z_r))'$:

$$\hat{f}(z_j) \pm m_{1-\alpha} \sigma \sqrt{s(z_j)'s(z_j)},$$

where the quantile $m_{1-\alpha}$ of Eq. (8.18) is determined via simulation.

The same formulae can be used when replacing σ^2 with a consistent estimate $\hat{\sigma}^2$.

We can summarize as follows: With the exception of the local median, all smoothing methods discussed thus far are in fact linear smoothers. In the following, we will therefore focus on linear smoothers and derive properties for this general class.

Confidence Intervals and Confidence Bands

In addition to point estimates for $f(z)$, information on the variability of the estimate or corresponding confidence intervals is typically required in practice. Considering the case of the function estimate at a fixed covariate value z , it is rather straightforward to derive a variance formula for linear smoothers. Since $\hat{f}(z) = s(z)'y$ and $\text{Var}(\epsilon) = \sigma^2 I$, it directly follows

$$\text{Var}(\hat{f}(z)) = \sigma^2 s(z)'s(z).$$

If we additionally assume normally distributed errors, a $(1 - \alpha)$ -confidence interval is given by

$$\hat{f}(z) \pm z_{1-\alpha/2} \sigma \sqrt{s(z)'s(z)}, \quad (8.15)$$

where $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ -quantile of the standard normal distribution. This construction is based on the assumption

$$\hat{f}(z) - f(z) \stackrel{a}{\sim} N(0, \sigma^2 \mathbf{s}(z)' \mathbf{s}(z)).$$

While both the variance of the estimate $\hat{f}(z)$ and the normal distribution follow from the distributional assumptions for the error terms, we also have to assume that $\hat{f}(z)$ is (approximately) unbiased. In general, this property will only be fulfilled asymptotically and therefore the coverage probability of the constructed confidence intervals is also asymptotic in nature. If the error terms are not assumed to be normally distributed, the normal distribution for the estimates will also hold only asymptotically.

One often uses the confidence intervals (8.15) to provide an impression about the variability of the estimated function in graphical representations. In addition to the estimated function, we also plot the lower and upper limit of the confidence interval, as functions of the covariate value z ; see Fig. 8.24. However, when interpreting such figures, we need to keep in mind that the confidence intervals have been constructed for a single covariate value z and therefore the coverage probability is only valid pointwise and not simultaneously for the complete function. More specifically, for a pointwise level $(1 - \alpha)$ -confidence interval $[L(z), U(z)]$, we have

$$P(L(z) \leq f(z) \leq U(z)) \geq 1 - \alpha,$$

i.e., the probability that the (random) interval covers the (fixed) true function value at the prespecified covariate value z is at least $1 - \alpha$. It is critical that the expression only applies for a fixed value z , and not for several covariate values $z_1, \dots, z_r$ at the same time. However, often such simultaneous expressions are of interest, and thus the construction of (simultaneous) *confidence bands* is necessary.

Such a simultaneous confidence band $[L(z), U(z)]$ of level $1 - \alpha$ should fulfill

$$P(L(z) \leq f(z) \leq U(z) \text{ for all } z \in \{z_1, \dots, z_r\}) \geq 1 - \alpha. \quad (8.16)$$

In this case, the coverage probability does not apply pointwise but rather simultaneously for all covariate values in $\{z_1, \dots, z_r\}$. A simple option to derive such a simultaneous confidence band is given by the *Bonferroni correction* of the level α . The Bonferroni correction is motivated by the fact that, for the simultaneous occurrence of events $A_1, \dots, A_r$, we have

$$P(A_1 \cap \dots \cap A_r) \leq \sum_{j=1}^r P(A_j).$$

In our case, A_j corresponds to the event “the function evaluation $f(z_j)$ is not contained in the interval $[L(z_j), U(z_j)]$.” Now the aim is to limit the probability that (at least) one of these events occurs by α . This can be achieved by combining pointwise confidence intervals with levels $\alpha_1, \dots, \alpha_r$, such that $\sum_j \alpha_j \leq \alpha$. The

most common approach for determining such levels α_j is to uniformly distribute the confidence level as $\alpha_j = \alpha/r$.

In summary, the Bonferroni correction provides a simple way to define simultaneous confidence bands. However, despite maintaining the desired confidence level, these confidence bands are often rather wide, and therefore the information provided by the confidence bands may be imprecise. We therefore explore a second option for constructing confidence bands, which is numerically more complex, but provides more exact results translating into narrower confidence bands. First, we determine the covariance matrix of the vector $\hat{\mathbf{f}}_r = (\hat{f}(z_1), \dots, \hat{f}(z_r))'$, which can be written as a linear smoother $\hat{\mathbf{f}}_r = \mathbf{S}_r \mathbf{y}$, where $\mathbf{S}_r$ consists of the vectors $\mathbf{s}(z_1), \dots, \mathbf{s}(z_r)$. The covariance matrix can then be expressed as

$$\text{Cov}(\hat{\mathbf{f}}_r) = \sigma^2 \mathbf{S}_r \mathbf{S}_r'$$

and therefore

$$\hat{\mathbf{f}}_r - \mathbf{f}_r \stackrel{a}{\sim} N(\mathbf{0}, \sigma^2 \mathbf{S}_r \mathbf{S}_r'). \quad (8.17)$$

From this joint distribution we obtain a simultaneous confidence band with level $1 - \alpha$ as

$$\hat{f}(z_j) \pm m_{1-\alpha} \sigma \sqrt{\mathbf{s}(z_j)' \mathbf{s}(z_j)},$$

where $m_{1-\alpha}$ defines the $(1 - \alpha)$ -quantile of the distribution of the random variable

$$\max_{1 \leq j \leq r} \left| \frac{\hat{f}(z_j) - f(z_j)}{\sigma \sqrt{\mathbf{s}(z_j)' \mathbf{s}(z_j)}} \right|. \quad (8.18)$$

In Eq. (8.18), we consider the maximum absolute standardized deviation between the true function value and the estimate. Since the distribution of this random variable and corresponding quantiles are difficult to obtain analytically, we approximate the distribution using simulation. To do so, we first draw N random vectors from the asymptotic distribution (8.17) and then calculate the respective N realizations from Eq. (8.18). If N is chosen large enough, we obtain an estimate for $m_{1-\alpha}$ based on the corresponding empirical $(1 - \alpha)$ -quantile.

In order to determine any of the three different confidence intervals or bands, we need the error variance σ^2 , which in practice is generally unknown and thus an estimate is needed. Since the coverage probability is only valid asymptotically, we can, however, replace σ^2 with a consistent estimate $\hat{\sigma}^2$. In the next section, we will present a possible estimate for σ^2 .

Example 8.4 Malnutrition in Tanzania—Confidence Bands for the Age Effect

To clarify the differences between pointwise confidence intervals and simultaneous confidence bands, we consider the age effect in the Tanzania data set. For estimation, we use a cubic P-spline with 20 interior knots and a second-order difference penalty, while fixing the smoothing parameter at $\lambda = 20$ and choosing a level of $\alpha = 0.05$. The goal is to construct a confidence band for the entire function, and thus we base our calculations of the confidence bands on all 60 observed distinct covariate values.

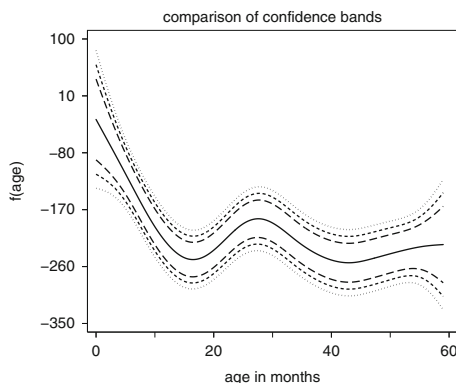


Fig. 8.24 Malnutrition in Tanzania: pointwise confidence intervals (—), simultaneous confidence band based on the joint distribution (- - -), and simultaneous confidence band based on the Bonferroni correction (···) for the age effect

Figure 8.24 shows the resulting estimated function, together with the three confidence areas. The pointwise confidence intervals correspond to the narrowest confidence region, since they only consider uncertainty in estimation at one given point. The other extreme results from the use of the Bonferroni correction, which leads to the widest confidence region. In this case, we distributed the level equally on all 60 observed covariate settings, yielding a level of $\alpha_j = 0.0008333$. Hence, the Bonferroni confidence bands are based on the 0.9995833-quantile of the standard normal distribution, which results in a value of 3.34. In comparison the pointwise confidence intervals are based on the 0.975-quantile of the standard normal distribution given by $z_{0.975} \approx 1.96$. The result is nearly a 1.7 times wider confidence area for the Bonferroni correction.

For the construction of a confidence region based on the joint distribution of the 60 estimated function values, we need to derive the corresponding quantile of the distribution in Eq. (8.18). A simulation with $N = 100,000$ repetitions provides $m_{0.95} = 2.66$ so that (as expected) the result is a confidence band in between the pointwise confidence intervals and the confidence band obtained with the Bonferroni correction.

△

Equivalent Degrees of Freedom (Effective Number of Parameters)

Thus far, all smoothing methods that we have discussed have in common that the smoothness of the estimated function is controlled by (at least) one smoothing parameter. For basis function methods without penalty, smoothness is mostly determined by the number of basis functions. For penalization procedures, an additional parameter was included to explicitly determine the impact of the penalty on the model fit. For nearest neighbor estimates, the number of the nearest neighbors or the order of the symmetric neighborhood determined the smoothness of the estimated function. Clearly, the smoothing parameters of different approaches are not directly comparable despite their common nature. Consequently, it would be desirable to have a general measure to evaluate the approximate dimension of the smoothness for an estimated function.

The derivation of such a measure is easiest by considering analogies to the linear model. The complexity of a linear model is provided by the number of regression

coefficients in the model. The number of parameters can, for example, be recovered by the trace of the prediction matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ since $\text{trace}(\mathbf{H}) = p$. Since the prediction matrix of the linear model corresponds to the smoothing matrix $\mathbf{S}$ of a linear smoother, the *equivalent degrees of freedom* are analogously defined as

$$\text{df}(\mathbf{S}) = \text{trace}(\mathbf{S}). \quad (8.19)$$

The equivalent degrees of freedom are also often interpreted as the *effective number of parameters* of a smoother. In fact, we reproduce the number of parameters in the model for basis function approaches without penalization, while in the case of penalization, the penalty effectively reduces the number of parameters.

Figure 8.25 shows the connection between the original smoothing parameter and the resulting effective degrees of freedom for a selection of smoothing approaches and the simulated data example. For the running mean, a larger neighborhood leads to a smoother estimate and, hence, to a smaller effective number of parameters. As a limiting case, we obtain only one parameter, when the neighborhood contains all observations. For polynomial splines, the effective number of parameters corresponds exactly to the number of basis functions, so that a linear relationship results. When considering penalized splines, the effective number of parameters decreases with an increasing smoothing parameter (note that a logarithmic scale for the smoothing parameter is used to facilitate presentation). For $\lambda = 0$, we default back to the unpenalized case, so that the number of parameters corresponds to the number of basis functions used. For a very large smoothing parameter, the effective number of parameters apparently approaches a limiting value, which depends on the order of differences used when constructing the penalty. We saw in Sect. 8.1.2 that a polynomial of degree $k - 1$ results as $\lambda \rightarrow \infty$ when using a k th order difference penalty. Such a polynomial is described by k parameters so that we obtain one or two parameters as limiting cases in Fig. 8.25.

If we want to determine the smoothing parameter corresponding to given effective degrees of freedom, we can in principle simply read them from Fig. 8.25. However, for polynomial splines and running means, the smoothing parameter is discrete and can only adopt a limited number of values. Thus, not every value of the effective degrees of freedom is actually achievable. For P-splines, however, we are able to choose every value for the effective degrees of freedom that lies between the difference order k and the number of basis functions used.

The equivalent degrees of freedom in Eq. (8.19) are of course not the only possibility to quantify the complexity of nonparametric function estimates. At least two further propositions exist, which can also be derived from analogies to the linear model but are less commonly used in practice. They are given by

$$\text{df}_{\text{var}}(\mathbf{S}) = \frac{1}{\sigma^2} \sum_{i=1}^n \text{Var}(\hat{f}(z_i)) = \text{trace}(\mathbf{S}\mathbf{S}')$$

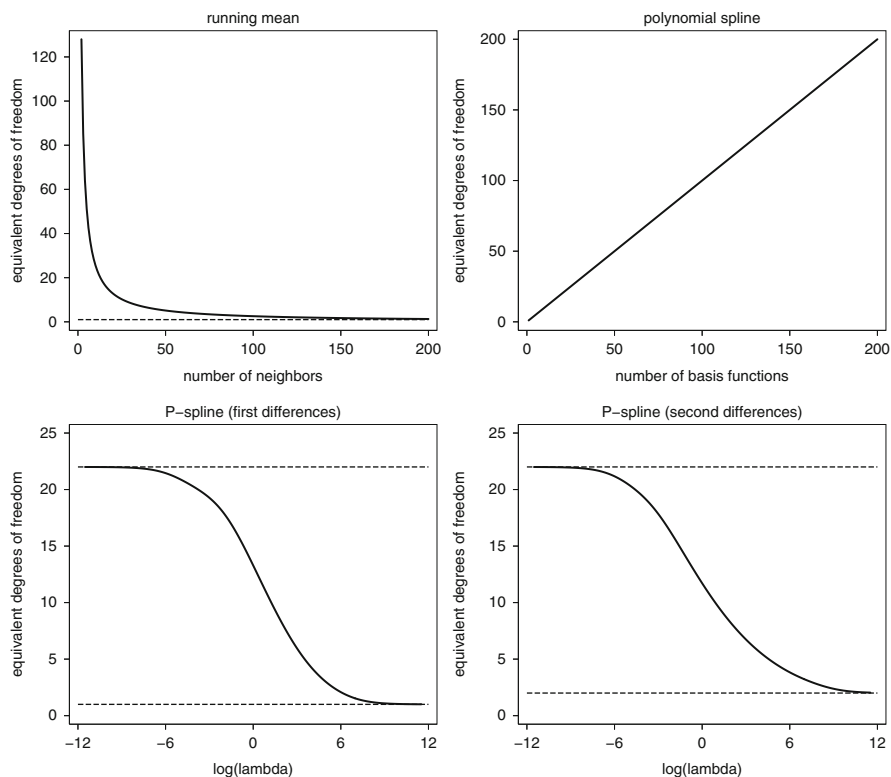


Fig. 8.25 Equivalent degrees of freedom as a function of the smoothing parameter. For P-splines, the log-transformed smoothing parameter is presented. The *dashed lines* indicate limiting values of the equivalent degrees of freedom

and

$$\text{df}_{\text{res}}(\mathbf{S}) = \text{trace}(\mathbf{2S} - \mathbf{S}\mathbf{S}'),$$

and result from considerations regarding the sum of variances of predicted values and the expected value of the mean squared error, respectively; see [Hastie and Tibshirani \(1990\)](#), Sect. 3.5 for a detailed derivation. If the smoothing matrix $\mathbf{S}$ is symmetric and idempotent (i.e., $\mathbf{S} = \mathbf{S}'$ and $\mathbf{S}^2 = \mathbf{S}$), the definitions of df , df_{var} , and df_{res} coincide. This is, for example, the case for simple basis function approaches, but does not apply for procedures involving penalization.

The simplicity of the calculation for the first definition of effective degrees of freedom usually makes it favored in practice. In particular, we can avoid the calculation of the complete $n \times n$ -matrix $\mathbf{S}$, since the trace operator is invariant to cyclical permutations of matrix products (see Theorem A.18 in Appendix A.4). For example, we obtain

$$\text{df}(\mathbf{S}) = \text{trace}(\mathbf{Z}(\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})^{-1}\mathbf{Z}') = \text{trace}(\mathbf{Z}'\mathbf{Z}(\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})^{-1})$$

in case of P-splines so that the trace can be computed from the product of two matrices with dimension given by the number of basis functions.

Estimates for the Error Variance

In the linear model, an unbiased estimate for the error variance σ^2 was given by

$$\hat{\sigma}^2 = \frac{1}{n-p} \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

In nonparametric regression, it seems plausible to replace the number of parameters p with the equivalent degrees of freedom, as outlined in the previous section. In fact, when assuming that at least the two first moments of the model are correctly specified, one can show that the expected value of the residual sum of squares can be written as

$$\mathbb{E} \left(\sum_{i=1}^n (y_i - \hat{f}(z_i))^2 \right) = (n - \text{tr}(2\mathbf{S} - \mathbf{S}\mathbf{S}'))\sigma^2 + \sum_{i=1}^n b_i^2,$$

where b_i defines the bias of the function estimate at the point z_i . If we can neglect the bias, i.e., if the smoother is approximately unbiased, we obtain an unbiased error variance estimate as

$$\hat{\sigma}^2 = \frac{1}{n - \text{df}_{\text{res}}} \sum_{i=1}^n (y_i - \hat{f}(z_i))^2.$$

Even though unbiasedness for $\hat{\sigma}^2$ is only assured when using df_{res} , the simpler definition df is typically used in practice for the estimation of the error variance.

Bias–Variance Trade-Off

In this section, we will more closely examine how varying specific parameters of a smoother affects the mean squared error of the function estimate. For the sake of simplicity, we first discuss the running mean estimate

$$\hat{f}(z_i) = \frac{1}{k} \sum_{j \in N(z_i)} y_j$$

that is based on the k -nearest neighbors. For this estimate, the expected value and the variance can be easily calculated as

$$\mathbb{E}(\hat{f}(z_i)) = \frac{1}{k} \sum_{j \in N(z_i)} f(z_j)$$

and

$$\text{Var}(\hat{f}(z_i)) = \frac{\sigma^2}{k}$$

with $k = |N(z)|$. Since MSE is additively composed of both the squared bias and the variance, we could try to choose the smoothing parameter k in such a way that bias and variance are simultaneously minimized. The formulae above, however, show that this approach is not feasible, since the variation of k has an opposite effect on the bias relative to the variance. If we increase the neighborhood, the variance clearly decreases. On the other hand, a larger bias results as more and more terms are contained in the expected value of $\hat{f}(z_i)$, which are different from $f(z_i)$. A neighborhood of $k = 1$ with $N(z_i) = \{i\}$ results in an unbiased estimate which, however, has the highest possible variance. We found a similar dilemma in Sect. 3.4 within the context of model choice in linear models and identified it as a classical example of the bias–variance trade-off.

In general, smoother function estimates resulting from the choice of a smaller number of effective degrees of freedom usually lead to reduced variability, but increasing bias. On the other hand, a more complex model with a larger number of effective degrees of freedom typically reduces the bias, while at the same time resulting in larger variability. We can also empirically identify this effect when revisiting the plots (e.g., Figs. 8.10 on p. 425, 8.16 on p. 440 or 8.22 on p. 463) that illustrate the impact of the smoothing parameter. In contrast to the running mean estimate, the explicit derivation of the expected value and the variance of most scatter plot smoothers are actually more complicated. Hence, we will only examine local polynomial models as a further example for the bias–variance trade-off, where at least asymptotic results are available; see [Fahrmeir and Tutz \(2001\)](#), Sect. 5.1.3, or [Fan and Gijbels \(1996\)](#).

The smoothness in local polynomial regression is mainly controlled by the bandwidth λ of the kernel function. As a consequence, we examine the influence of λ on bias and variance of the resulting estimate. We limit ourselves to asymptotic results, i.e., we will examine the behavior of the estimate for a large sample size, $n \rightarrow \infty$. Note that in order to obtain meaningful asymptotic expressions, we have to impose conditions not only on the sample size itself but also on the smoothing parameter. The underlying reason is that an increasing sample size also yields more precise estimates and therefore requires a smaller bandwidth. More specifically, we examine the limiting case $\lambda \rightarrow 0$ and $n\lambda \rightarrow \infty$, i.e. even though we let the bandwidth approach zero, the sample size increases at a rate such that $n\lambda$ still diverges to infinity. Under these conditions, we obtain

$$E(\hat{f}(z) - f(z)) \approx \frac{\lambda^{l+1} f^{(l+1)}(z)}{(l+1)!} \mu_{l+1}(K)$$

as an approximation for the asymptotic bias with odd polynomial degree l and

$$E(\hat{f}(z) - f(z)) \approx \left(\frac{\lambda^{l+2} f^{(l+1)}(z) d'_z(z)}{(l+1)! d_z(z)} + \frac{\lambda^{l+2} f^{(l+2)}(z)}{(l+2)!} \right) \mu_{l+2}(K)$$



in case of an even polynomial degree. In the latter case, $d_z(z)$ and $d'_z(z)$ define the density of the distribution of z and its first derivative, respectively, and $\mu_q(K) = \int u^q K(u) du$ is the q -th moment for the chosen kernel. For the variance, we obtain the asymptotic approximation

$$\text{Var}(\hat{f}(z)) \approx \frac{\sigma^2}{n\lambda d_z(z)} \int K^2(u) du.$$

These two results, provide us with very interesting and useful information, which again illustrate the mentioned trade-off between bias and variance associated with nonparametric regression estimates. Regarding the bias, we can identify the following main points:

- The bias of $\hat{f}(z)$ decreases with decreasing bandwidth λ .
- For even l , the bias depends on the distribution of z (expressed through the density $d_z(z)$). The bias decreases when $d_z(z)$ is large, i.e., when a large number of observations are expected in the neighborhood of z . On the contrary, the bias increases in regions with a lower expected number of observations (i.e., a lower density $d_z(z)$).
- For odd l , the bias is independent of the distribution of the covariates.
- For $l = 0$, the bias depends on the first and second derivative of the function $f(z)$. A larger bias results in regions with a steep gradient $f'(z)$. Since the second derivative is a measure for the curvature of the function, we additionally obtain more bias in regions having large $f''(z)$. More precisely, we find underestimation (a negative bias) in local maxima and overestimation (a positive bias) in local minima.
- For even l , we get an additional term in the bias formula. This generally yields a preference for an odd polynomial degree.

For the variance, we have the following main results:

- An increase of the smoothing parameter leads to reduced variability in estimation.
- For both even or odd l , the variance depends on the distribution of the covariates. Analogous to the bias, we observe a decrease in the variance for large $d_z(z)$, and an increase for small $d_z(z)$. However, the precise form of the effect differs for bias and variance.



In summary, it is again not possible to decrease both variance and bias simultaneously. The use of the mean squared error is an appropriate compromise, which especially could be used to choose an optimal smoothing parameter. However, since the MSE generally depends on the unknown true function $f(z)$, this approach is typically not feasible. Nevertheless it provides the basis for strategies that rely on an approximation of the mean squared error.

8.1.9 Choosing the Smoothing Parameter

Our considerations of nonparametric regression models thus far yield one important remaining question: How do we “optimally” choose the smoothing parameter to

properly describe the data? Based on our findings related to the bias–variance trade-off presented in the previous section, we can equivalently formulate this question as the problem of how to obtain an appropriate compromise between bias and variance of the function estimate. The mean squared error offers an immediate option, and we will first focus on this criterion in choosing smoothing parameters. Moreover, we will briefly discuss the use of AIC to determine optimal smoothing parameters. The second and third part of this section focus on penalization approaches and derive general possibilities to estimate the smoothing parameter in this context. Specifically, on the one hand, we will rely on the close connection between penalization approaches and mixed models, and on the other hand, we will derive Bayesian Markov chain Monte Carlo simulation procedures.

Choosing the Smoothing Parameter Based on Optimality Criteria

Our first option for choosing the smoothing parameter results from our considerations on the bias–variance trade-off. We discovered that both the bias and variance associated with a fitted smooth function depend on the smoothing parameter, and that both cannot be simultaneously decreased. Thus, an appropriate compromise has to be found, for example, based on the mean squared error

$$\text{MSE}(\hat{f}(z)) = \text{E} \left((\hat{f}(z) - f(z))^2 \right) = \left(\text{E}(\hat{f}(z) - f(z)) \right)^2 + \text{Var}(\hat{f}(z)),$$

which consists of both the squared bias and variance. From the pointwise MSE, we obtain a measure for the quality of the entire estimated function by averaging over the observed covariate values, i.e.,

$$\frac{1}{n} \text{E} \left(\sum_{i=1}^n (\hat{f}(z_i) - f(z_i))^2 \right).$$

A naive approximation is the residual sum of squares

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(z_i))^2.$$

However, this choice is not useful to determine an optimal smoothing parameter, as it can be minimized by $\hat{f}(z_i) = y_i$. The optimization of the residual sum of squares would therefore lead to nothing more than a trivially interpolating estimate; see also Sect. 3.4.

Consequently, the squared prediction error for new observations y^* is preferred when choosing the smoothing parameter. Typically, such new data are not available, and we therefore resort to a squared error approximation that results from cross validation. The procedure is as follows: We first fix the smoothing parameter, remove one of the observations from the data, and estimate the smooth function using the remaining $n - 1$ observations. Using this estimated function, we next

predict the function value $f(z_i)$ at the location of the eliminated observation. Let $\hat{f}^{(-i)}(z)$ denote the estimated function obtained when removing observation (y_i, z_i) . Cycling over all the observation, we then obtain the cross validation criterion

$$CV = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}^{(-i)}(z_i))^2.$$

Note that the cross validation criterion depends on the chosen smoothing parameter and therefore choosing the smoothing parameter to minimize the cross validation criterion provides a way to determine an optimal smoothing parameter minimizing the prediction error. Furthermore, the use of the cross validation criterion is also theoretically justified due to the fact that

$$E(CV) \approx \frac{1}{n} \sum_{i=1}^n (y_{n+i} - \hat{f}(z_i))^2,$$

where y_{n+i} define new observations at the points z_i . Hence, the mean squared prediction error results as the expectation of the CV criterion.

At first glance, it appears that for computation of the cross validation criterion, we have to perform n separate nonparametric regression model fits (for each value of the smoothing parameter). However, we can actually obtain the CV score by only performing one fit using all the data (similar as in the linear model in Sect. 3.4). Utilizing the diagonal elements s_{ii} of the smoother matrix $\mathbf{S}$, CV can in fact be calculated via

$$CV = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{f}(z_i)}{1 - s_{ii}} \right)^2.$$

Although this statement does not hold in full generality for all types of smoothers, it is still routinely used to determine an approximation to CV. The calculation of the smoothing matrix and its diagonal elements can, however, still be numerically complex (especially for large data sets). For this reason one often replaces the diagonal elements with their average, yielding the generalized cross validation criterion (GCV):

$$GCV = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{f}(z_i)}{1 - \text{tr}(\mathbf{S})/n} \right)^2.$$

The sum of the diagonal elements exactly corresponds to our first definition of the equivalent degrees of freedom, namely the trace of the smoothing matrix. As we saw for P-splines, it is straightforward to calculate this trace, since products of matrices can be cyclically permuted without affecting the trace (see p. 475). In addition to a simpler calculation, the GCV criterion, in comparison to the CV criterion, also has the theoretical advantage to be invariant under orthogonal transformations of the data; compare Wood (2006), Sect. 4.5.2.

As an alternative to the generalized cross validation criterion, we can also use other criteria developed in the context of model choice for the determination of smoothing parameters. A typical example is Akaike's information criterion AIC,

which was discussed in Sect. 3.4. The AIC includes the number of parameters as a form of penalization to correct the likelihood for the model complexity; see also Sect. B.4.5. In nonparametric regression, the effective number of parameters replaces the actual number of parameters leading to the criterion

$$\text{AIC} = n \log(\hat{\sigma}^2) + 2(\text{df} + 1),$$

which needs to be minimized with respect to the smoothing parameter. As in the linear model, $\hat{\sigma}^2 = \sum (y_i - \hat{f}(z_i))^2 / n$ refers to the ML estimate for the error variance.

Figure 8.26 shows GCV and AIC as functions of the smoothing parameter for a cubic P-spline with 20 inner knots and a second-order difference penalty, using the Tanzania data together with the resulting optimal estimated functions. To improve readability of the GCV and AIC plots, the minimal value of AIC or GCV was set to zero so that the curves reflect differences relative to the optimal model. It appears that GCV and AIC show a very similar relationship with the smoothing parameter λ . In fact, both criteria are actually asymptotically equivalent although in specific applications clear differences may be observed. For the Tanzania data, both criteria provide similar, although not identical, optimal smoothing parameters and therefore also yield very similar function estimates.

How can we concretely determine the optimal smoothing parameter for a given criterion? So far, we only examined the simplest case, i.e., univariate nonparametric regression. Thus it is possible to directly optimize the criterion of interest, for example, using a grid search over λ (or using more efficient methods for numerical minimization). To do so, we generate a grid of candidate smoothing parameters, calculate the criterion for each of these candidates, and then choose the value corresponding to the minimal criterion. It can also be useful to iteratively refine the grid based on a first rough discretization to assure that the optimal value is obtained precisely.

In more complex problems, as the ones that we will examine in Chap. 9, direct optimization is no longer possible or at least very time consuming, since several smoothing parameters have to be determined simultaneously. However, even in these settings, algorithms exist for an efficient minimization of the GCV or the AIC. Such approaches are based on the Newton method, i.e., rely on the derivative of the criterion with respect to the smoothing parameters. Since these methods are mathematically and algorithmically complex, we do not discuss them here in detail but rather refer to the corresponding literature, especially Wood (2000, 2004, 2006, 2008). The R package `mgcv` provides implementations.

Mixed Model Representation of Penalization Approaches

The methods based on model choice criteria that we discussed in the previous section can be generally used for the choice of smoothing parameters for any scatter plot smoother. In the following two paragraphs, we rather focus on penalization approaches, with penalties

$$\lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}.$$

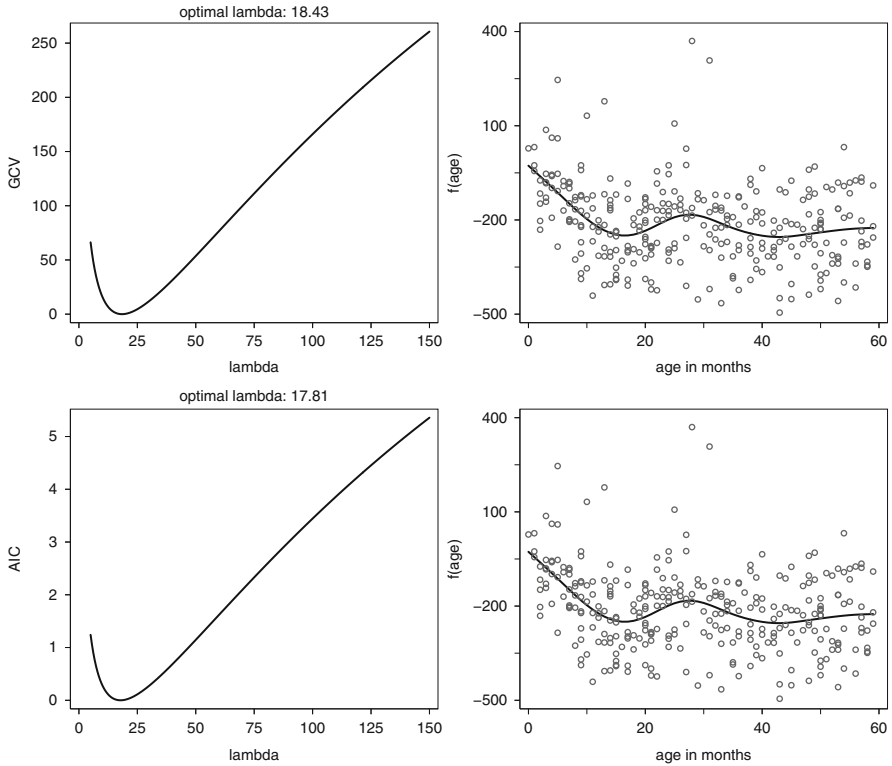


Fig. 8.26 Malnutrition in Tanzania: GCV and AIC (*left panel*) and cubic P-splines fits resulting with the corresponding optimal smoothing parameters (*right panel*)

As seen, the penalty is a quadratic form in the parameters and depends on a (symmetric, positive semidefinite) penalty matrix $\mathbf{K}$. This limitation is, however, not very restrictive, because it includes all methods examined in Sects. 8.1.2–8.1.6, especially penalized splines, smoothing splines, random walks, and kriging.

We start our considerations with P-splines in truncated power series representation. In this case, the PLS criterion is given by

$$\text{PLS}(\lambda) = (\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma})'(\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma}) + \lambda \sum_{j=l+2}^d \gamma_j^2$$

(see p. 432). To establish a connection to the mixed models of Chap. 7, we first partition the parameter vector $\boldsymbol{\gamma}$ into $\boldsymbol{\beta} = (\gamma_1, \dots, \gamma_{l+1})'$ and $\tilde{\boldsymbol{\gamma}} = (\gamma_{l+2}, \dots, \gamma_d)'$. The vector $\boldsymbol{\beta}$, thus, consists of the non-penalized parameters corresponding to the global polynomial, whereas $\tilde{\boldsymbol{\gamma}}$ contains the parameters of the truncated power functions that are included as squared values in the penalty. If we split the design matrix accordingly into $\mathbf{X}$ and $\mathbf{U}$ so that $\mathbf{Z} = [\mathbf{X} \ \mathbf{U}]$, the PLS criterion can be

reexpressed as

$$(y - X\beta - U\tilde{\gamma})'(y - X\beta - U\tilde{\gamma}) + \lambda\tilde{\gamma}'\tilde{\gamma}. \quad (8.20)$$

In Sect. 7.3.1, we found that in the mixed model formulation,

$$y = X\beta + U\tilde{\gamma} + \varepsilon, \quad \varepsilon \sim N(\mathbf{0}, \sigma^2 I), \quad \tilde{\gamma} \sim N(\mathbf{0}, \tau^2 I),$$

estimates for the regression coefficients can also be obtained by minimizing the PLS criterion

$$(y - X\beta - U\tilde{\gamma})'(y - X\beta - U\tilde{\gamma}) + \frac{\sigma^2}{\tau^2}\tilde{\gamma}'\tilde{\gamma}.$$

If we now compare the penalized least squares criterion arising from mixed models with Eq. (8.20), we find that β can be interpreted as a vector of fixed effects while $\tilde{\gamma}$ corresponds to a vector of random effects. The variance of the random effects is then defined by $\tau^2 = \sigma^2/\lambda$ and, as a consequence, we can interpret the smoothing parameter λ as the ratio of error variance and random effects variance similar to our considerations regarding Bayesian P-splines (p. 441ff).

Embedding P-splines in the scope of mixed models has the advantage that we can use the estimation procedures discussed in Chap. 7 to determine estimates of σ^2 and τ^2 and therefore also the smoothing parameter λ . Specifically, we can use either maximum likelihood or restricted maximum likelihood estimation as introduced in Sect. 7.3.2. Based on the estimates $\hat{\sigma}^2$ and $\hat{\tau}^2$, we obtain the optimal smoothing parameter $\hat{\lambda} = \hat{\sigma}^2/\hat{\tau}^2$. Although smoothing parameter selection via restricted maximum likelihood is usually done in the context of mixed models a direct approach without resorting to the connection with mixed models is possible; see Wood (2011).

In summary, a close connection exists between P-splines in TP-representation and mixed models yielding an alternative perspective on P-splines. We next examine this connection more generally for arbitrary penalization approaches. Starting with the penalized sum of squared residuals

$$\text{PLS}(\lambda) = (y - Z\gamma)'(y - Z\gamma) + \lambda\gamma'K\gamma,$$

it may be tempting to define the corresponding mixed model as

$$y = Z\gamma + \varepsilon, \quad \varepsilon \sim N(\mathbf{0}, \sigma^2 I), \quad \gamma \sim N(\mathbf{0}, \tau^2 K^{-1}) \quad (8.21)$$

with $\lambda = \sigma^2/\tau^2$. However, in this direct approach, we encounter the problem that the inverse K^{-1} generally does not exist, since the penalty matrix K often does not have full rank (e.g., in case of P-splines based on B-splines). For kriging, the formulation in Eq. (8.21) is immediately applicable since the penalty matrix has full rank.

To relate arbitrary penalization approaches to mixed models, we therefore have to proceed differently than we did for P-splines with a TP basis. The basic problem is that the density of the random effects resulting from representation (8.21),



$$p(\boldsymbol{\gamma}) \propto \exp\left(-\frac{1}{2\tau^2}\boldsymbol{\gamma}'\mathbf{K}\boldsymbol{\gamma}\right)$$

cannot be normalized. Such a density is also called *improper*. More precisely, we obtain a partially improper density, since $\text{rk}(\mathbf{K}) > 0$, yet $\text{rk}(\mathbf{K}) \neq \dim(\boldsymbol{\gamma})$. In a mixed model, this partial impropriety is avoided by separating the density into a noninformative distribution for fixed effects and a nonsingular normal distribution with a normalized density for the random effects. It is our goal to derive such a separation for any penalization approach (see also Sect. B.3.2 in Appendix B).

To do so, we need to split $\boldsymbol{\gamma}$ into two subvectors $\boldsymbol{\beta}$ and $\tilde{\boldsymbol{\gamma}}$ with dimensionalities depending on the rank deficiency of $\mathbf{K}$. Let $r = \text{rk}(\mathbf{K})$ denote the rank of the penalty matrix and $d = \dim(\boldsymbol{\gamma})$ the dimension of the coefficient vectors. We then consider reparameterizations

$$\boldsymbol{\gamma} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \tilde{\mathbf{U}}\tilde{\boldsymbol{\gamma}}$$

with $d \times (d-r)$ - and $d \times r$ -dimensional design matrices $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{U}}$ and corresponding $(d-r)$ and r -dimensional parameter vectors $\boldsymbol{\beta}$ and $\tilde{\boldsymbol{\gamma}}$. Our aim is to choose the design matrices $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{U}}$ so that we can rewrite the penalty $\lambda\boldsymbol{\gamma}'\mathbf{K}\boldsymbol{\gamma}$ as $\lambda\tilde{\boldsymbol{\gamma}}'\tilde{\boldsymbol{\gamma}}$. As such $\boldsymbol{\beta}$ can be considered a vector of fixed effects while $\tilde{\boldsymbol{\gamma}}$ corresponds to a vector of independent and identically distributed (i.i.d.) random effects. More precisely, this requires the following properties of the design matrices:

1. The composite matrix $[\tilde{\mathbf{X}}, \tilde{\mathbf{U}}]$ should have full rank to yield a one-to-one transformation.
2. $\tilde{\mathbf{X}}'\mathbf{K} = \mathbf{0}$, so that $\mathbf{K}$ does not penalize $\boldsymbol{\beta}$.
3. $\tilde{\mathbf{U}}'\mathbf{K}\tilde{\mathbf{U}} = \mathbf{I}$, so that $\tilde{\boldsymbol{\gamma}}$ consists of i.i.d. random effects.

For the penalty, we then have

$$\begin{aligned}\boldsymbol{\gamma}'\mathbf{K}\boldsymbol{\gamma} &= (\tilde{\mathbf{X}}\boldsymbol{\beta} + \tilde{\mathbf{U}}\tilde{\boldsymbol{\gamma}})'\mathbf{K}(\tilde{\mathbf{X}}\boldsymbol{\beta} + \tilde{\mathbf{U}}\tilde{\boldsymbol{\gamma}}) \\ &= \boldsymbol{\beta}'\underbrace{\tilde{\mathbf{X}}'\mathbf{K}}_{=\mathbf{0}}\tilde{\mathbf{X}}\boldsymbol{\beta} + 2\boldsymbol{\beta}'\underbrace{\tilde{\mathbf{X}}'\mathbf{K}}_{=\mathbf{0}}\tilde{\mathbf{U}}\tilde{\boldsymbol{\gamma}} + \tilde{\boldsymbol{\gamma}}'\underbrace{\tilde{\mathbf{U}}'\mathbf{K}\tilde{\mathbf{U}}}_{=\mathbf{I}}\tilde{\boldsymbol{\gamma}} \\ &= \tilde{\boldsymbol{\gamma}}'\tilde{\boldsymbol{\gamma}}.\end{aligned}$$

The model can therefore be rewritten as a mixed model, yielding

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon} = \mathbf{Z}(\tilde{\mathbf{X}}\boldsymbol{\beta} + \tilde{\mathbf{U}}\tilde{\boldsymbol{\gamma}}) + \boldsymbol{\varepsilon} = \mathbf{X}\boldsymbol{\beta} + \mathbf{U}\tilde{\boldsymbol{\gamma}} + \boldsymbol{\varepsilon}$$

with design matrices $\mathbf{X} = \mathbf{Z}\tilde{\mathbf{X}}$ and $\mathbf{U} = \mathbf{Z}\tilde{\mathbf{U}}$, as well as fixed effects $\boldsymbol{\beta}$ and random effects $\tilde{\boldsymbol{\gamma}} \sim \mathbf{N}(\mathbf{0}, \tau^2\mathbf{I}_r)$. As for P-splines in a TP-representation, we are now able to use mixed models methodology to determine the variance parameters, and thus the smoothing parameter.

The remaining question is the construction of design matrices $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{U}}$ that actually fulfill conditions (1) and (2) listed above. For the design matrix of the

fixed effects, we can easily obtain the desired orthogonality to the penalty matrix by using a basis of the null space of $\mathbf{K}$ for the columns of $\tilde{\mathbf{X}}$; see Definition A.16 in Appendix A.2). For P-splines with a B-spline basis, this null space can also be easily characterized, since polynomials of degree $k - 1$ remain unpenalized by the difference penalty. Thus, we can define $\tilde{\mathbf{X}}$ as

$$\tilde{\mathbf{X}} = \begin{pmatrix} 1 & \kappa_1 & \dots & \kappa_1^{k-1} \\ \vdots & \vdots & & \vdots \\ 1 & \kappa_d & \dots & \kappa_d^{k-1} \end{pmatrix},$$

where $\kappa_1, \dots, \kappa_d$ are the knots of the spline basis.

The design matrix of the random effects can be derived from the spectral decomposition of the penalty matrix (see Theorem A.25 in Appendix A). Using the decomposition $\mathbf{K} = \mathbf{\Gamma} \mathbf{\Omega}_+ \mathbf{\Gamma}'$, where $\mathbf{\Omega}_+$ is the matrix of the positive eigenvalues and $\mathbf{\Gamma}$ is the corresponding orthonormal matrix of eigenvectors, we can define an appropriate $\tilde{\mathbf{U}}$ by $\tilde{\mathbf{U}} = \mathbf{L}(\mathbf{L}'\mathbf{L})^{-1}$ with $\mathbf{L} = \mathbf{\Gamma} \mathbf{\Omega}_+^{1/2}$. It follows then that

$$\tilde{\mathbf{U}}' \mathbf{K} \tilde{\mathbf{U}} = (\mathbf{L}'\mathbf{L})^{-1} \mathbf{L}' \mathbf{L} \mathbf{L}' \mathbf{L} (\mathbf{L}'\mathbf{L})^{-1} = \mathbf{I}.$$

In some specific situations, we can avoid the use of the spectral decomposition. For P-splines based on a B-spline basis, for example, we can also choose $\mathbf{L} = \mathbf{D}'$ with difference matrix $\mathbf{D}$. This also indicates that the mixed model decomposition of $\boldsymbol{\gamma}$ is not unique.

In summary, we have found a possibility to transfer basically any penalization approach into a mixed model and, in doing so, to obtain an optimal smoothing parameter using ML or REML estimation. In the following Sect. 8.2, we will further apply this knowledge to more penalization methods in the context of bivariate smoothing. A further advantage of the representation as a mixed model that is often claimed is that standard mixed model software can be used. We must, however, be aware of the fact that mixed models derived from penalization approaches differ significantly in structure from most of the mixed models discussed in Chap. 7, even though they can formally be written in the same way. The mixed models arising in the context of smoothing do not have any grouping structure, which is one reason why numerical problems may occur when using standard software. Thus, generally speaking, estimation using specialized software that is based on the mixed model representation should be preferred. This is especially the case with the more complex models that will be discussed in Chap. 9.

Finally, we will now consider a methodological problem that arises when estimating the smoothing parameter using mixed models. From a frequentist perspective, the parameters $\boldsymbol{\gamma}$ to be estimated in a penalization approach are fixed, unknown coefficients. In the reformulation as a mixed model, one part of the vector $\boldsymbol{\gamma}$ will be converted into random effects and does, consequently, no longer formally represent a fixed parameter, but rather a random variable. According to the frequentist interpretation, however, $\tilde{\boldsymbol{\gamma}}$ should not be interpreted as random. Strictly speaking, any representation as a mixed model is then to be understood

as an algorithmic trick and not as an actual reformulation of the model. From a Bayesian perspective, this problem does not occur, since all parameters are viewed as random anyway and either representation defines an equivalent formulation of the same prior assumption. According to the distinction made in Sect. 7.4, the estimates resulting from the Bayesian interpretation of mixed model-based smoothing have to be interpreted as empirical Bayes estimates since $\boldsymbol{y}$ is viewed as random while the variance parameters are estimated from a likelihood and therefore in a frequentist setting.

The difference between the Bayesian and the frequentist understanding of penalization methods is also evident in the covariance matrix of the estimates; see also the different covariance matrices that were obtained in Sect. 7.3.3 (p. 378) for general mixed models. Inserting the specific quantities

$$\boldsymbol{R} = \sigma^2 \boldsymbol{I}, \quad \boldsymbol{G} = \tau^2 \boldsymbol{I}, \quad \boldsymbol{C} = (\boldsymbol{X}, \boldsymbol{U}), \quad \boldsymbol{B} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & 1/\tau^2 \boldsymbol{I} \end{pmatrix},$$

in Eq. (7.30) of Sect. 7.3.3, we obtain the Bayesian covariance matrix

$$\text{Cov} \begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{y}} \end{pmatrix} = \sigma^2 \boldsymbol{H}^{-1},$$

with

$$\boldsymbol{H} = \begin{pmatrix} \boldsymbol{X}'\boldsymbol{X} & \boldsymbol{X}'\boldsymbol{U} \\ \boldsymbol{U}'\boldsymbol{X} & \boldsymbol{U}'\boldsymbol{U} + \lambda \boldsymbol{I} \end{pmatrix}.$$

Note that $\boldsymbol{H}$ is required for the estimation of the regression coefficients anyway. The frequentist version (7.32) of the covariance is given by

$$\text{Cov} \begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{y}} \end{pmatrix} = \boldsymbol{H}^{-1} \begin{pmatrix} \boldsymbol{X}' \\ \boldsymbol{U}' \end{pmatrix} \text{Cov}(\boldsymbol{y}) (\boldsymbol{X} \ \boldsymbol{U}) \boldsymbol{H}^{-1} = \sigma^2 \boldsymbol{H}^{-1} \boldsymbol{H}_1 \boldsymbol{H}^{-1},$$

with $\text{Cov}(\boldsymbol{y}) = \sigma^2 \boldsymbol{I}$ and the cross product matrix

$$\boldsymbol{H}_1 = \begin{pmatrix} \boldsymbol{X}'\boldsymbol{X} & \boldsymbol{X}'\boldsymbol{U} \\ \boldsymbol{U}'\boldsymbol{X} & \boldsymbol{U}'\boldsymbol{U} \end{pmatrix}.$$

In the frequentist approach, we obtain the “sandwich” form for the covariance estimate that was already discussed in Sect. 7.3.3. In contrast, a simpler covariance matrix results from the Bayesian approach, which leads to somewhat wider confidence intervals.



Bayesian Smoothing Parameter Choice Based on MCMC

After defining empirical Bayes estimates for the smoothing parameter based on mixed models, we now discuss a fully Bayesian alternative relying on MCMC

simulations. We again restrict ourselves to penalization approaches with quadratic penalties. For Bayesian P-splines, we already found that such a penalty can be derived equivalently by specifying an appropriate smoothness prior. More generally, the following equivalence holds: Frequentist regression models of the form $\mathbf{y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$ with quadratic penalty

$$\frac{\sigma^2}{\tau^2} \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}$$

correspond to Bayesian regression models of the same structure with multivariate normal prior distribution

$$p(\boldsymbol{\gamma} \mid \tau^2) \propto \left(\frac{1}{\tau^2} \right)^{\text{rk}(\mathbf{K})/2} \exp \left(-\frac{1}{2\tau^2} \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma} \right)$$

for the regression coefficients. Note that, in general, this density cannot be normalized, since $\mathbf{K}$ is often not of full rank and therefore $p(\boldsymbol{\gamma} \mid \tau^2)$ is partially improper. Taking the logarithm of the density shows that it is equivalent (up to a change in the sign and an additive constant) to the penalty in the PLS criterion. Therefore, the posterior mode $\hat{\boldsymbol{\gamma}}$ is equivalent to the PLS estimate for given smoothing parameter (as was the case with Bayesian P-splines). In analogy to the penalization of certain function types, we can interpret the prior distribution as a smoothness prior, which enforces prior beliefs regarding the smoothness of the function to be estimated.

Rather than examining the posterior mode estimate, we will now use MCMC simulation techniques to determine the posterior mean. Therefore we require a fully Bayesian formulation of the nonparametric regression problem where adequate prior distributions are provided for all unknown parameters. This includes the variance parameters, which were estimated with ML or REML in the empirical Bayes approach. Since the inverse gamma distribution $\text{IG}(a, b)$ is the conjugate prior distribution in the normal setting, the choice of IG priors for σ^2 and τ^2 yields feasible models; see also Sect. 4.4 on Bayesian linear models. More specifically, we assume

$$\sigma^2 \sim \text{IG}(a, b) \quad \text{and} \quad \tau^2 \sim \text{IG}(a_1, b_1).$$

Assuming (conditional) independence a priori, the joint posterior distribution is given by

$$\begin{aligned} p(\boldsymbol{\gamma}, \sigma^2, \tau^2 \mid \mathbf{y}) &\propto p(\mathbf{y} \mid \boldsymbol{\gamma}, \sigma^2) p(\boldsymbol{\gamma} \mid \tau^2) p(\sigma^2) p(\tau^2) \\ &\propto (\sigma^2)^{-\frac{n}{2}} \exp \left(-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma})' (\mathbf{y} - \mathbf{Z}\boldsymbol{\gamma}) \right) \\ &\quad \cdot (\tau^2)^{-\frac{\text{rk}(\mathbf{K})}{2}} \exp \left(-\frac{1}{2\tau^2} \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma} \right) \\ &\quad \cdot \frac{1}{(\sigma^2)^{a+1}} \exp \left(-\frac{b}{\sigma^2} \right) \frac{1}{(\tau^2)^{a_1+1}} \exp \left(-\frac{b_1}{\tau^2} \right). \end{aligned}$$

Due to the conjugate IG priors, we obtain known distributions for all full conditional distributions so that a Gibbs sampler can be used for estimation. More precisely, we obtain a normal distribution

$$\boldsymbol{\gamma} \mid \mathbf{y}, \sigma^2, \tau^2 \sim N(\boldsymbol{\mu}_{\boldsymbol{\gamma}}, \boldsymbol{\Sigma}_{\boldsymbol{\gamma}}),$$

for $\boldsymbol{\gamma}$ with expected value and covariance matrix

$$\begin{aligned}\boldsymbol{\mu}_{\boldsymbol{\gamma}} &= E(\boldsymbol{\gamma} \mid \cdot) = \left(\frac{1}{\sigma^2} \mathbf{Z}' \mathbf{Z} + \frac{1}{\tau^2} \mathbf{K} \right)^{-1} \frac{1}{\sigma^2} \mathbf{Z}' \mathbf{y} \\ \boldsymbol{\Sigma}_{\boldsymbol{\gamma}} &= \text{Cov}(\boldsymbol{\gamma} \mid \cdot) = \left(\frac{1}{\sigma^2} \mathbf{Z}' \mathbf{Z} + \frac{1}{\tau^2} \mathbf{K} \right)^{-1}.\end{aligned}$$

For the variance parameters, the full conditional distributions are of the inverse gamma type, and we find

$$\begin{aligned}\tau^2 \mid \mathbf{y}, \boldsymbol{\gamma}, \sigma^2 &\sim \text{IG}(a_1 + 0.5\text{rk}(\mathbf{K}), b_1 + 0.5\boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}), \\ \sigma^2 \mid \mathbf{y}, \boldsymbol{\gamma}, \tau^2 &\sim \text{IG}(a + 0.5n, b + 0.5(\mathbf{y} - \mathbf{Z} \boldsymbol{\gamma})'(\mathbf{y} - \mathbf{Z} \boldsymbol{\gamma})).\end{aligned}$$

Based on these full conditionals, we can now derive an MCMC sampler by iteratively drawing random numbers with current parameters inserted. However, $\boldsymbol{\gamma}$ may be of relatively high dimension in nonparametric regression models; a naive implementation can be rather time-consuming as we have to solve high-dimensional systems of equations in every iteration to sample from $\boldsymbol{\gamma} \mid \mathbf{y}, \sigma^2, \tau^2$. To considerably shorten the computing times, efficient algorithms for drawing multivariate normal random numbers can be used. These are based on the fact that both the penalty matrix $\mathbf{K}$ and the cross product of the design matrix $\mathbf{Z}' \mathbf{Z}$ have a very specific and typically sparse structure that can be exploited in the calculations; see [Lang, Umlauf, Wechselberger, Harttgen, and Kneib \(2012\)](#) for details.

Besides point estimates $\hat{f}(z)$ obtained as the median or mean of the samples

$$f^{(t)}(z) = \sum_{j=1}^d \gamma_j^{(t)} B_j(z),$$

the MCMC samples also form the basis for defining appropriate credible regions, i.e., the Bayesian analogues to confidence intervals. Pointwise credible intervals are typically computed by taking empirical quantiles of the samples, e.g., for obtaining a 95 % pointwise credible interval for $f(z)$, the 2.5 % and the 97.5 % quantile of the samples $f^{(t)}(z)$, $t = 1, \dots, T$, define the boundaries of the interval. Simultaneous credible bands as discussed on p. 471 in the frequentist context are more difficult to obtain since there is no common ordering of the samples for a vector of function evaluations $\mathbf{f}_r = (f(z_1), \dots, f(z_r))'$, where $z_1, \dots, z_r$ are a

representative selection of covariate values (e.g., the subset of all distinct covariate values or a grid of equidistant values).

An early suggestion for constructing simultaneous credible bands has been made by [Besag, Green, Higdon, and Mengersen \(1995\)](#) credible bands based on the order statistics of the samples. An alternative approach based on assuming posterior normality has been proposed by [Crainiceanu, Ruppert, Carroll, Adarsh, and Goodner \(2007\)](#). Let $\hat{f}_r$ denote the posterior mean and $\hat{s}_i$, $i = 1, \dots, r$, the posterior standard deviation for each element of f_r computed from the samples. Then after assuming approximate posterior normality and deriving the $(1 - \alpha)$ sample quantile $m_{1-\alpha}$ of

$$\max_{i=1, \dots, r} \left| \frac{f(z_i)^{(t)} - \hat{f}(z_i)}{\hat{s}_i} \right|, \quad t = 1, \dots, T, \quad (8.22)$$

a simultaneous credible band for f_r is given by the hyperrectangular

$$\left[\hat{f}(z_i) - \hat{s}_i m_{1-\alpha}, \hat{f}(z_i) + \hat{s}_i m_{1-\alpha} \right], \quad i = 1, \dots, r.$$

In fact, this credible band is the sampling-based analogue to the simultaneous confidence band derived from Eq. (8.18) on p. 472. It implicitly depends on the assumption of posterior normality and uses the posterior standard deviation as a measure of uncertainty (also assuming symmetry of the posterior distribution) and the posterior mean as a point estimate. As a consequence, the posterior information contained in the samples is not fully utilized but reduced to measures of location and scale.

[Krivobokova, Kneib, and Claeskens \(2010\)](#) propose another alternative definition of simultaneous Bayesian credible bands that form a kind of compromise between the approaches by [Besag et al. \(1995\)](#) and [Crainiceanu et al. \(2007\)](#) and thereby avoids their disadvantages. As a starting point, the pointwise credible intervals derived from the $\alpha/2$ and $1 - \alpha/2$ quantiles of the samples $f(z_1)^{(t)}, \dots, f(z_r)^{(t)}$, $t = 1, \dots, T$, are used to measure pointwise uncertainty in the estimated function instead of using the pointwise standard deviation as in [Crainiceanu et al. \(2007\)](#). Then these pointwise credible intervals are scaled with a constant factor until a fraction of $(1 - \alpha)$ of all sampled curves is contained in the credible band. This can be achieved by monitoring for each of the sampled curves whether it is already contained in the credible band and by modifying the scaling factor until the desired confidence level $1 - \alpha$ is achieved. By construction, this new credible band is still based on pointwise measures of uncertainty (reflected by the pointwise credible intervals) but avoids the assumption of posterior normality and in particular the assumption of symmetry for the posterior distribution.

[Krivobokova et al. \(2010\)](#) also perform a simulation study comparing the three suggested credible bands. It turns out that the approach of [Besag et al. \(1995\)](#) often does not yield the desired coverage properties but understates uncertainty

and therefore yields coverages below the nominal level. In contrast, both the approaches by [Crainiceanu et al. \(2007\)](#) and [Krivobokova et al. \(2010\)](#) work well in case of normally distributed responses. Since this is the ideal situation for the credible intervals based on posterior normality, the similar performance is not too surprising. However, in general, the approach by [Krivobokova et al. \(2010\)](#) requires less assumptions and is therefore probably preferable in practice. It also has the advantage that no additional computations for deriving the quantiles in Eq. (8.22) are required.

Example 8.5 Malnutrition in Tanzania—Credible Bands for the Age Effect

We demonstrate the difference between pointwise credible intervals and simultaneous credible bands by analyzing the age effect in the Tanzania example, providing a Bayesian supplement to the frequentist results achieved in Example 8.4 (p. 472). Figure 8.27 shows the posterior mean together with a pointwise 95 % credible interval and a simultaneous 95 % credible band computed according to the approach of [Krivobokova et al. \(2010\)](#). The scaling factor required to move from the pointwise to the simultaneous credible band is about 1.6, which seems to be a common value in many analyses. As can be seen, the simultaneous band picks up local variability in the pointwise bands but scales them to a larger magnitude to cover the overall variability of the curve samples.

△

8.1.10 Adaptive Smoothing Approaches



In Sect. 8.1.1, we first introduced polynomial splines without penalization and then motivated the need of appropriate regularization implemented via different penalization approaches. Alternatively, algorithms can be constructed for estimating polynomial splines without penalty but with optimally selected number and position of the knots. In this setting, regularization is achieved directly through model choice strategies, rather than indirectly through the addition of a penalty term. Such approaches are then often referred to as *adaptive procedures*, since the construction of the polynomial spline is adaptively controlled by the observed response values. Note that with penalization approaches, the number and position of the knots are chosen based on either the covariate values alone or some partition of the covariate domain.

Various different adaptive procedures exist in both Bayesian and frequentist formulations, and we will not be able to explain all of them in detail here. Instead, we restrict the presentation to a broad overview and develop the main ideas while providing additional references in Sect. 8.4.

Multivariate Adaptive Regression Splines

Many frequentist approaches for adaptive nonparametric regression can be developed analogously to the variable selection ideas that have been discussed in Sect. 3.4. As a representative example, we describe a variant of the popular Multivariate Adaptive Regression Splines (MARS) algorithm. The basis function

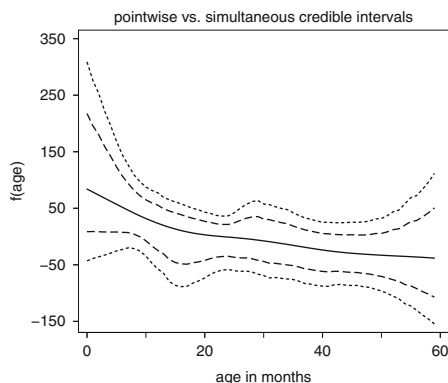


Fig. 8.27 Malnutrition in Tanzania: pointwise credible intervals (—) and simultaneous confidence band (---) for the age effect

representation

$$y_i = f(z_i) + \varepsilon_i = \sum_{j=1}^d \gamma_j B_j(z_i) + \varepsilon_i,$$

serves as the foundation for the MARS algorithm. However, the basis functions are no longer fixed but are rather selected from a large set of possible candidates. The model fit is then evaluated using one of the model choice criteria already discussed, such as GCV or AIC. A possible search strategy for determining the optimal model is presented in the following algorithm:

1. Start with the minimal model, i.e., the smallest basis to be considered.
2. Iteratively extend the model until the full model is reached:
 - For all basis functions B_j that are not yet included in the model, calculate the score statistic for the test on $\gamma_j = 0$ and add the basis function with the highest score statistic.
 - Estimate the new model and save the respective model fit criterion.
3. Based on the full model, iteratively eliminate basis functions until the minimal model is reached:
 - For each basis function B_j still included in the model, calculate the Wald statistic for the test on $\gamma_j = 0$ and remove the basis function with the smallest Wald statistic from the model.
 - Estimate the new model and save the respective model fit criterion.
4. From the resulting sequence of models choose the one which optimizes the model fit criterion as the best model.

The algorithm mainly consists of two steps in which a forward selection and a backward elimination are conducted through the space of possible models. In doing so, different criteria are used for the inclusion of new terms and for the elimination of terms already included in the model so that two different sequences of models result. The test criteria are chosen to ease calculation: While the score statistic for the inclusion of a new basis function can be determined without actually estimating the extended model, the Wald statistic can be calculated without having

to estimate the reduced model. Thus, the execution of the algorithm is feasible even for a large number of basis functions.

For demonstration purposes, we analyze the simulated data example using a MARS implementation available in function `polymars` from the R package `polyspline`. The implementation uses a linear TP basis and starts with the intercept as the minimal model. All observed covariate values are candidate knot positions resulting in a maximum number of 17 basis functions. To evaluate the model fit, we chose the GCV criterion. Figure 8.28 (left) shows the resulting GCV for the sequence of selected models. Large values appear at both the beginning and the end of the model sequence corresponding to simple models with a small number of basis functions that clearly fit the data poorly. A similar statement holds for very complex models that overfit the data and therefore also yield larger values for the GCV. In addition, the GCV sequence shows that the forward and backward search actually result in different model sequences and, as a consequence, the GCV curve is not completely symmetric. In more complex models with a larger number of regressor variables, even stronger deviations from symmetry are to be expected. Figure 8.28 (right) shows the optimal function estimate with eight basis functions obtained from the MARS algorithm. In the example, a satisfactory model fit results even with linear splines. In general, a basis of higher degree will be recommended to ensure smoother fits, but such basis functions are not available in `polymars`.

Regression Trees

As a second example for adaptive smoothing, we examine a procedure that approximates $f(z)$ using piecewise constant functions. In this approach, the domain of the covariate z is partitioned in such a way that the observations within the resulting groups are as homogeneous as possible (with respect to the response variable) while the groups themselves are as heterogeneous as possible. Within the scope of nonparametric regression, this problem can be formulated as follows: The function $f(z)$ is expressed as

$$f(z) = \sum_{j=1}^d \gamma_j I(\kappa_{j-1} < z \leq \kappa_j),$$

i.e., the domain of z will be divided into intervals $R_j = (\kappa_{j-1}, \kappa_j]$ based on the cut points $\kappa_0 < \kappa_1 < \dots < \kappa_d$, and the function f takes the constant value γ_j on such an interval. The cut points $\kappa_0, \dots, \kappa_d$, as well as the function values γ_j , are then to be chosen to minimize the residual sum of squares $\sum_i (y_i - f(z_i))^2$. Without further restrictions, this will lead to a perfect fit, in which every observation is represented by one individual group. As a consequence, additional constraints are usually imposed on $f(z)$, as we will discuss in more detail in the rest of this section.

For a piecewise constant function, the residual sum of squares can be written as

$$\sum_{j=1}^d \sum_{i: z_i \in R_j} (y_i - \gamma_j)^2, \quad (8.23)$$

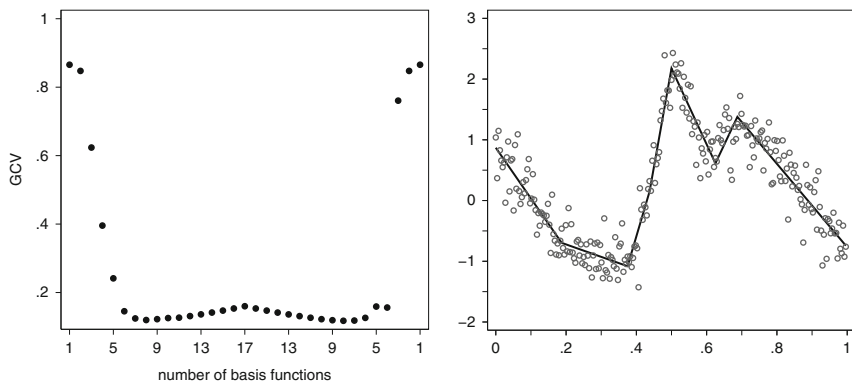


Fig. 8.28 MARS estimates for the simulated data: GCV criterion as a function of basis functions (*left*) and corresponding optimal estimate with eight basis functions (*right*)

and therefore estimates for the function values γ_j are simply given by the local means of the response variable for $z_i \in R_j$, i.e., $\hat{\gamma}_j = \bar{y}(R_j)$. The optimal determination of the cut points is, however, more difficult and not directly attainable. Thus, one often resorts to recursive algorithms, which successively split the covariate domain into intervals. Figure 8.29 illustrates this approach using the malnutrition example (and also motivates the notion *regression tree*). In a first step, a split of the observations is performed based on a single cut point κ so that the variability within the resulting two groups is minimized. In our example, we obtain an optimal cut point of $\kappa = 7.5$, as denoted in the top branch of Fig. 8.29. The same splitting principle is then iteratively applied to the resulting subgroups and finally yields a decomposition of the covariate domain. The resulting structure can then be represented as a binary tree, in which every branch corresponds to a specific cut point. In Fig. 8.29, the cut points are provided on top of each branch. In addition, the lower part of the tree provides information about the number of observations in the final leaves of the tree and the estimated function value $\hat{\gamma}_j = \bar{y}(R_j)$ assigned to the group.

We now define regression trees in a more formal way: The tree is initialized based on a decomposition in the intervals

$$R_1 = \{z : z \leq \kappa\}, \quad R_2 = \{z : z > \kappa\},$$

where κ is chosen to minimize the criterion

$$\sum_{\{i: z_i \in R_1\}} (y_i - \bar{y}(R_1))^2 + \sum_{\{i: z_i \in R_2\}} (y_i - \bar{y}(R_2))^2.$$

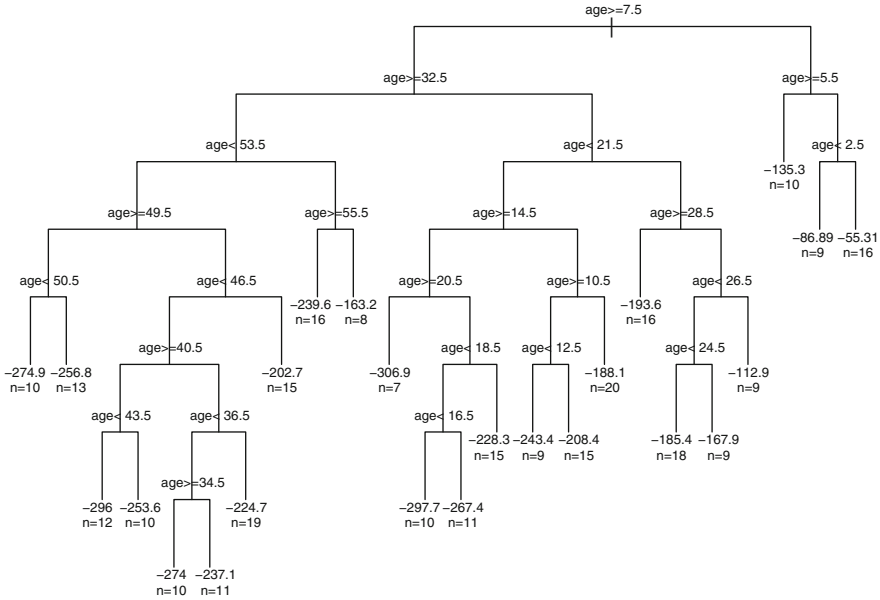


Fig. 8.29 Malnutrition in Tanzania: regression tree. No further splits were performed once a leaf of the tree contained less than 20 observations

Since we only need to determine one single split, the numerical determination of κ is rather simple. We can choose from all unique observed covariate levels z_i so that a direct search is possible. In the next step, the same principle is applied to the two resulting intervals R_1 and R_2 and proceeding iteratively through the intervals finally yields a decomposition of the form (8.23).

The specification of a regression tree is completed by defining a stopping criterion that determines whether or not additional branches are to be included. Two simple strategies to do so are as follows:

1. The given interval is only eligible for a further split if it contains at least a pre-specified number of observations.
2. The given interval is only eligible for a further split if a certain (absolute or relative) reduction of the residual sum of squares can be achieved.

We used the first criterion with a minimal number of 20 observations for the example in Fig. 8.29.

In general, regression trees that result from these simple breakdown criteria have too many branches, so that it is necessary to use an additional strategy to reduce model complexity. To do so, we first define a complex tree with a large number of branches, and then, with the help of a complexity measure, we eliminate superfluous branches (pruning). A complexity-adjusted model fit criterion is given by

$$\text{PLS}_\lambda(T) = \text{LS}(T) + \lambda|T|,$$

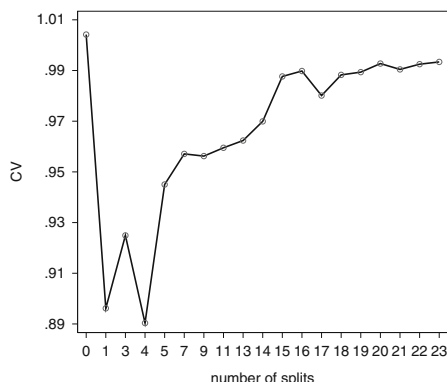


Fig. 8.30 Malnutrition in Tanzania: cross validation criterion as a function of the number of splits

where $LS(T)$ defines the residual sum of squares of tree T , and $|T|$ is the number of branches in the tree. This construction corresponds to our previous considerations regarding penalization methods and, in particular, λ can be interpreted as a smoothing or complexity parameter. In varying λ , we obtain trees of different complexity with increasing complexity resulting from smaller values of λ . The optimal value for λ can then be determined using cross validation criteria.

Figure 8.30 plots the cross validation criterion for the Tanzania data set against the number of branches in the tree. Note that the value of the smoothing parameter λ implicitly defines the number of branches. In fact, not all numbers of branches are actually observed since a change in λ may lead to a model with more than just one additional branch. It appears that four splits are optimal while one split also leads to relatively small value of the cross validation criterion, with a slight preference towards the more complex model. Figure 8.31 provides a visualization of this model with four splits. We again recognize the main features of the data, namely a higher Z-score, i.e., a lower malnutrition risk for younger children, and a pronounced increase in f after the 24th month when the reference population is changing.

One of the main advantages of regression trees is their rather simple interpretation arising from the piecewise constant approximation. In more complex models, regression trees also allow to detect interactions fairly easily. The assumption of a piecewise constant function, however, can be problematic in many applications. Moreover, regression trees are unstable, i.e., a slight change of the data set can lead to an entirely different optimal tree. In addition, such an instability can also strongly influence the choice of an optimal smoothing parameter determined through cross validation when considering small variations in the cross validation folds. Modern extensions of tree-based models rectify this problem by using resampling methods in combination with weighted averaging (random forests). Further comments on such approaches are given in Sect. 8.4.

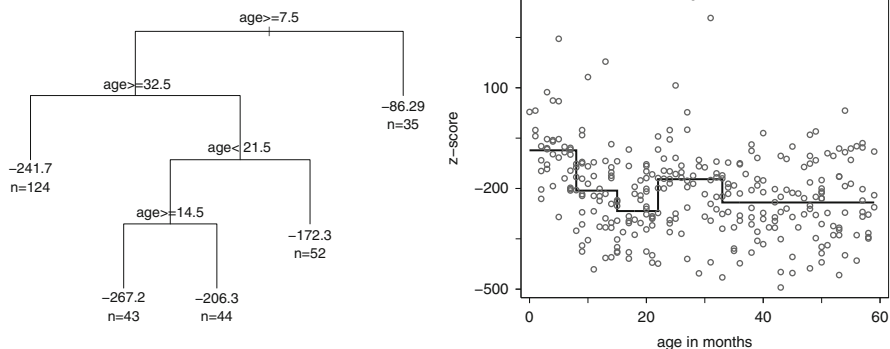


Fig. 8.31 Malnutrition in Tanzania: pruned regression tree and corresponding piecewise constant function estimate

Bayesian Adaptive Smoothing I: Model Averaging

To motivate Bayesian adaptive smoothing approaches, we start with the basis function formulation

$$y_i = f(z_i) + \varepsilon_i = \sum_{j=1}^d \gamma_j B_j(z_i) + \varepsilon_i,$$

using a large number of candidate basis functions, from which an adequate subset shall be chosen. In other words, we are interested in determining which of the coefficients γ_j differ from zero and are therefore associated with basis functions that have non-negligible impact on the response. Exactly the same question has already been discussed in Sect. 4.4.3 where we developed Bayesian linear models with built-in variable selection. In fact, the same models can be used for adaptive knot selection (with minor modifications). In the following, we briefly review some of the main ideas of Sect. 4.4.3 in the context of basis function selection.

In a Bayesian approach, the question of basis function selection can be approached by inserting indicator variables $\delta = (\delta_1, \dots, \delta_d)'$, such that

$$\delta_j = \begin{cases} 1 & \text{if } \gamma_j \neq 0, \\ 0 & \text{if } \gamma_j = 0. \end{cases}$$

When no prior information regarding the plausibility of the models exists, a uniform prior is a straightforward choice. In our example, every possible indicator vector δ would then obtain the prior probability $1/2^d$. Compare, however, the remarks in Sect. 4.4.3 regarding the prior expected model size and the difficulties arising from this (naive) prior specification.

The original model can be rewritten as

$$f(z) = \sum_{j=1}^d \delta_j \gamma_j B_j(z_i).$$

For a given vector of indicators, define $\boldsymbol{\gamma}_\delta$ as the regression coefficient vector, containing only the components that remain after eliminating coefficients associated with zero indicators. Similarly, let $\mathbf{Z}_\delta$ denote the corresponding design matrix resulting from the basis functions with $\delta_j = 1$. In matrix notation, we now obtain the vector of function evaluations $\mathbf{f}_\delta = \mathbf{Z}_\delta \boldsymbol{\gamma}_\delta$ that is based on the model defined by δ .

If we neglect the error variance σ^2 for the moment, the posterior distribution for the vector $\mathbf{f}$ (without conditioning on a specific model) can be written as

$$p(\mathbf{f} | \mathbf{y}) = \sum_{\delta \in \{0,1\}^d} p(\mathbf{f}_\delta | \delta, \mathbf{y}) p(\delta | \mathbf{y}), \quad (8.24)$$

i.e., we obtain a mixture of all possible configurations of the indicator vector δ , where the posterior distributions $p(\mathbf{f}_\delta | \delta, \mathbf{y})$ are weighted by the posterior probabilities $p(\delta | \mathbf{y})$. In particular, the marginal posterior expectation of $\mathbf{f}$ is also a mixture of the conditional posterior expected values:

$$\mathbb{E}(\mathbf{f} | \mathbf{y}) = \sum_{\delta \in \{0,1\}^d} \mathbb{E}(\mathbf{f}_\delta | \delta, \mathbf{y}) p(\delta | \mathbf{y}). \quad (8.25)$$

This is a special example of the concept of *model averaging*; see Appendix B.5.5 for an introduction and Sect. 4.4.3 for details in the context of variable selection in linear models.

In our model choice problem, we obtain normal posterior distributions for $\boldsymbol{\gamma}_\delta$ and the function evaluations $\mathbf{f}_\delta$ when using normal prior distributions. Direct access through Eq. (8.24) is, however, limited by another difficulty. For a given number of basis functions, we have to consider 2^d models so that, in practice, the determination of all posterior probabilities cannot be realized, even for a modest number of basis functions. This is the reason why we generally do not use the explicit formulae for model averaging, but an approximation based on MCMC simulations. To do so, first, the indicators δ are updated to obtain a current model, at each iteration. Based on this current model, we are able to simulate $\mathbf{f}_\delta$. More precisely, we obtain the following algorithm, which also contains the estimation of the error variance:

1. Choose adequate starting values $\delta^{(0)}$, $\boldsymbol{\gamma}^{(0)}$, and $\sigma^{2(0)}$ as well as a maximal number of iterations T and set $t = 1$.
2. For $j = 1, \dots, d$ generate $\delta_j^{(t)}$ from the marginal full conditional density $p(\delta_j | \delta_{-j}^{(t-1)}, \mathbf{y})$, where

$$\boldsymbol{\delta}_{-j}^{(t-1)} = (\delta_1^{(t-1)}, \dots, \delta_{j-1}^{(t-1)}, \delta_{j+1}^{(t-1)}, \dots, \delta_d^{(t-1)})$$

is the current index vector without the j th entry.

3. Simulate $\mathbf{f}_\delta^{(t)}$ according to the full conditional density $p(\mathbf{f}_\delta | \boldsymbol{\delta}^{(t)}, \sigma^{2(t-1)}, \mathbf{y})$.
4. Simulate $\sigma^{2(t)}$ from the full conditional density $p(\sigma^2 | \boldsymbol{\delta}^{(t)}, \mathbf{f}_\delta^{(t)}, \mathbf{y})$.
5. If $t < T$, set $t = t + 1$ and go back to 2.

The use of the marginal full conditional densities for the simulation of δ_j (and therefore a distribution that is independent from $\boldsymbol{\gamma}_\delta$ and σ^2) generally leads to a more favorable behavior of the generated Markov chain. If we use conjugate prior distributions, i.e., a normal prior for the regression parameters and an inverse gamma prior for σ^2 , steps 3 and 4 can be realized with the help of Gibbs sampling. A more detailed discussion of adequate prior distributions and a description of the resulting MCMC algorithm can be found in [Smith and Kohn \(1996\)](#). Note also that the algorithm differs from the MC3 algorithm of Box 4.8 (p. 247). In contrast to the work by [Smith and Kohn \(1996\)](#) an accessible implementation of the MC3 algorithm in form of the R package BMS is available. The implementation is, however, not specialized to nonparametric regression. Hence the use in the context of basis function selection is tedious.

Based on realizations of the MCMC algorithm, the posterior expected value (8.25) can be approximated through

$$\frac{1}{T} \sum_{t=1}^T \mathbf{f}_\delta^{(t)}.$$

Since models with a high posterior probability are visited more often within the run of the MCMC algorithm, the weighting of the mean value corresponds (at least approximately) to the theoretical weighting in Eq. (8.25). The empirical frequencies

$$\frac{1}{T} \sum_{t=1}^T \boldsymbol{\delta}^{(t)}$$

also provide estimates for the posterior probabilities $P(\delta_j = 1 | \mathbf{y})$, i.e., for the posterior inclusion probabilities for the individual basis functions. If we apply a threshold to these probabilities, we also gain the possibility to choose an adequate individual model based on the MCMC output, which does only contain basis function important in terms of their posterior probabilities.

Bayesian Adaptive Smoothing II: Reversible Jump MCMC

A second Bayesian approach yielding adaptive smoothing in nonparametric regression results from including the number and the location of the knots as additional parameters to be estimated. Therefore, we first need to find adequate prior distributions for these new parameters. The number of knots m can, for example,

be assigned to a Poisson distribution with parameter λ , which represents prior information on the model size. For a large parameter λ , more complex models are preferred while a small parameter favors simpler models with a smaller number of knots. In practice, the Poisson distribution is often truncated at an upper limit $m_{\max}$, i.e., the maximal number of basis functions is bounded upwards so that $m \leq m_{\max}$. A uniform distribution on $\{0, \dots, m_{\max}\}$ is an alternative to the Poisson distribution. The prior distribution for the position of knots is often defined on a large number of candidate knots, from which a selection is made during the model definition. All knot positions can be assigned equal prior probability. An alternative prior distribution is obtained from the distribution of the order statistic of an i.i.d. sample from the uniform distribution on the domain of z .

The resulting model formulation, enhanced by the new parameters corresponding to the number and location of the knots, can now again be transformed into an MCMC algorithm. However, the construction of such an algorithm is complicated by the fact that the dimension of the parameter vector to be estimated varies from iteration to iteration. Therefore standard MCMC methods have to be extended to so-called *reversible jump MCMC* (RJMCMC) approaches that include new steps for simulating the number and the position of knots in addition to the steps already known to be necessary when simulating the regression coefficients and the error variance at a given knot configuration. In an RJMCMC algorithm, this simulation is achieved indirectly, by adequately modifying the current values instead of directly proposing new values. There are three possibilities to do so:

1. *Creation of a new knot (birth step)*: Randomly choose a knot that so far has not been included in the model and add this knot to the knot set.
2. *Elimination of an existing knot (death step)*: Randomly choose a knot currently included in the model and eliminate it from the knot set.
3. *Relocation of a present knot*: Randomly choose a knot currently in the model and relocate it within the range defined by the two adjacent knots.

While the first two approaches modify the number of knots, the third approach allows for variation in the knot positions without actually changing their number. Within the RJMCMC algorithm, we have to randomly decide, at each iteration, whether we simulate new coefficients from the current model specification or whether to vary the knot configuration using one of the three available options. When modifying the knot configuration, the probability of acceptance must also be adjusted. We will not discuss this in further detail, but refer to the appropriate literature ([Green, 1995](#)) for the basic RJMCMC algorithm, as well as [Biller \(2000\)](#) and [Denison, Mallick, and Smith \(1998\)](#) for applications within the scope of Bayesian adaptive smoothing.

Similar to Bayesian model averaging, a sample of possible models results from the RJMCMC simulation, weighted according to their posterior probabilities. In addition, we obtain the posterior distribution for the number and position of knots so that these can be analyzed using standard approaches for MCMC algorithms.



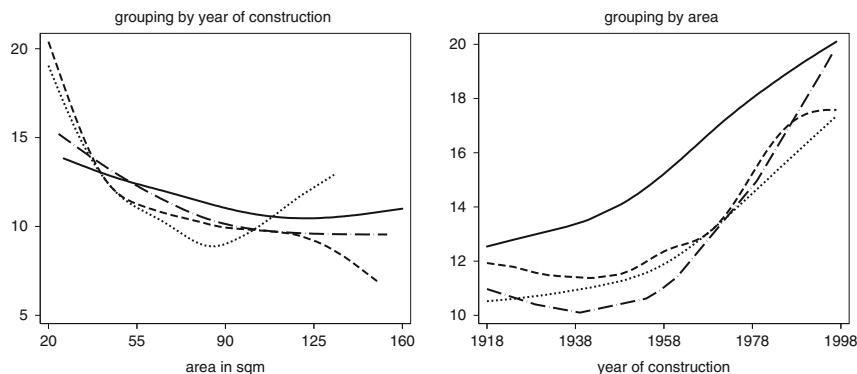


Fig. 8.32 Munich rent index: separate estimates for the effect of living area grouped according to year of construction (*left*) and separate estimates for the effect of year of construction grouped according to living area (*right*). The grouping structure has been determined as follows: $yearc \leq 1938$ (—), $1938 < yearc \leq 1958$ (---), $1958 < yearc \leq 1978$ (···), $yearc > 1978$ (-·-·); $area \leq 55$ (—), $55 < area \leq 90$ (---), $90 < area \leq 135$ (···), $area > 135$ (-·-·)

8.2 Bivariate and Spatial Smoothing

Thus far, we investigated the effect of *one single* continuous regressor in nonparametric regression. In this section, we examine approaches for bivariate smoothing (i.e., models with two continuous regressors) and for modeling spatial effects. To get an impression of the different possible problems and data situations, we start with various representative examples.

Example 8.6 Munich Rent Index—Interaction Between Living Area and Year of Construction

In Chap. 3 (Example 3.5), we saw that the year of construction and the living area have possibly nonlinear effects on the net rent per square meter. With the methods discussed in Sect. 8.1, we would only be able to examine each nonparametric effect of the two variables separately. However, when doing so, we assume that no interaction exists between the two effects. To evaluate this assumption, we grouped the data according to year of construction and living area, respectively, and estimated the effects in the corresponding groups using nonparametric methods. The results shown in Fig. 8.32 suggest that interaction effects occur between the two covariates, since the effects differ clearly for the individual groups. A flexible model for the description of such interaction effects has the form

$$rentsqm_i = f(area_i, yearc_i) + \varepsilon_i,$$

where f is a smooth function that, as in Sect. 8.1, is to be estimated flexibly from the given data, but now depending on two covariates. Since the interaction between living area and year of construction is modeled by a surface, we also refer to such a model as an interaction surface.

△

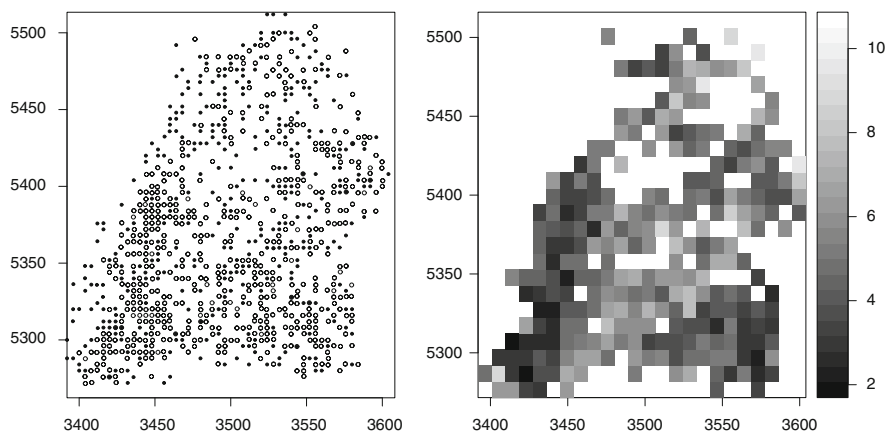


Fig. 8.33 Forest health status: *Left panel*, open circles indicate locations with known calcium concentration while predictions for the calcium concentration are required at locations represented as solid circles. *Right panel*, local averages based on a division in regular rectangles

Example 8.7 Forest Health Status—Spatial Smoothing of Chemical Concentration

This example shows an analysis of forest health in Baden–Württemberg, one of the federal states of Germany. The health status of trees is to be related to the concentration of different chemicals in the soil. However, the chemical concentrations have not been determined at all locations where the forest condition was measured. Figure 8.33 illustrates this situation with the concentration of calcium. In the left panel, every point represents a location where the forest condition was evaluated. Open circles indicate locations where calcium concentrations were also determined, whereas solid circles represent locations where calcium concentrations were not measured. To consider all observations in the analysis, we use the model

$$Ca_i = f(x_i, y_i) + \varepsilon_i$$

to predict the calcium concentration even at locations where it was not determined. In this model, Ca denotes the measured calcium concentration, whereas x and y correspond to the locations of the measurements in terms of longitude and latitude. In fact, we are then in the same situation as in Example 8.6, i.e., a continuous response variable is flexibly modeled in relation to two continuous explanatory variables. However, the problem differs from Example 8.6 for two reasons: First, the covariates now represent spatial information so that special methodology of spatial statistics can be used. Secondly, we need a representation of f , which can be evaluated at arbitrary locations. The latter is difficult to achieve, e.g., through data grouping in a regular grid and local averaging, because the data show relatively large gaps (again see Fig. 8.33). However, the figure illustrates that the calcium concentration changes continuously, and thus a smooth surface seems to be an appropriate model.

△

Example 8.8 Human Brain Mapping

Human brain mapping aims at identifying areas within the brain that are associated with specific tasks such as processing a visual stimulus. Therefore, a number of subjects were examined in an experiment using functional magnetic resonance imaging (fMRI), in

which periods of visual stimulation are alternated with rest periods. Figure 8.34 shows a single layer of the brain at two different time points of visual stimulation (and therefore a very similar spatial activation structure). Regions with an increased activation are represented by the lighter colored pixels. We note that these light pixels are primarily located towards the back portion of the brain, i.e., the location of the visual cortex, which is responsible for processing visual stimuli. With the help of spatial smoothing methods, we aim at filtering the noise out of the data in an effort to more clearly differentiate between activated and deactivated areas. Hence, we again find a spatial smoothing problem where a bivariate (or spatial) function is to be estimated. Although, in a strict sense, spatial information in the human brain mapping example is only available discretely, the large number of pixels allows us to disregard the discreteness and to identify every pixel with the coordinates at its center.

△

In the examples presented thus far, the covariates or spatial coordinates were always measured on a continuous scale or could at least be interpreted on a quasi-continuous scale. Within the scope of spatial statistics, we will also refer to these coordinates as *continuous location variables*. With other typical spatial statistics data, only discrete information is actually provided, and in such cases we refer to the covariates as *discrete location variables*. An example of such discrete spatial information could be the location of a household or residence, e.g., at a county level. It is then our goal to use the spatial allocation of the counties in order to estimate a spatially smooth function. We already encountered similar situations in Sect. 2.8. We next consider another spatial example using the rent index in Munich.

Example 8.9 Munich Rent Index—Examining Spatial Dependence

The rent index data also contain information regarding the subquarters in Munich in which the apartments are located. Figure 8.35 shows the average net rent per square meter for each of these subquarters separately. For example, the graphical presentation reveals somewhat reduced rents in the northern part of Munich. The goal of a spatial analysis in this context is to more clearly show such spatially structured effects by taking into account the spatial proximity of the subquarters in a smoothing approach.

△

Generally speaking, we can distinguish two different problems:

- Estimation of two-dimensional surfaces to model interactions or spatial effects of location variables measured on a continuous scale.
- Estimation of spatial effects based on discrete spatial information, represented, for example, in form of regions or spatial locations on a discrete grid.

Sections 8.2.1–8.2.3 focus on the former problem. We first extend the univariate penalization approaches discussed in Sect. 8.1 to the bivariate setting, especially penalized splines and the kriging method. Approaches for modeling discrete spatial information, which actually can be viewed as extensions of the random walk models discussed in Sect. 8.1.5, will be further investigated in Sect. 8.2.4. In Sect. 8.2.6, we briefly discuss some bivariate extensions of local and adaptive methods.

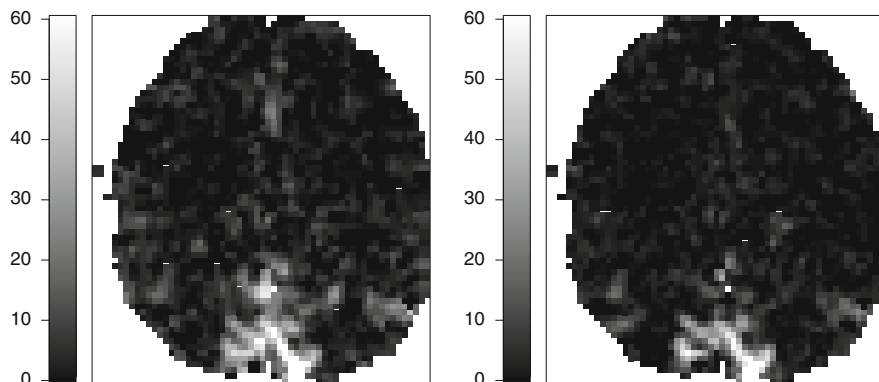


Fig. 8.34 Human brain mapping: pixelwise activity as a response to visual stimulation. The *left panel* shows the activity at time point $t = 18$ while the *right panel* refers to activity at time point $t = 38$

Fig. 8.35 Munich rent index: average net rent per square meter per subquarter. *Striped regions* correspond to subquarters without any observations (such as parks or industrial areas)



8.2.1 Tensor Product P-Splines

Tensor Product Bases

In Sect. 3.1.3, we have been modeling interactions between covariates as products of the corresponding design vectors. When using basis functions in nonparametric regression, this idea can be extended to bivariate interaction surfaces through the use of *tensor product bases*. Such a bivariate basis is obtained by considering all pairwise products of two univariate bases constructed for univariate smooths.

More specifically, we consider the following situation: The response variable y is to be described in terms of a two-dimensional surface $f(z_1, z_2)$, where z_1 and z_2 can be continuous covariates, as well as coordinates in the case of a spatial model. We then first construct the univariate bases for z_1 and z_2 , yielding the basis functions $B_j^{(1)}(z_1)$, $j = 1, \dots, d_1$, and $B_r^{(2)}(z_2)$, $r = 1, \dots, d_2$. The tensor product basis then consists of all basis functions of the form

$$B_{jr}(z_1, z_2) = B_j^{(1)}(z_1) \cdot B_r^{(2)}(z_2), \quad j = 1, \dots, d_1, \quad r = 1, \dots, d_2,$$

so that we obtain the following representation for $f(z_1, z_2)$:

$$f(z_1, z_2) = \sum_{j=1}^{d_1} \sum_{r=1}^{d_2} \gamma_{jr} B_{jr}(z_1, z_2).$$

For the case of polynomial splines, we refer to the tensor product basis as *tensor product splines* or bivariate polynomial splines. To illustrate the construction of a tensor product basis, Fig. 8.36 shows linear tensor product splines based on the univariate TP basis functions

$$B_1^{(1)}(z_1) = 1, \quad B_2^{(1)}(z_1) = z_1, \quad B_3^{(1)}(z_1) = (z_1 - \kappa_1)_+$$

and

$$B_1^{(2)}(z_2) = 1, \quad B_2^{(2)}(z_2) = z_2, \quad B_3^{(2)}(z_2) = (z_2 - \kappa_2)_+.$$

The constant function on the upper left results from the product of the two univariate constant basis functions $B_1^{(1)}$ and $B_1^{(2)}$. The first row and the first column are then obtained by multiplying the constant basis function in z_1 -direction with the basis functions in z_2 -direction or vice versa. The remaining four basis functions correspond to the products of the remaining univariate basis functions.

For regularizing tensor product TP-splines, we again use penalties derived from squared coefficients, similar as in the univariate setting discussed in Sect. 8.1.3. However, the numerical difficulties discussed there are even more pronounced in the bivariate setting, and we will therefore resort to tensor products of the numerically more stable B-spline basis. Figure 8.37 shows individual tensor product B-splines for the degrees $l = 0, 1, 2$, and 3. We again notice that a higher spline degree leads to more smoothness. In particular, tensor product splines of degree $l = 0$ are not continuous, while tensor product splines of degree $l = 1$ are continuous but not differentiable (see the definition of bivariate polynomial splines in Dierckx (1993) for a more detailed description of continuity and differentiability properties of tensor product splines). Figure 8.38 shows a larger number of cubic B-spline basis functions. In order to get a clearer picture, not all basis functions of a complete basis are mapped, as a considerably strong overlapping would occur, similar to what we saw for univariate B-splines in Sect. 8.1.1.

If we look at the contour plots (i.e., a graphical representation of the contour lines) of tensor product B-splines (Fig. 8.39), we find that the contour lines clearly deviate from circles, especially for the lower spline degrees. Consequently, tensor product B-splines are not radial. Radial basis functions will be discussed in Sects. 8.2.2 and 8.2.3.

Although at first sight, tensor product approaches appear to be much more complex than univariate basis function approaches, it is nevertheless straightforward to represent them in the form of large linear models. To do so, we define the design

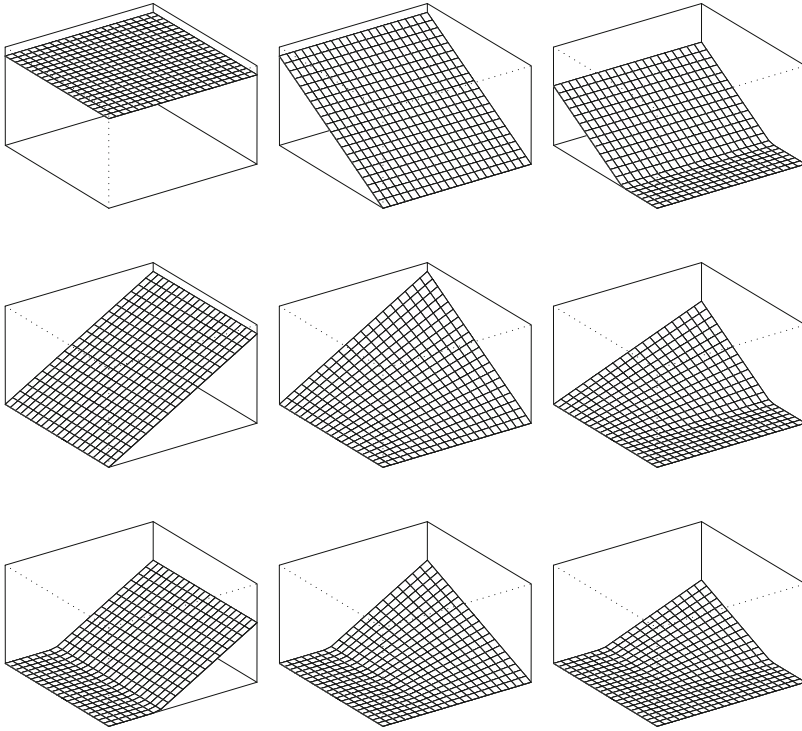


Fig. 8.36 Tensor product basis obtained from univariate linear TP bases

matrix $\mathbf{Z}$ with rows

$$\mathbf{z}'_i = (B_{11}(z_{i1}, z_{i2}), \dots, B_{d_1 1}(z_{i1}, z_{i2}), \dots, B_{1d_2}(z_{i1}, z_{i2}), \dots, B_{d_1 d_2}(z_{i1}, z_{i2}))$$

and express the vector of the corresponding regression coefficients as

$$\boldsymbol{\gamma} = (\gamma_{11}, \dots, \gamma_{d_1 1}, \dots, \gamma_{1d_2}, \dots, \gamma_{d_1 d_2})'.$$

We finally obtain the standard regression equation

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}. \quad (8.26)$$

In principle, bivariate smoothing approaches can therefore also be estimated within the scope of linear models. However, in comparison to the univariate case, the number of the parameters to be estimated is typically much larger. Numerically efficient computation of $\hat{\boldsymbol{\gamma}}$ is thus even more important, which can be implemented utilizing the sparse structure of the design matrix for tensor product B-splines (see also the comments on p. 430 and p. 488).

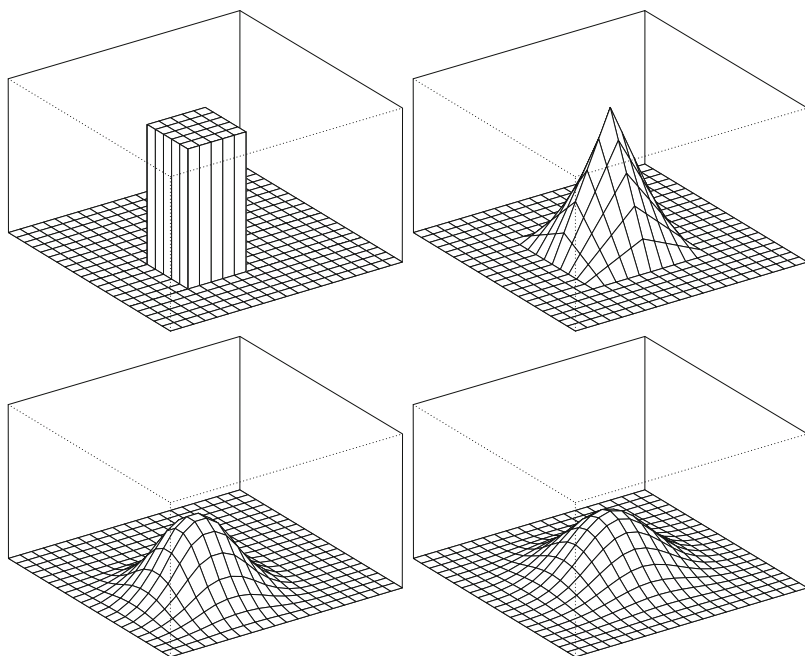


Fig. 8.37 Tensor product basis functions obtained from univariate B-splines of degrees $l = 0, 1, 2$, and 3

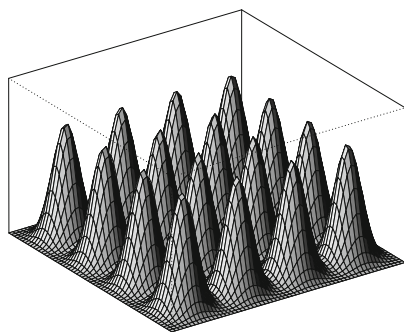


Fig. 8.38 A partial tensor product basis obtained from cubic univariate B-splines

As with univariate polynomial splines, we need to determine the optimal number and position of knots to construct tensor product splines. Moreover, we often encounter the problem that certain data regions may not have any observations. In such cases, it is impossible to estimate the coefficients of the basis functions associated with these regions. This is the case in the human brain mapping example. With tensor product splines, we construct basis functions over the entire area, $[\min(z_1), \max(z_1)] \times [\min(z_2), \max(z_2)]$. However, due to the shape of the brain,

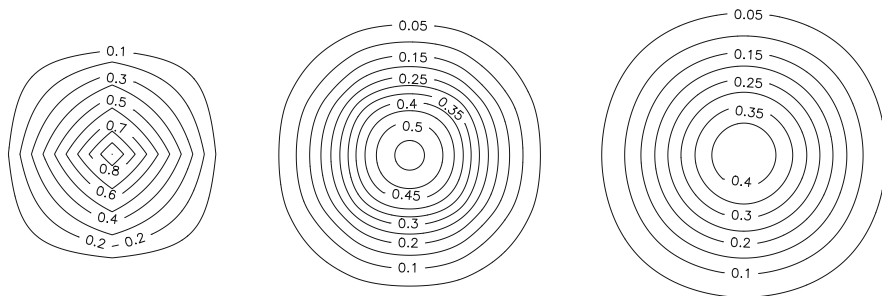


Fig. 8.39 Contour plots of tensor product B-splines of degree $l = 1, 2$, and 3

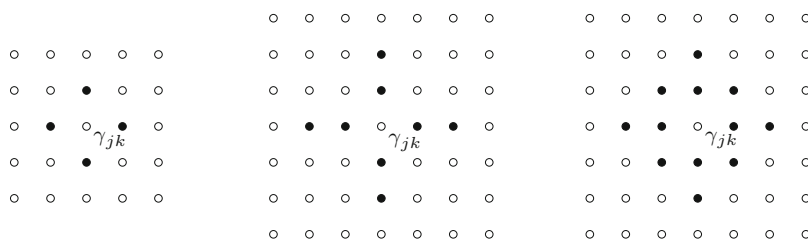


Fig. 8.40 Spatial neighborhoods on a regular lattice: The neighbors of γ_{jk} are marked as *filled dots*

no observations can be made in the corners of the resulting rectangle. Consequently, we are not able to estimate the corresponding regression coefficients. The same problems can occur, in principle, for univariate B-splines, e.g., when the covariate data has large gaps. Such problems are not quite as common in the univariate setting compared to the two-dimensional case. Both the problem of determining the number and position of the knots and the non-identifiability resulting from gaps in the data can be eliminated by adding a penalty for regularizing the estimation problem.

2D Penalties

Even though we could choose a ridge-type penalty for the tensor product TP basis, the spatial alignment of the basis functions and the associated regression coefficients may need to be incorporated appropriately when using tensor product B-splines. For univariate B-splines, penalties have been constructed based on squared differences of coefficients associated with neighboring basis functions. To transfer this concept to the two-dimensional case, we first define appropriate spatial neighborhoods. Figure 8.40 shows possible neighborhood definitions for four, eight, and twelve neighbors, respectively. We will now introduce several penalties that are based on these neighborhoods.

We start with the simplest case of a neighborhood consisting of four neighbors. A reasonable choice of the penalty is based on the squared differences between γ_{jk} and these four neighbors. Therefore, let $\mathbf{D}_1$ and $\mathbf{D}_2$ denote the univariate difference matrices of first order in z_1 - and z_2 -direction, respectively. Row-wise first-order differences are then obtained by applying the expanded difference matrix $\mathbf{I}_{d_2} \otimes \mathbf{D}_1$ to the vector $\boldsymbol{\gamma}$, where $\mathbf{I}_d$ is the d -dimensional identity matrix and $\otimes$ denotes the Kronecker product (see Definition A.10 in Appendix A.1). In fact, applying this difference matrix to the vector of regression coefficients yields

$$\boldsymbol{\gamma}'(\mathbf{I}_{d_2} \otimes \mathbf{D}_1)'(\mathbf{I}_{d_2} \otimes \mathbf{D}_1)\boldsymbol{\gamma} = \sum_{r=1}^{d_2} \sum_{j=2}^{d_1} (\gamma_{jr} - \gamma_{j-1,r})^2,$$

which is the sum of all squared row-wise differences. Analogously, we obtain the squared column-wise differences

$$\boldsymbol{\gamma}'(\mathbf{D}_2 \otimes \mathbf{I}_{d_1})'(\mathbf{D}_2 \otimes \mathbf{I}_{d_1})\boldsymbol{\gamma} = \sum_{j=1}^{d_1} \sum_{r=2}^{d_2} (\gamma_{jr} - \gamma_{j,r-1})^2.$$

Summing up all squared row-wise and column-wise differences then finally yields the penalty

$$\lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma} = \lambda \boldsymbol{\gamma}' [(\mathbf{I}_{d_2} \otimes \mathbf{D}_1)'(\mathbf{I}_{d_2} \otimes \mathbf{D}_1) + (\mathbf{D}_2 \otimes \mathbf{I}_{d_1})'(\mathbf{D}_2 \otimes \mathbf{I}_{d_1})] \boldsymbol{\gamma}. \quad (8.27)$$

Based on properties of Kronecker products (see Theorem A.4 in Appendix A.1), one can show that the penalty can equivalently be defined as

$$\lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma} = \lambda \boldsymbol{\gamma}' [\mathbf{I}_{d_2} \otimes \mathbf{K}_1 + \mathbf{K}_2 \otimes \mathbf{I}_{d_1}] \boldsymbol{\gamma},$$

with the univariate penalty matrices $\mathbf{K}_1 = \mathbf{D}_1' \mathbf{D}_1$ and $\mathbf{K}_2 = \mathbf{D}_2' \mathbf{D}_2$.

A Bayesian derivation of this penalty can be given as follows: As in the univariate case, we can define the difference penalties that were constructed row by row or column by column by using first-order random walks. Consequently, we can interpret $\mathbf{K}$ as the precision matrix of the entire distribution of the vector $\boldsymbol{\gamma}$ when assuming a two-dimensional random walk of first order. More precisely, we obtain the prior distribution for $\boldsymbol{\gamma}$ as

$$p(\boldsymbol{\gamma} \mid \tau^2) \propto \left(\frac{1}{\tau^2} \right)^{\text{rk}(\mathbf{K})/2} \exp \left(-\frac{1}{2\tau^2} \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma} \right). \quad (8.28)$$

Based on this joint prior, we can calculate the conditional distribution of γ_{jr} given all other coefficients as

$$\gamma_{jr} \mid \cdot \sim N\left(\frac{1}{4}(\gamma_{j-1,r} + \gamma_{j+1,r} + \gamma_{j,r-1} + \gamma_{j,r+1}), \frac{\tau^2}{4}\right)$$

(apart from points at the boundary of the coefficient space). Thus, a spatial form of the Markov property holds for the vector $\boldsymbol{\gamma}$, since the conditional distribution of γ_{jr} only depends on the four nearest (spatial) neighbors. For the expected value of the conditional distribution, we obtain the local mean computed from the four nearest neighbors. Hence, the penalty for two-dimensional surfaces generalizes the properties that we discussed in Sect. 8.1.2, p. 441ff for univariate P-splines.

The principle for building two-dimensional penalties, based on row-wise or column-wise differences, can also be applied to difference matrices of a higher order. We then obtain penalties of the form

$$\lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma} = \lambda \boldsymbol{\gamma}' \left[\mathbf{I}_{d_2} \otimes \mathbf{K}_1^{(k_1)} + \mathbf{K}_2^{(k_2)} \otimes \mathbf{I}_{d_1} \right] \boldsymbol{\gamma}$$

with univariate penalty matrices $\mathbf{K}_1^{(k_1)}$ and $\mathbf{K}_2^{(k_2)}$ of orders k_1 and k_2 . For example, with $k_1 = k_2 = 2$, we obtain a penalty based on squared second-order differences whose neighborhood structure consists of the eight nearest neighbors along the coordinate axis (see Fig. 8.40).

In conclusion, we find that the approaches discussed in this section for bivariate penalties (and also some extensions) in combination with tensor product B-splines can be expressed in the general form seen in Sect. 8.1: The vector of function evaluations can be represented as a large linear model $\mathbf{Z} \boldsymbol{\gamma}$ with a quadratic penalty $\lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}$ or an equivalent Gaussian prior for $\boldsymbol{\gamma}$ with density (8.28). It follows that the estimation procedures discussed in Sect. 8.1.9 are again applicable. Since the number of parameters in the two-dimensional model is, however, much larger than in the univariate case, numerically efficient implementations are even more crucial, especially when considering MCMC algorithms.

For mixed model-based inference, we also need a partition of the regression coefficients as

$$\boldsymbol{\gamma} = \tilde{\mathbf{X}} \boldsymbol{\beta} + \tilde{\mathbf{U}} \tilde{\boldsymbol{\gamma}}.$$

Recall from Sect. 8.1.9 that $\tilde{\mathbf{X}}$ results from a basis of the null space of the matrix $\mathbf{K}$. The null space is determined by the space of functions that remains unpenalized by the penalty matrix $\mathbf{K}$. When using a Kronecker product penalty based on first-order differences, the addition of a constant term does not affect either the row-wise or the column-wise penalty. One can actually show that indeed $\text{rk}(\mathbf{K}) = d_1 d_2 - 1$ still holds, so that it is only the constant term that remains unpenalized, and the null space of the matrix $\mathbf{K}$ is given by a $d_1 d_2$ -dimensional vector of ones. In general, the null space of a Kronecker product penalty matrix $\mathbf{K}$ is given by the tensor product



8.12 Tensor Product P-Splines

The goal is to estimate the bivariate function f in the model

$$y = f(z_1, z_2) + \varepsilon$$

based on continuous covariates z_1 and z_2 . Tensor product splines are constructed through forming all pairwise products of univariate polynomial splines for z_1 and z_2 , i.e.,

$$B_{jr}(z_1, z_2) = B_j^{(1)}(z_1) \cdot B_r^{(2)}(z_2), \quad j = 1, \dots, d_1, \quad r = 1, \dots, d_2.$$

This yields the representation

$$f(z_1, z_2) = \sum_{j=1}^{d_1} \sum_{r=1}^{d_2} \gamma_{jr} B_{jr}(z_1, z_2)$$

and therefore the model

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon},$$

where $\mathbf{Z}$ consists of the evaluated basis functions and $\boldsymbol{\gamma}$ are the corresponding regression coefficients.

Penalties can be constructed from Kronecker products of univariate penalty matrices:

$$\mathbf{K} = \mathbf{I}_{d_2} \otimes \mathbf{K}_1 + \mathbf{K}_2 \otimes \mathbf{I}_{d_1}.$$

This results in the quadratic penalty

$$\lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma},$$

and therefore the methods discussed in Sect. 8.1.9 can be used for estimating λ .

of the null spaces of the univariate penalty matrices. Thus, second-order differences lead to a four-dimensional null space, containing the constant, linear effects in z_1 - and z_2 -direction, and the interaction $z_1 \cdot z_2$. This null space basis can be represented based on the univariate knots $\kappa_1^{(1)}, \dots, \kappa_{d_1}^{(1)}$ and $\kappa_1^{(2)}, \dots, \kappa_{d_2}^{(2)}$, leading to

$$\begin{pmatrix} 1 & \kappa_1^{(1)} & \kappa_1^{(2)} & \kappa_1^{(1)}\kappa_1^{(2)} \\ 1 & \kappa_2^{(1)} & \kappa_2^{(2)} & \kappa_2^{(1)}\kappa_2^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \kappa_{d_1}^{(1)} & \kappa_{d_1}^{(2)} & \kappa_{d_1}^{(1)}\kappa_{d_1}^{(2)} \\ 1 & \kappa_1^{(1)} & \kappa_2^{(2)} & \kappa_1^{(1)}\kappa_2^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \kappa_{d_1}^{(1)} & \kappa_2^{(2)} & \kappa_{d_1}^{(1)}\kappa_2^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \kappa_1^{(1)} & \kappa_{d_2}^{(2)} & \kappa_1^{(1)}\kappa_{d_2}^{(2)} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \kappa_{d_1}^{(1)} & \kappa_{d_2}^{(2)} & \kappa_{d_1}^{(1)}\kappa_{d_2}^{(2)} \end{pmatrix}.$$

For the construction of $\tilde{U}$, we can proceed as in Sect. 8.1.9, i.e., we define $\tilde{U}$ based on the spectral decomposition of $\mathbf{K}$. Note that we actually need to use the spectral decomposition in this case, since it is not possible to represent $\mathbf{K}$ as $\mathbf{K} = \mathbf{D}'\mathbf{D}$.

Example 8.10 Munich Rent Index—Interaction Between Year of Construction and Living Area



We use tensor product splines with different penalties to examine the interaction effect between year of construction and living area (see Example 8.6, p. 500) while neglecting all other covariate information and in particular the spatial information on the location of the flats in specific subquarters of Munich. Figure 8.41 shows such estimates based on cubic B-splines with 20 inner knots, using first- and second-order differences.

Both estimates basically reflect the same structure, i.e., higher estimated net rents per square meter for new and small apartments. It is rather clear that an interaction effect exists, and thus estimation based on univariate functions of year of construction and living area would be insufficient. If we compare the smoothness of the estimated surfaces, the use of higher differences in the penalty, as already seen in the univariate case, tends to lead to smoother functions.

△

Example 8.11 Forest Health Status—Spatial Smoothing of Chemicals

As a second example, we analyze the spatial distribution of calcium concentrations in Baden–Württemberg; see Example 8.7. Figure 8.42 shows the estimated surface using cubic tensor product B-splines with 15 inner knots for each of the longitudinal and latitudinal coordinates, and a penalty based on first-order differences. The estimated surface does reproduce the regional trend found in the descriptive analysis very well (Fig. 8.33 on p. 501) but does have the particular advantage that it can be visualized in a very high resolution, and it can also be extrapolated into regions without observations. Augustin, Lang, Musio, and von Wilpert (2007) use these extrapolated values in their analysis of forest health in Baden–Württemberg. Note, however, that with tensor product splines, we always estimate the unknown surface over a rectangular domain which in case of the calcium concentrations means that we are sometimes extrapolating very far from the observed data (e.g., in the

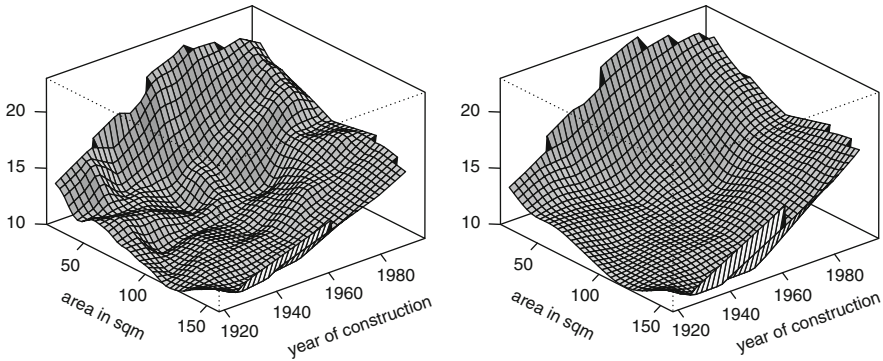


Fig. 8.41 Munich rent index: estimated interaction surface between living area in square meters and year of construction based on cubic B-splines with 20 inner knots and first (*left*) and second (*right*) order difference penalty

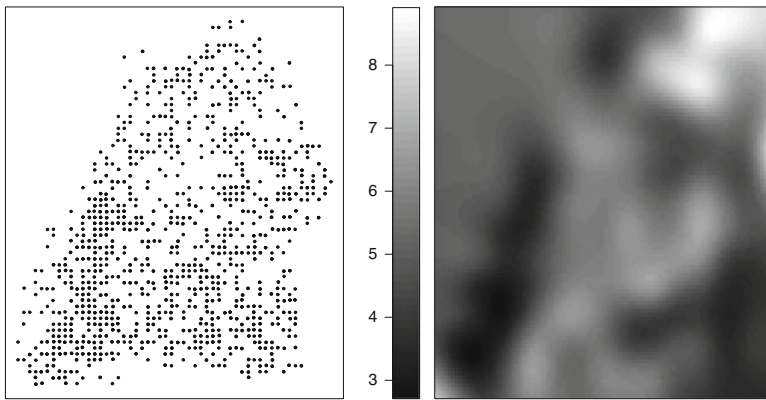


Fig. 8.42 Forest health status: estimated spatial calcium concentration based on cubic tensor product B-splines with 15 inner knots and first-order difference penalty

upper left corner of the right panel in Fig. 8.42). In these cases, information on the estimated surface is very scarce, which will typically be reflected by very wide confidence intervals. $\triangle$

8.2.2 Radial Basis Functions and Thin Plate Splines

An alternative way of constructing bivariate basis functions is obtained with *radial bases*. Generally speaking, a radial basis function is defined as a function of the Euclidian distance between a knot $\kappa = (\kappa_1, \kappa_2)$ and an observation point $z = (z_1, z_2)$, i.e.,

$$B_{\kappa}(z) = B(\|z - \kappa\|) = B(r),$$

with a suitably chosen scalar function B and Euclidean distance $r = \|\mathbf{z} - \boldsymbol{\kappa}\| = ((z_1 - \kappa_1)^2 + (z_2 - \kappa_2)^2)^{0.5}$. The term radial basis function results from the fact that, due to their construction, the contour plots of radial basis functions consist of circular contour lines.

All radial basis functions have the same functional form and, in contrast to tensor product B-splines, are attributed to one specific knot. Typically, the knots of a radial basis are a subset of the observation points, i.e., $\{\boldsymbol{\kappa}_1, \dots, \boldsymbol{\kappa}_d\} \subset \{\mathbf{z}_1, \dots, \mathbf{z}_n\}$. Thus, the distribution of the radial basis functions adapts automatically to the distribution of the data, whereas for tensor product bases, a large number of basis functions can be placed in regions in which no observations were made. For example, in case of the Baden–Württemberg forest health data, a large number of knots were actually placed outside the region of interest when using tensor products, since the knots are equally distributed over $[\min(z_1), \max(z_1)] \times [\min(z_2), \max(z_2)]$. However, this extended knot set also provides us with the possibility to extrapolate into these regions, whereas the extrapolation region for radial basis functions is more limited.

The most well-known example of radial basis functions can be obtained through optimizing the criterion

$$\sum_{i=1}^n (y_i - f(\mathbf{z}_i))^2 + \lambda \iint \left[\left(\frac{\partial^2}{\partial^2 z_1} + 2 \frac{\partial^2}{\partial z_1 \partial z_2} + \frac{\partial^2}{\partial^2 z_2} \right) f(\mathbf{z}_1, \mathbf{z}_2) \right]^2 d\mathbf{z}_1 d\mathbf{z}_2 \rightarrow \min_f \quad (8.29)$$

over the class of all twice continuously differentiable functions $f(\mathbf{z})$. In this case,

$$\iint \left[\left(\frac{\partial^2}{\partial^2 z_1} + 2 \frac{\partial^2}{\partial z_1 \partial z_2} + \frac{\partial^2}{\partial^2 z_2} \right) f(\mathbf{z}_1, \mathbf{z}_2) \right]^2 d\mathbf{z}_1 d\mathbf{z}_2 \quad (8.30)$$

is the bivariate analogue of the integrated squared second derivative, which we discussed in the context of smoothing splines. Hence, this approach aims at generalizing smoothing splines to bivariate surface smoothing. As a solution, we obtain the *thin plate splines*, a generalization of natural cubic splines that also fulfills the natural boundary conditions, i.e., it behaves linearly outside of the observation domain. A thin plate spline can be represented as

$$f(\mathbf{z}_1, \mathbf{z}_2) = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \sum_{j=1}^n \gamma_j B_j(\mathbf{z}_1, \mathbf{z}_2)$$

with

$$B_j(\mathbf{z}_1, \mathbf{z}_2) = B(\|\mathbf{z} - \mathbf{z}_j\|) = \|\mathbf{z} - \mathbf{z}_j\|^2 \log(\|\mathbf{z} - \mathbf{z}_j\|)$$

and subject to an identifiability restriction on the coefficients γ_j that we will work out below. Thus, the thin plate spline relies on linear effects in the directions z_1 and z_2 and radial basis functions

$$B(r) = r^2 \log(r)$$

centered at the n covariate values; see [Green and Silverman \(1993\)](#) for more details on thin plate splines and a proof of their optimality property. In summary, the thin plate spline is a bivariate generalization of the smoothing spline which would result as a special case when restricting the surface to a univariate regression function.

In matrix notation, the above representation of the thin plate spline induces the model equation

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \varepsilon,$$

where $\mathbf{X}$ is the design matrix containing a constant as well as the linear effects of z_1 and z_2 , $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)'$ collects the corresponding regression coefficients,

$$\mathbf{Z}[i, j] = B_j(z_{i1}, z_{i2})$$

contains the radial basis functions evaluated at the observed covariate values, and $\boldsymbol{\gamma}$ is the vector of the basis coefficients. When counting the number of regression coefficients in this model, we obtain $n + 3$ parameters and therefore the model is overspecified. To obtain an identifiable version, the restriction $\mathbf{X}'\boldsymbol{\gamma} = \mathbf{0}$ has to be imposed, basically ensuring that the linear part of the model is orthogonal to the part represented by the radial basis functions. In addition, it can be shown that the integral penalty (8.30) can equivalently be represented as

$$\boldsymbol{\gamma}'\mathbf{Z}\boldsymbol{\gamma},$$

where the penalty matrix coincides with the design matrix. Hence, the overall estimation problem (8.29) can be rewritten as

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\boldsymbol{\gamma})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\boldsymbol{\gamma}) + \lambda\boldsymbol{\gamma}'\mathbf{Z}\boldsymbol{\gamma} \rightarrow \min_{\boldsymbol{\beta}, \boldsymbol{\gamma}}$$

subject to the constraint $\mathbf{X}'\boldsymbol{\gamma} = \mathbf{0}$.

Similar as in the context of smoothing splines, the number of regression coefficients associated with thin plate splines is often too large in practice since $(n + 3) \times (n + 3)$ systems of equations have to be solved. It is therefore of interest to obtain low rank approximations to the thin plate spline while still remaining as close to the optimal solution as possible. While simple rules for choosing a subset of observations as knots may often also produce sensible solutions, [Wood \(2003\)](#) proposed an optimal approximation based on a spectral decomposition of the design matrix $\mathbf{Z}$ that is also implemented in the R package `mgcv`. Basically, we first compute the spectral decomposition

$$\mathbf{Z} = \boldsymbol{\Gamma}\boldsymbol{\Omega}\boldsymbol{\Gamma}',$$

where $\boldsymbol{\Gamma}$ is an orthonormal matrix of eigenvectors and $\boldsymbol{\Omega}$ contains the corresponding (nonnegative) eigenvalues in descending order. It can then be shown that

$$\mathbf{Z}_d = \boldsymbol{\Gamma}_d\boldsymbol{\Omega}_d\boldsymbol{\Gamma}_d',$$

where $\mathbf{\Gamma}_d$ and $\mathbf{\Omega}_d$ are the submatrices of $\mathbf{\Gamma}$ and $\mathbf{\Omega}$ associated with the d largest eigenvalues, is the best rank d approximation to $\mathbf{Z}$ in the sense of the spectral norm $\|\mathbf{Z} - \mathbf{Z}_d\|$ (with $\|\mathbf{A}\|$ corresponding to the square root of the largest eigenvalue of the positive semidefinite matrix $\mathbf{A}$). The idea is now to replace $\mathbf{Z}$ with $\mathbf{Z}_d$ which basically projects the original estimation problem into the optimal d -dimensional subspace where d can be chosen such that the approximation error is small. In fact, fast numerical algorithms for truncated spectral decompositions exist that avoid the need to compute the full spectral decomposition.

Other commonly used radial basis functions are, for example,

$$B(r) = r^l, \quad l \text{ odd},$$

or

$$B(r) = \sqrt{r^2 + c^2} \quad \text{for a constant } c > 0.$$

In this case, the penalty matrix results from the integrated squared second derivatives of the basis functions, similar to those in the univariate case. In the following section, we will see that the stationary Gaussian fields that we previously discussed in Sect. 8.1.6 can also be interpreted as radial basis function approaches.

8.2.3 Kriging: Spatial Smoothing with Continuous Location Variables

In Sect. 8.1.6 we already discovered that temporal correlations can be described using stationary Gaussian processes and parametric correlation functions. We will now transfer this approach to spatial correlations using Gaussian random fields. In contrast to the previously discussed approaches for modeling interaction surfaces based on basis functions, we now consider a probabilistic modeling framework, i.e., spatial effects are described in terms of stochastic processes. However, as in the univariate case, we will see that the probabilistic model formulation is in fact equivalent to a special basis function approach.

In general, a Gaussian field $\{\gamma(s), s \in \mathbb{R}^2\}$ is characterized by the expectation function $\mu(s) = E(\gamma(s))$, the variance function $\tau^2(s) = \text{Var}(\gamma(s))$, and the correlation function $\rho(s, t) = \text{Corr}(\gamma(s), \gamma(t))$. For *stationary* Gaussian fields, the expected value and variance are spatially constant ($\mu(s) \equiv \mu$ and $\tau^2(s) \equiv \tau^2$), and the correlation function only depends on the difference $s - t$. Thus, we obtain $\rho(s, t) = \rho(s - t) = \rho(h)$ with $h = s - t$. When analyzing spatial correlations, we often limit ourselves to the special case of *isotropic* correlation functions, where

$$\rho(s, t) = \rho(\|s - t\|) = \rho(r)$$

with $r = \|s - t\|$. It follows that the correlation between two points s and t only depends on their Euclidean distance but not on the exact position of the points or on the direction of the vector between the two points. When assuming isotropy, we can use all parametric classes discussed in Sect. 8.1.6 (p. 453) for the correlation

function ρ , since the actual bivariate spatial correlation function only depends on the scalar r .

An easy way to incorporate anisotropy is to replace the Euclidean distance

$$||s - t|| = \sqrt{(s - t)'(s - t)}$$

with

$$\sqrt{(s - t)'R(\psi)'D(\delta)R(\psi)(s - t)}, \quad (8.31)$$

where $R(\psi)$ defines a rotation matrix

$$R(\psi) = \begin{pmatrix} \cos(\psi) & \sin(\psi) \\ -\sin(\psi) & \cos(\psi) \end{pmatrix},$$

with anisotropy angle $\psi \in [0, 2\pi]$ and $D(\delta)$ defines a prolongation matrix

$$D(\delta) = \begin{pmatrix} \delta^{-1} & 0 \\ 0 & 1 \end{pmatrix}$$

with anisotropy ratio $\delta \geq 1$. Figure 8.43 shows the resulting contour lines for various combinations of ψ and δ . Note that Eq. (8.31) can only be used to produce correlation functions with elliptic contour lines. More general approaches to the construction of anisotropic correlation functions also allow other deviations from radial contour lines.

Classical Geostatistics

In spatial statistics, kriging was developed for smoothing or interpolating spatial phenomena. The classical geostatistical model is defined as

$$y(s) = \mathbf{x}(s)'\boldsymbol{\beta} + \gamma(s) + \varepsilon(s),$$

where

$\mathbf{x}(s)'\boldsymbol{\beta}$ is the spatial trend parameterized by covariates $\mathbf{x}$,

$\gamma(s)$ is a stationary Gaussian process with expected value 0, variance τ^2 , and correlation function $\rho(h)$, and

$\varepsilon(s)$ is the usual i.i.d. error $\varepsilon(s) \sim N(0, \sigma^2)$ (independent of $\gamma(s)$).

If the spatial trend is constant, i.e., $\mathbf{x}(s)'\boldsymbol{\beta} \equiv \beta_0$, the model simplifies to the case of ordinary kriging, whereas the general case is referred to as universal kriging.

In matrix notation, the kriging model can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon},$$

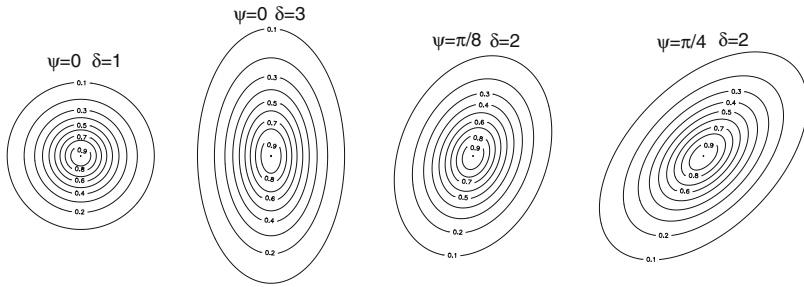


Fig. 8.43 Contour lines of anisotropic correlation functions constructed based on the distance measure (8.31)

where $\boldsymbol{\gamma} = (\boldsymbol{\gamma}(s_{(1)}), \dots, \boldsymbol{\gamma}(s_{(d)}))'$ defines the values of the stationary Gaussian process at the d unique observed spatial locations $s_{(1)}, \dots, s_{(d)}$. The matrix $\mathbf{Z}$ corresponds to an incidence matrix with

$$\mathbf{Z}[i, j] = \begin{cases} 1 & \text{if } y_i \text{ is observed at point } s_{(j)} \text{ (i.e., } s_i = s_{(j)}), \\ 0 & \text{otherwise.} \end{cases}$$

Based on the model specification, the covariance matrix of the response vector $\mathbf{y}$ becomes

$$\text{Cov}(\mathbf{y}) = \tau^2 \mathbf{Z} \mathbf{R} \mathbf{Z}' + \sigma^2 \mathbf{I}_n,$$

where the correlation matrix of the spatial effects $\boldsymbol{\gamma}$,

$$\mathbf{R} = (\text{Corr}(\boldsymbol{\gamma}(s_{(i)}), \boldsymbol{\gamma}(s_{(j)})) = (\rho(s_{(i)} - s_{(j)})),$$

introduces spatial correlation and is embedded in $\text{Cov}(\mathbf{y})$.

The goal of classical geostatistics is to obtain optimal predictions for $\boldsymbol{\gamma}(s_0)$ or $\boldsymbol{\eta}(s_0) = \mathbf{x}(s_0)' \boldsymbol{\beta} + \boldsymbol{\gamma}(s_0)$ at new locations s_0 . Using the definitions $\mathbf{r} = (\rho(s_1 - s_0), \dots, \rho(s_n - s_0))'$, $\eta_0 = \boldsymbol{\eta}(s_0)$ and $\mu_0 = \mathbf{x}(s_0)' \boldsymbol{\beta}$, the optimal prediction can be derived from the joint normal distribution

$$\begin{pmatrix} \mathbf{y} \\ \eta_0 \end{pmatrix} \sim \mathcal{N} \left(\begin{bmatrix} \mathbf{X} \boldsymbol{\beta} \\ \mu_0 \end{bmatrix}, \begin{bmatrix} \tau^2 \mathbf{Z} \mathbf{R} \mathbf{Z}' + \sigma^2 \mathbf{I}_n & \tau^2 \mathbf{r}' \\ \tau^2 \mathbf{r} & \tau^2 \end{bmatrix} \right)$$

(see Theorem B.6 in Appendix B.3). The mean squared error optimal prediction is simply the conditional expectation of η_0 given $\mathbf{y}$, i.e.,

$$\hat{\eta}_0 = \mathbb{E}(\eta_0 | \mathbf{y}) = \mu_0 + \tau^2 \mathbf{r}' (\tau^2 \mathbf{Z} \mathbf{R} \mathbf{Z}' + \sigma^2 \mathbf{I}_n)^{-1} (\mathbf{y} - \mathbf{X} \boldsymbol{\beta}).$$

The conditional variance of $\hat{\eta}_0$ can also be derived from the joint normal distribution, yielding

$$\text{Var}(\eta_0 | \mathbf{y}) = \tau^2 - \tau^2 \mathbf{r}' (\tau^2 \mathbf{Z} \mathbf{R} \mathbf{Z}' + \sigma^2 \mathbf{I}_n)^{-1} \mathbf{r} \tau^2.$$

Even without the explicit assumption of normality, one can show that $\hat{\eta}_0$ still fulfills certain optimality properties. Similar to the estimates of random effects presented in Sect. 7.3.1 (p. 371), $\hat{\eta}_0$ is the best linear unbiased predictor.

Kriging as a Basis Function Approach

With the use of classical geostatistics, we obtain optimal predictions of the spatial effects at arbitrary locations s_0 . However, in contrast to the basis function approaches discussed thus far, we do not obtain a compact representation for such predictions. A possible scenario arises from the developments presented in Sect. 8.1.6 (p. 453), where we reexpressed the stochastic kriging approach using basis functions derived from the correlation function $\rho(\cdot)$. This reparameterization can also be used for the spatial kriging model, so that the classical geostatistics model can be expressed as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \tilde{\mathbf{Z}}\tilde{\boldsymbol{\gamma}} + \boldsymbol{\varepsilon},$$

with design matrix

$$\tilde{\mathbf{Z}}[i, j] = \rho(s_i, s_{(j)}).$$

For a single observation, this yields the model specification

$$y(s) = \mathbf{x}(s)'\boldsymbol{\beta} + f_{geo}(s) + \varepsilon(s)$$

with spatial effect

$$f_{geo}(s) = \sum_{j=1}^d \tilde{\gamma}_j B_j(s)$$

and basis functions

$$B_j(s) = \rho(s, s_{(j)}),$$

obtained from the correlation function.

For isotropic correlation functions, we obtain the special case of radial basis functions $B_j(s) = \rho(\|s - s_{(j)}\|)$. Similar to Sect. 8.1.6, the penalty for $\tilde{\boldsymbol{\gamma}}$ results from the correlation matrix $\mathbf{R}$ as

$$\lambda \tilde{\boldsymbol{\gamma}}' \mathbf{K} \tilde{\boldsymbol{\gamma}} = \frac{\sigma^2}{\tau^2} \tilde{\boldsymbol{\gamma}}' \mathbf{R} \tilde{\boldsymbol{\gamma}}.$$

If we choose Matérn correlation functions, we obtain the class of *Matérn splines*. Figure 8.44 shows two isotropic Matérn spline basis functions, which result from the hyperparameters $\kappa = 0.5$ and $\kappa = 1.5$. Contour plots, which illustrate the radially of the basis functions, are provided as well.

From a Bayesian point of view, the kriging approach in the basis function representation is equivalent to a certain smoothing prior, which is given by

$$\tilde{\boldsymbol{\gamma}} \sim \mathbf{N}(\mathbf{0}, \tau^2 \mathbf{R}^{-1}). \quad (8.32)$$

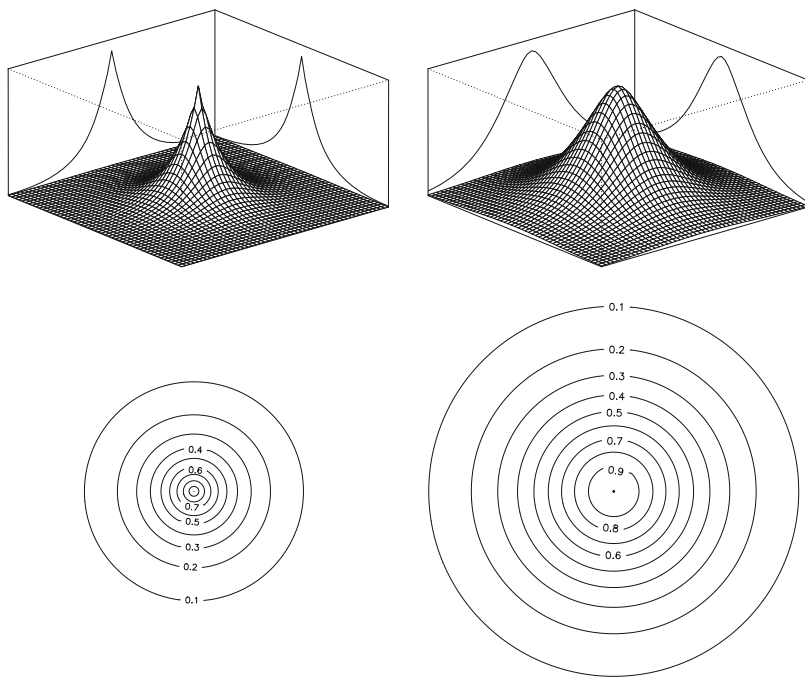


Fig. 8.44 Perspective plots and contour lines of Matérn spline basis functions with $\kappa = 0.5$ (left) and $\kappa = 1.5$ (right)

In contrast to the tensor product methods that we discussed for polynomial splines, the knots for the kriging approach are determined from the observed data locations. As the number of knots is usually close to the sample size, there are typically a large number of coefficients to estimate. Therefore, to alleviate computing demands, one therefore often considers only a subset of the observed locations as knots, i.e.,

$$\mathcal{D} = \{\kappa_1, \dots, \kappa_m\} \subset \mathcal{C} = \{s_{(1)}, \dots, s_{(d)}\}.$$

Based on adequate criteria, this subset can be determined as representative for the original observations; see [Johnson, Moore, and Ylvisaker \(1990\)](#) and [Nychka and Saltzman \(1998\)](#) for details. Through $\tilde{\mathbf{Z}}\tilde{\boldsymbol{\gamma}}$, we have an approximate kriging surface, where the design matrix is defined by

$$\tilde{\mathbf{Z}}[i, j] = \rho(s_i - \kappa_j), \quad (8.33)$$

and the m -dimensional coefficient vector has the distribution

$$\tilde{\boldsymbol{\gamma}} \sim \mathbf{N}(\mathbf{0}, \tau^2 \mathbf{R}^{-1}) \quad \text{with} \quad \mathbf{R}[i, j] = \rho(\kappa_i, \kappa_j). \quad (8.34)$$

8.13 Kriging

Classical Geostatistics Model

$$y(s) = \mu(s) + \gamma(s) + \varepsilon(s), \quad s \in \mathbb{R}^2,$$

where $\mu(s) = \mathbf{x}(s)' \boldsymbol{\beta}$ defines the spatial trend and $\gamma(s)$ is a (stationary) Gaussian field with expectation 0, variance τ^2 , and parametric correlation function $\rho(h)$.

Estimation

For given variances, the best linear unbiased predictions for $\eta_0 = \mathbf{x}(s_0)' \boldsymbol{\beta} + \gamma(s_0)$ result from the joint normal distribution of $\mathbf{y}$ and η_0 . Since $\gamma(s)$ corresponds to a spatially correlated random effect, estimation can be approached more generally using the mixed model methods discussed in Chap. 7.

Basis Function Representation

As with the univariate kriging method, we also obtain a bivariate basis function representation with basis functions

$$B_j(s) = \rho(s, s_{(j)}) \quad \text{or} \quad B_j(s) = \rho(\|s - s_{(j)}\|).$$

This yields the model specification

$$\mathbf{y} = \mathbf{X} \boldsymbol{\beta} + \tilde{\mathbf{Z}} \tilde{\boldsymbol{\gamma}} + \boldsymbol{\varepsilon},$$

with $\tilde{\mathbf{Z}}[i, j] = \rho(s_i, s_{(j)})$ or $\tilde{\mathbf{Z}}[i, j] = \rho(\|s_i - s_{(j)}\|)$. The penalty for $\tilde{\boldsymbol{\gamma}}$ is determined by the correlation matrix $\mathbf{R}$ with elements $\mathbf{R}[i, j] = \rho(s_{(i)}, s_{(j)})$ yielding

$$\lambda \tilde{\boldsymbol{\gamma}}' \mathbf{K} \tilde{\boldsymbol{\gamma}} = \frac{\sigma^2}{\tau^2} \tilde{\boldsymbol{\gamma}}' \mathbf{R} \tilde{\boldsymbol{\gamma}}.$$

An alternative would be to use a low rank approximation based on the spectral decomposition of $\mathbf{Z}$ similar as discussed for thin plate splines.

Estimation of Kriging Models

Estimation of kriging models can be achieved in one of two ways resulting from the different perspectives on the geostatistical model. If we consider the geostatistical model in the original stochastic interpretation, the assumption of a Gaussian field

for the spatial effect basically corresponds to the assumption of a spatially correlated random effect. The methods discussed in Chap. 7 can therefore be used to determine ML or REML estimates of all parameters of the correlation function (including for example the range parameter).

When interpreting kriging as a basis function method, all parameters of the correlation function are typically fixed at prespecified values or chosen based on simple rules (as done in Example 8.3). The variance τ^2 then remains as the only unknown parameter, which acts as a smoothing parameter for the basis function method, while the penalty matrix $\mathbf{R}$ is completely specified. To estimate τ^2 , we can use the methods discussed in Sect. 8.1.9, in particular the mixed model representation or MCMC-based algorithms.

Example 8.12 Human Brain Mapping—Kriging

We next apply the spatial kriging method in the human brain mapping example with a visual stimulus (see Example 8.8 on p. 501). Figure 8.45 shows two estimates for this activation profile based on Matérn splines with parameter $\kappa = 1.5$ using either 100 or 200 knots. The range parameter of the correlation function was chosen according to the rule of thumb, which we already used in Example 8.8.

For both 100 and 200 knots, strong activation is clearly visible within the visual cortex. The basis function approach does not only allow for pixelwise identification of activation, but in fact provides a continuous activation surface. Comparing the two knot sets (100 or 200 knots), we find that the higher knot density leads to a somewhat rougher estimate. However, in general, the estimated function is not highly sensitive to the number of knots. $\triangle$

8.2.4 Markov Random Fields: Spatial Smoothing with Discrete Location Variables

To this point, we discussed spatial effects based on spatial coordinates, i.e., cases where continuous spatial information is available. As we saw in Example 8.9 (p. 502), it is also possible to have discrete spatial information, e.g., spatial regions $s \in \{1, \dots, d\}$. Discrete spatial information also arises naturally in case of data arranged on a regular grid. Note that the human brain mapping example actually consists of such gridded data. However, with a large number of grid points, it is also possible to analyze such data using a continuous spatial model, identifying each pixel with its respective coordinates.

Spatial Neighborhoods

Whereas in the case of continuous spatial information we can easily determine the distance between two locations using, for example, the Euclidean distance, this is no longer the case with discrete spatial data. Consequently, we use a different concept to describe the spatial arrangement of the data, which relies on the definition of adequate forms of neighborhoods. Such neighborhoods can actually be constructed in different ways:

- For spatial data where the spatial covariate s defines the membership of an observation to a particular region s , neighborhoods are usually defined by

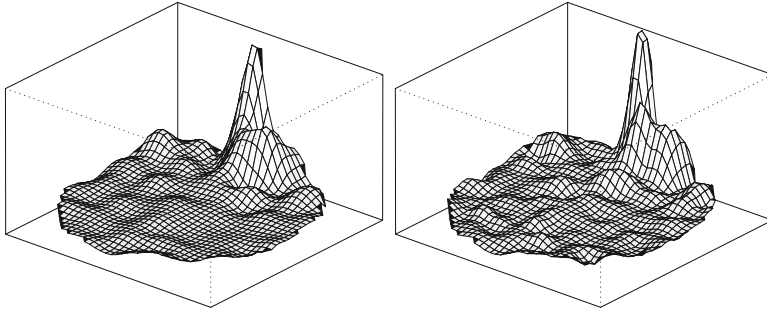


Fig. 8.45 Human brain mapping: kriging estimates for the activation at time point $t = 38$ obtained from Matérn splines with $\kappa = 1.5$ and 100 (left) or 200 (right) knots

common boundaries (see Fig. 8.46). Modifications of this definition may be necessary when some regions are islands, or when the observation domain is divided into separate subdomains.

- On regular grids, one often uses nearest neighbors on the grid, e.g., the four or eight nearest neighbors. In this case, it is also possible to define neighborhoods of a higher order, as we have already seen when discussing bivariate polynomial splines. However, in the following, we limit ourselves to simple, direct neighborhoods.

In the following we will use the notation $s \sim r$ to denote that regions s and r are neighbors.

Similarly, as we proceeded in the case of random walks in Sect. 8.1.5, every region s is assigned its own regression coefficient $f_{geo}(s) = \gamma_s$, $s = 1, \dots, d$. Due to the large number of coefficients that usually results from this approach, we need an appropriate structure to regain smooth spatial effects and thus to reduce the effective number of parameters. In obtaining smooth spatial effects, the coefficients of nearby regions should not divert too strongly from each other. Thus, we create a penalty that is based on squared differences between the parameters of nearby regions. Consider the PLS criterion

$$\text{PLS}(\lambda) = \sum_{i=1}^n (y_i - f_{geo}(s_i))^2 + \lambda \sum_{s=2}^d \sum_{r \in N(s), r < s} (\gamma_r - \gamma_s)^2, \quad (8.35)$$

where $N(s)$ defines the set of neighbors for region s . The penalty consists of squared differences of all possible combinations of neighboring regions, where each combination is considered only once. This in fact yields a penalty that discourages large deviations of effects associated with neighboring regions.

To include this approach within the scope of general penalization methods, we first rewrite the PLS criterion in matrix notation. To do so, we define the design matrix $\mathbf{Z}$ with the entries

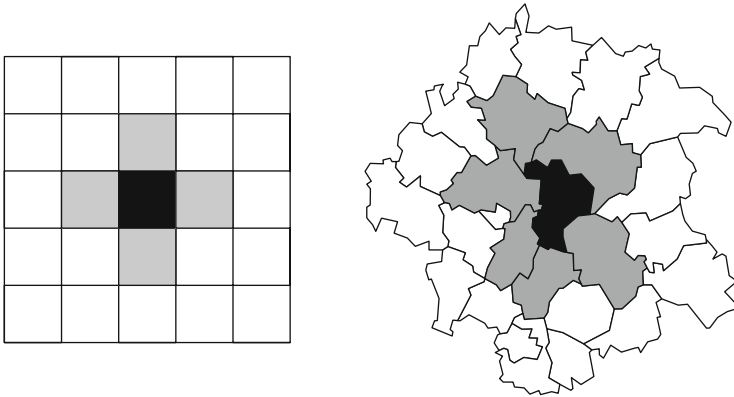


Fig. 8.46 First-order neighborhoods on a regular grid (*left*) and for irregular regional data (*right*). The neighbors of the *black* region are shaded in *grey*

$$\mathbf{Z}[i, s] = \begin{cases} 1 & \text{if } y_i \text{ was observed in region } s \text{ (i.e., } s_i = s \text{) and} \\ 0 & \text{otherwise.} \end{cases}$$

With this definition, we can express the vector of the function evaluations $\mathbf{f}_{geo} = (f_{geo}(s_1), \dots, f_{geo}(s_n))'$ as a linear model $\mathbf{Z}\boldsymbol{\gamma}$. In turn, the penalty consists of squared differences and can be compactly written as the quadratic form $\lambda\boldsymbol{\gamma}'\mathbf{K}\boldsymbol{\gamma}$ with

$$\mathbf{K}[s, r] = \begin{cases} -1 & s \neq r, s \sim r, \\ 0 & s \neq r, s \not\sim r, \\ |N(s)| & s = r. \end{cases}$$

By minimizing the PLS criterion, we then again obtain the PLS estimate $\hat{\boldsymbol{\gamma}} = (\mathbf{Z}'\mathbf{Z} + \lambda\mathbf{K})^{-1}\mathbf{Z}'\mathbf{y}$.

According to its definition, the penalty matrix $\mathbf{K}$ has the structure of an adjacency matrix, since the entries $\mathbf{K}[s, r]$ only differ from zero when s and r are neighbors. Efficient numerical methods for storing and processing $\mathbf{K}$ can be used since $\mathbf{K}$ is a sparse matrix. However, we should be aware that the order of the regions within the vector $\boldsymbol{\gamma}$ plays a crucial role for the structure of $\mathbf{K}$. Figure 8.47 illustrates this phenomenon for a large number of regions in two different orderings. In the left panel, the regions are simply arranged according to their number, whereas in the right panel, the arrangement results from the attempt to reorder the regions such that the nonzero elements are as close as possible to the diagonal, yielding a matrix that is close to a band matrix with small bandwidth. This ordering was achieved with the Cuthill–McKee algorithm; see, for example, [George and Liu \(1981\)](#), who also offer further details regarding numerically efficient algorithms for sparse matrices. [Rue and Held \(2005\)](#) also discuss some of these algorithms from a statistical point of view.

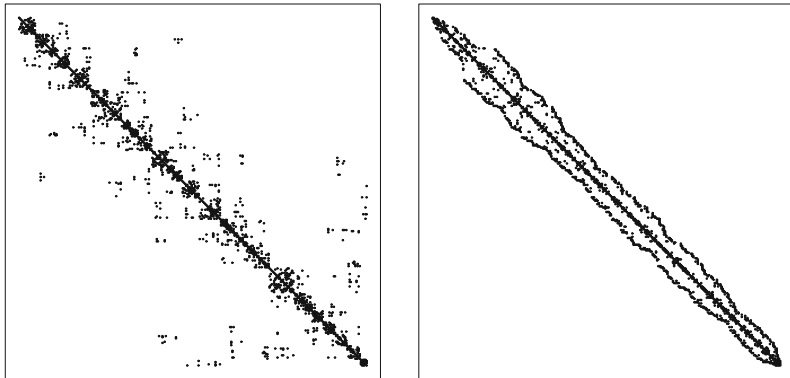


Fig. 8.47 Adjacency matrix for a large set of regions in two different orderings

Bayesian Model Formulation

As with the other penalization approaches discussed thus far, the above model formulation can also be interpreted in a Bayesian context. This yields so-called Markov random fields, which we will now introduce in a somewhat more general context. In the literature, penalization methods for discrete spatial locations are commonly motivated in this Bayesian framework.

We start with a general definition of Markov random fields (MRF) that generalize the temporal Markov property to the spatial situation that we discussed for random walks in Sect. 8.1.5 (p. 452). Let $D = \{1, \dots, s, \dots, d\}$ denote the set of all regions. Then, $\boldsymbol{\gamma} = \{\gamma_s, s \in D\}$ follows an MRF if the conditional distribution of γ_s given all other effects $\gamma_r, r \neq s$, depends only on its neighbors. The corresponding (conditional) density can then be written as

$$p(\gamma_s | \gamma_r, r \neq s) = p(\gamma_s | \gamma_r, r \in N(s)).$$

If we consider the model specification

$$y_i = f_{geo}(s_i) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2),$$

the aim is to assign an MRF to the distribution of $f_{geo}(s) = \gamma_s$. Gauß–Markov random fields (GMRF), i.e., MRF where the conditional distributions (and also the joint distribution) are normal, provide an adequate class of latent MRF to describe such spatial effects. One then often proceeds as follows: Start by specifying the conditional distributions $\gamma_s | \gamma_r, r \in N(s)$ and then derive the joint distribution of the entire vector $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_d)'$ from these conditional distributions. However, we must be aware that not all specifications of conditional distributions are actually allowed to obtain a valid joint distribution. At this point, however, we do not intend to discuss the theoretical considerations that are necessary for the formulation of

adequate classes of conditional distributions, since we are able to derive a valid specification directly from the PLS criterion.

This particular class has the form

$$\gamma_s \mid \gamma_r, r \in N(s) \sim N \left(\frac{1}{|N(s)|} \sum_{r:r \sim s} \gamma_r, \frac{\tau^2}{|N(s)|} \right), \quad (8.36)$$

where $|N(s)|$ is the number of neighbors for region s . According to this specification, the conditional expectation of the spatial effect in region s is given by the arithmetic mean of the neighboring regional effects. The variance τ^2 controls how much the spatial effect γ_s can deviate from this expectation. Thus, we obtain a stochastic formulation of the desired property that neighboring regions show a similar spatial effect. The joint distribution of all spatial effects can now be derived from the conditional distributions (which in this case actually define a system of consistent distributions) yielding the density

$$p(\boldsymbol{\gamma} \mid \tau^2) \propto \left(\frac{1}{\tau^2} \right)^{(d-1)/2} \exp \left(-\frac{1}{2\tau^2} \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma} \right). \quad (8.37)$$

This density is only defined upon proportionality since, in fact, the joint distribution is partially improper. The precision matrix $\mathbf{K}$ exactly corresponds to the penalty matrix from the previously given PLS criterion (8.35), and thus the Bayesian formulation (8.36) in fact leads to an equivalent model. In the literature, GMRFs with a partially improper distribution are also referred to as *intrinsic* GMRF (IGMRF). It is, however, also possible to define MRF with a proper joint distribution; see, for example, [Banerjee, Carlin, and Gelfand \(2003, Chap. 3\)](#).

Thus far, we have only used the spatial neighborhood information when constructing an MRF. It can also be meaningful to generalize this definition to some extent and appropriately weigh the influence of the neighbors. Therefore we extend Eq. (8.36) to

$$\gamma_s \mid \gamma_r, r \in N(s) \sim N \left(\sum_{r:r \sim s} \frac{w_{sr}}{w_{s+}} \gamma_r, \frac{\tau^2}{w_{s+}} \right),$$

with symmetric weights $w_{sr} = w_{rs}$ and $w_{s+} = \sum_{r:r \sim s} w_{sr}$. Now, the conditional expectation of γ_s is given by the weighted average of the neighboring coefficients. The following strategies are commonly used to define weights w_{sr} :

- Use the same weight for all neighbors, i.e., $w_{sr} = 1$. This specification yields our original MRF definition.
- Use weights that are inversely proportional to the distance of the centroids, e.g., $w_{sr} \propto \exp(-d(s, r))$, where $d(s, r)$ defines the Euclidean distance between the centroids of the regions s and r .
- Use weights proportional to the length of the common boundary of regions s and r .

If weights are to be considered, then the penalty or precision matrix $\mathbf{K}$ has to be modified, taking the form

$$\mathbf{K}[s, r] = \begin{cases} -w_{sr} & s \neq r, s \sim r, \\ 0 & s \neq r, s \not\sim r, \\ w_{s+} & s = r. \end{cases}$$

Example 8.13 Munich Rent Index—Analysis of Spatial Dependence

Figure 8.48 shows the MCMC-based estimate of an unweighted IGMRF for the Munich rent index data. If we compare the estimated effect with the separate means obtained in Fig. 8.35 (p. 503), we clearly recognize the smoothing effect that is introduced by the IGMRF assumption. This approach has the advantage that, for districts where only a few apartments were sampled, the estimated net rent is stabilized and less variable due to the integration of neighborhood information. Moreover, the classification of subquarters into areas with rent above or below average is more sensible based on smoothed effects.

Surprisingly, the results shown in Fig. 8.48 also indicate that the rent per square meter for apartments in the city center is smaller than for those in the suburbs. This demonstrates that the sole use of a regional effect can lead to results that are difficult to interpret due to the fact that other important explanatory variables are not taken into consideration. In our example, the reason for the surprising result is that the apartments in the city center are typically older than those in the suburbs. If we take this effect into consideration, as we will in some examples of Chap. 9, the effect changes and the apartments in the city center will actually turn out to be more expensive than those in the suburbs after adjusting for year of construction. This shows that the *simultaneous* consideration of covariate effects and regional effects is necessary to obtain improved estimates.

△

Spatially Autoregressive Processes

In econometrics, MRFs are widely applied in a variety of models other than the one we have considered thus far. In applications, autoregressive processes in time are generalized such that the MRF is assumed for the response variable y instead of a latent spatial process. This leads to spatially autoregressive processes (SAR), with the simplest special case given by

$$y_s = \alpha \sum_{r \in N(s)} w_{sr} y_r + \varepsilon_s, \quad \varepsilon_s \sim N(0, \sigma^2),$$

with autoregressive parameter $\alpha \in [0, 1)$ and symmetric weights w_{sr} fulfilling $w_{ss} = 0$. We then obtain the conditional distributions

$$y_s | y_r, r \neq s \sim N \left(\alpha \sum_{r \in N(s)} w_{sr} y_r, \sigma^2 \right),$$

such that the vector of the response variable $\mathbf{y} = \{y_s, s = 1, \dots, d\}$ appears to follow an MRF. If covariate effects are to be included in a SAR model, this is

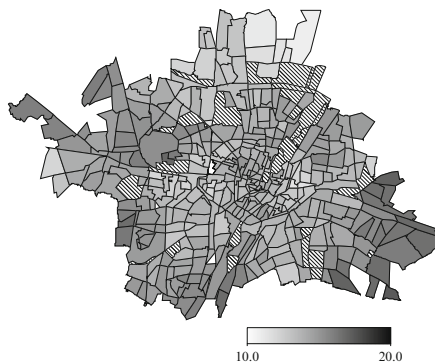


Fig. 8.48 Munich rent index: spatial effect obtained with an IGMRF without weights estimated with MCMC in a fully Bayesian framework

typically achieved by applying the SAR specification to the residuals $y_s - \mathbf{x}'_s \boldsymbol{\beta}$. Alternatively, it is also possible to add covariate effects directly to the SAR model, yielding

$$y_s = \mathbf{x}'_s \boldsymbol{\beta} + \alpha \sum_{r \in N(s)} w_{sr} y_r + \varepsilon_s, \quad \varepsilon_s \sim N(0, \sigma^2).$$

However, the interpretation of covariate effects differs greatly between the two model specifications: The covariates either effect the response adjusted by the regional effects or the response variable directly.

Since SAR models do not fit into the general scope of penalization methods that we are considering, we will not discuss them in further detail (see, e.g., [Anselin \(1988\)](#) or [Banerjee et al. \(2003\)](#), Chap. 3, for more information).

8.2.5 Summary of Roughness Penalty Approaches

We now provide a brief summary of the bivariate smoothing methods that have been discussed thus far. As we have seen, bivariate basis function methods, stationary Gaussian fields, and GMRFs can all be subsumed in the general framework introduced in Sect. 8.1.3. In every case, we obtained a linear model of the form

$$\mathbf{y} = \mathbf{Z} \boldsymbol{\gamma} + \boldsymbol{\varepsilon},$$

with a large number of regression coefficients, representing either the coefficients of the basis functions or the spatial effects. To regularize estimation of $\boldsymbol{\gamma}$, we introduced quadratic penalties

$$\lambda \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}.$$

Depending on the method, the penalties were constructed based on either differences of coefficients, derivative operators, correlations, or neighborhood structures. Due to the general formulation we can use the methods discussed in Sect. 8.1.9 to determine a data-driven choice for the smoothing parameter.

8.14 Gauß–Markov Random Fields

Model

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\epsilon}$$

with

$$\mathbf{Z}[i, s] = \begin{cases} 1 & \text{if } y_i \text{ was observed in region } s \text{ (i.e., } s_i = s) \text{ and} \\ 0 & \text{otherwise} \end{cases}$$

and $\boldsymbol{\gamma} = (f_{geo}(s_1), \dots, f_{geo}(s_d))'$.

IGMRF for the Spatial Effect

$$\gamma_s \mid \gamma_r, r \in N(s) \sim N\left(\sum_{r:r \sim s} \frac{w_{sr}}{w_{s+}} \gamma_r, \frac{\tau^2}{w_{s+}}\right)$$

with $\gamma_s = f_{geo}(s)$ and $w_{sr} = w_{rs}$. The (partially improper) joint distribution is

$$p(\boldsymbol{\gamma} \mid \tau^2) \propto \left(\frac{1}{\tau^2}\right)^{(d-1)/2} \exp\left(-\frac{1}{2\tau^2} \boldsymbol{\gamma}' \mathbf{K} \boldsymbol{\gamma}\right)$$

with

$$\mathbf{K}[s, r] = \begin{cases} -w_{sr} & s \neq r, s \sim r, \\ 0 & s \neq r, s \not\sim r, \\ w_{s+} & s = r. \end{cases}$$

In the bivariate case, differences between the approaches are more apparent than in the univariate case. First of all, we generally distinguish between continuous and discrete regional information, where specific approaches were developed in either context. In our case, these are basis function approaches and stationary Gaussian fields on the one hand, and Markov random fields on the other hand. However, the distinction is not as obvious as it might seem. Methods which require continuous spatial information can be used, e.g., for regional data, where the centroids of the regions are used as the locations in the analysis. Models that require discrete spatial information can also be used in the continuous case, for example, by discretizing the observation domain on a regular grid.

Secondly, if we compare the different approaches regarding their numerical properties, the results vary depending on which inferential concept is used. If we use MCMC methods, it is necessary to solve a high-dimensional system of equations

for the simulation of $\boldsymbol{\gamma}$ in every iteration. With the help of sophisticated methods for sparse matrices, these calculations can be performed efficiently. Sparse cross product matrices $\mathbf{Z}'\mathbf{Z}$ and penalty matrices $\mathbf{K}$ result, in particular, for Gauß–Markov random fields and bivariate P-splines. However, if stationary Gaussian fields or radial bases are used, then $\mathbf{Z}'\mathbf{Z}$ and $\mathbf{K}$ are typically dense. Recently, [Lang et al. \(2012\)](#) proposed a highly efficient Gibbs sampler even in the case of dense matrices. The alternative sampling scheme works with a transformed parameterization such that the cross product of the design matrix and the penalty matrix are *diagonal* resulting in a diagonal posterior precision matrix.

If inference is based on a mixed model representation, then the structure of the penalty matrix $\mathbf{K}$ is less important since the penalty matrix only needs to be evaluated once during the reparameterization. Rather, the dimension of the parameter vector $\boldsymbol{\gamma}$ is more important. This is the reason why radial basis function approaches and stationary Gaussian fields, without an adequate strategy for choosing a small set of knots, are less useful for this method than bivariate P-splines and Gauß–Markov random fields. If we, however, use a strategy for reducing the number of knots, it is often possible to reduce the dimension of $\boldsymbol{\gamma}$ such that every estimation method shows a similar level of complexity.

8.2.6 Local and Adaptive Smoothing

Local smoothers are also often used for the analysis of interactions and spatial effects, relying on extensions of the methods discussed in Sect. 8.1.7. This extension is particularly straightforward for nearest neighbor estimates, since the k -nearest neighbors can also be determined in bivariate problems using, for example, the Euclidean distance. Consequently, it is also possible to generalize the loess procedure, as well as the Nadaraya–Watson estimate that is based on kernel density estimates, to bivariate smoothing. The choice of an optimal smoothing parameter can be chosen either via a grid search or with the help of adequate numerical procedures; see, for example, [Kauermann and Opsomer \(2004\)](#). However, the use of local smoothing procedures as building blocks in more complex models is difficult when combining, for example, nonparametric effects and random effects. This is the reason why we will not discuss local methods in detail, but instead refer to the literature listed in Sect. 8.4.

Adaptive procedures can also be efficiently used for bivariate smoothing and more complex models. These procedures have the particular advantage that not only are they able to determine how to model the joint effect of z_1 and z_2 but they also are able to detect whether or not an interaction effect is in fact present. This can be especially useful for the extensions presented in Chap. 9, where potentially even more than two effects need to be determined simultaneously.

For the MARS algorithm, we obtain a bivariate extension by examining, in every iteration, the univariate basis functions for z_1 and z_2 , as well as their corresponding interactions for potential improvements of the model fit. This results in a sequence of models, which can comprise complex interaction models, as well as simple

models containing only univariate covariate effects. With the help of the model choice criterion, we adaptively decide which of the models is the most adequate for modeling the effects of the covariates.

For regression trees, extensions for more complex models result when considering, in every iteration, partitions in all variables. Hence, in each iteration step, we optimally choose both the variable to be split and the split point for this variable. Thus, we obtain adaptive splits of the data into homogeneous sets of similar response values, which correspond to a piecewise constant function estimate for $f(z_1, z_2)$.

In a similar way, we can also generalize the Bayesian adaptive procedures. Since we will concentrate on penalization methods in Chap. 9, we will not discuss adaptive methods in more detail at this point. Further information about adaptive procedures will be provided at the end of this chapter in Sect. 8.4.

8.3 Higher-Dimensional Smoothing

In principle, the ideas developed in Sect. 8.2 can also be applied for modeling higher-dimensional surfaces, i.e., in models of the form

$$y = f(z_1, \dots, z_q) + \varepsilon.$$

For example, higher-dimensional tensor product splines result from considering all possible interactions of univariate splines for $z_1, \dots, z_q$. The respective penalty matrices can then be constructed analogous to those in Sect. 8.2.1 based on Kronecker products leading to

$$\mathbf{I}_{d_3} \otimes \mathbf{I}_{d_2} \otimes \mathbf{K}_1 + \mathbf{I}_{d_3} \otimes \mathbf{K}_2 \otimes \mathbf{I}_{d_1} + \mathbf{K}_3 \otimes \mathbf{I}_{d_2} \otimes \mathbf{I}_{d_1},$$

in the special case of $q = 3$, where $\mathbf{K}_1$, $\mathbf{K}_2$, and $\mathbf{K}_3$ denote univariate penalty matrices and $\mathbf{I}_d$ are identity matrices of appropriate dimension. For radial bases and kriging terms that are based on isotropic correlation functions, the construction of higher-dimensional surfaces is even simpler. In such cases, the basis functions rely only on the Euclidean distance between two points, which is defined in $\mathbb{R}^q$, as well as in the special case of $\mathbb{R}^2$. By extending the concept of neighborhoods, the construction of Markov random fields can also immediately be applied to problems in higher dimensions.

Nevertheless, regardless of the chosen approach, problems occur when estimating higher-dimensional functions nonparametrically. In general, such functions rely on a large number of parameters and the respective estimation procedures are computationally expensive. For example, a tensor product spline described by a 20-dimensional basis in each dimension has already $20^3 = 8,000$ parameters in $\mathbb{R}^3$. Even though a penalty reduces the effective number of parameters, we are still required to solve an 8,000-dimensional system of equations (often multiple times) for obtaining the estimates.

Another difficulty arising in higher-dimensional approaches, is the “curse of dimensionality.” Consider the following problem: Suppose we are given a q -dimensional unit cube (i.e., a cube with edges of length 1), where all data are uniformly distributed within this cube. What will be the edge length l of a partial cube that contains a fraction r of the data? For $q = 1$, we obviously have $l = r$, whereas for $q = 2$, we have $l = \sqrt{r}$. In general, it follows that

$$l = r^{\frac{1}{q}},$$

and therefore for $r = 0.1$

$$\begin{array}{ll} q = 1 & l = 0.1 \\ q = 2 & l \approx 0.3 \\ q = 3 & l \approx 0.47 \\ \vdots & \vdots \\ q = 10 & l \approx 0.8. \end{array}$$

For a three-dimensional estimation problem, this implies that the partial cube must already have an edge length of about 0.5 to contain 10 % of the data. Obviously, our intuition of “localness” no longer works in higher dimensions. Moreover, the curse of dimensionality suggests that in higher-dimensional smoothing problems, we only obtain a good coverage of the observation domain when the number of parameters for each dimension is actually greater than what we used in univariate smoothing problems. Consequently, we again face the problem that a large number of parameters need to be estimated.

This is the reason why, in practice, we rarely consider surfaces that have a dimension higher than $q = 2$. Rather, we assume an additive structure for the function $f(z_1, \dots, z_q)$, leading to

$$f(z_1, \dots, z_q) = f_1(z_1) + \dots + f_q(z_q).$$

This yields the class of additive models and their extensions, which we will examine in detail in Chap. 9.

8.4 Bibliographic Notes

In the following, we give an overview of selected literature, which will provide either more detailed or more general information regarding some aspects of Chap. 8. Many of the references listed will also be of interest for the extensions to be discussed in Chap. 9.

[Hastie and Tibshirani \(1990\)](#) introduce and systematically examine univariate smoothing procedures within the scope of additive models and, in particular, discuss general questions regarding linear smoothing procedures in detail. [Fahrmeir and Tutz \(2001, Chap. 5\)](#) provide a good overview of modern extensions. The overview by [Wand \(2000\)](#) compares some univariate smoothing procedures via simulation,

and also includes adaptive procedures. Wood (2006) describes nonparametric regression models based on penalization approaches and splines, along with an introduction to the R package *mgcv*. Mathematical properties of splines are discussed in De Boor (2001) and Dierckx (1993); see also Eilers and Marx (2010) for a nice (and for most practical purposes sufficient) introduction to splines. Detailed descriptions of local regression models are given in Härdle (1990), Fan and Gijbels (1996), Loader (1999), and Härdle, Müller, Sperlich, and Werwatz (2004).

Extensive discussions of the connection between penalized splines and mixed models used in Sect. 8.1.9 can be found in Ruppert et al. (2003), Wand (2003), or Kauermann (2006). Fahrmeir, Kneib, and Lang (2004) describe this connection for general penalization approaches and in particular for smoothing procedures developed in spatial statistics. Extensions for survival models are discussed in Kauermann and Khomski (2006) or Kneib and Fahrmeir (2007). Bayesian approaches for the selection of smoothing parameters that are based on MCMC simulations are discussed in detail in Lang and Brezger (2004), Brezger and Lang (2006), and Lang et al. (2012).

Friedman (1991), Stone, Hansen, Kooperberg, and Truong (1997), and Hansen and Kooperberg (2002) discuss adaptive approaches that are based on frequentist model choice strategies. The articles by Biller and Fahrmeir (2001), Biller (2000), Yau, Kohn, and Wood (2003), Denison et al. (1998), or DiMatteo, Genovese, and Kass (2001) describe corresponding Bayesian approaches. An introduction to regression trees is provided in Tutz (2011, Chap. 9); the classical reference in the field of regression and classification trees is Breiman, Friedman, Stone, and Olshen (1984). A conceptually powerful extension of regression trees is proposed in Hothorn, Hornik, and Zeileis (2006). Here the leaves of a tree are not constant but full linear or generalized linear models. As already mentioned in Sect. 8.1.10, regression trees are unstable due to the fact that even small changes within the data structure can result in completely different trees. Resampling methods, which build not only one but an entire ensemble of regression trees, mitigate this instability. Examples are bagging (Breiman, 1996) or random forests (Breiman, 2001). We can actually employ similar methods in regression and smoothing procedures based on extensions of boosting algorithms; see, for example, Bühlmann and Yu (2003, 2006), Tutz and Binder (2006), or Bühlmann and Hothorn (2007).

Finally, we want to refer to some approaches which we were unable to discuss in detail in this chapter. Wavelets are a smoothing procedure similar in spirit to Fourier analysis, which can be used especially for the representation of functions with pronounced (local) variability. In principle, we can also understand wavelets as a basis function approach, but estimation relies on efficient algorithms, which employ special properties of wavelet basis functions, and therefore differ greatly from the methods that we discussed. Wavelets are especially interesting from a numerical point of view but have the disadvantage that complex mathematical tools are required. Hastie et al. (2009) offer a brief introduction to wavelets, while Ogden (1997) or Gençay, Selçuk, and Whitcher (2002) provide more challenging descriptions.

Another class of nonparametric regression models is considered in the context of functional data analysis. In such a setting, the parameters and/or the response variable are viewed as functions, and methods are used that are especially designed for this situation. [Ramsay and Silverman \(2005\)](#) provide a detailed introduction while [Ramsay and Silverman \(2002\)](#) describe functional data analysis within the scope of some case studies.

[Banerjee et al. \(2003\)](#) and [Schabenberger and Gotway \(2005\)](#) are introductory textbooks on spatial statistics, i.e., the methods discussed in Sect. 8.2. [Rue and Held \(2005\)](#), as well as [Chiles and Delfiner \(1999\)](#) and [Stein \(1999\)](#), elaborate on aspects of Gauß–Markov random fields and kriging, respectively. [Ecker and Gelfand \(2003\)](#) and [Zimmermann \(1993\)](#) discuss possibilities as how to consider anisotropy in geostatistical models. [Nychka \(2000\)](#) offers a rigorous mathematical access to the relationship between kriging and smoothing procedures that are based on radial basis functions, while [Eilers and Marx \(2003\)](#) and [Dierckx \(1993\)](#) discuss bivariate polynomial splines in more detail.